THE TWO-MINUS ADJOINT-CHARACTER CASE OF GINIBRE'S Q_3 CONDITION PARTS I–II: CLASSIFICATION AND EXACT CERTIFICATES

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ABSTRACT. Let G be a compact connected Lie group with simple Lie algebra, let χ_{ad} be its adjoint character, and define

$$Q_3^G(n) = \int_{G\times G} (\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^2 (\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n \, d\,\mathrm{Haar}(g)\, d\,\mathrm{Haar}(h).$$

We prove $Q_3^G(n) \geq 0$ for every such G and every integer $n \geq 0$. This is the specialization of Ginibre's condition (Q3) in which every function is χ_{ad}, exactly two factors carry a minus sign, and the other n factors carry plus signs; it is not a proof of the full cone-valued Q3 condition. The moment formula turns the odd-exponent problem into inequalities for invariant multiplicities in tensor powers of the adjoint representation. The proof follows the classification: type A uses transition ratios; types B, C, D use stable moments, finite correction prefixes, and uniform trace tails; and the exceptional types use exact character-ring prefixes and rectangular analytic tails. All finite decisions are made with exact integer or rational arithmetic or with outward-rounded intervals. The computer-assisted steps are stated as finite mathematical contracts, with their active domains and proof dependencies separated from certificate generation and transcript authentication.

Remark 0.1 (Companion architecture)**.** This journal manuscript consolidates Parts I–II of the proof: the classification reduction, family closures, compact correction-prefix contract, its exact finite-verification proof, the local half-stable bridge, and the accepted-certificate boundary. No theorem imports a result from `paper_full.pdf`; that optional derivation archive expands the offset-by-offset combinatorics and certificate ledgers from the same source. Part III, `full_q3_extension.pdf`, imports the theorem proved here and establishes the full signed-product hierarchy on the adjoint-generated cone.

Remark 0.2 (Publication proof spine and author self-audits)**.** The load-bearing dependency spine for Parts I–III is listed in `PUBLICATION_PROOF_SPINE.md` and, for the present theorem, in the final-facing dependency map below. The optional derivation archive and all explicitly marked diagnostic and superseded routes remain provenance and are not consumed. The archived historical report and the programs under `referee_audits/` are author-controlled self-audits. They provide redundant checks and source mappings; they are not independent peer review and are not premises of any theorem.

1. Introduction

Ginibre's condition (Q3) asks that, for a probability measure σ and every finite family $f_1, \ldots, f_r$ in a specified multiplicative convex cone, each signed product integral

$$\iint \prod_{i=1}^{r} \bigl(f_i(x) \pm f_i(y)\bigr)\, d\sigma(x)\, d\sigma(y)$$

be nonnegative; see [2, §1, condition (Q3), equation (1.2)]. The present theorem addresses one precise noncommutative specialization. We take normalized Haar measure on a compact connected group with simple Lie algebra, set every f_i equal to the adjoint character, choose exactly two minus signs, and choose n plus signs. The resulting integral is $Q_3^G(n)$. We do not assert Q3 for a cone of functions, for unequal characters, or for four or more minus factors.

Even n is immediate from pointwise nonnegativity. Odd n is not: the adjoint character takes negative values, and expansion produces differences of large invariant-tensor multiplicities. The moment formula in lemma 2.3 converts the problem into exact inequalities for $m_k^G = \mathrm{mult}(\mathbf{1}, \mathrm{ad}^{\otimes k})$. The proof then separates into the classification families. Type A is governed by stable derangement moments and finite-rank transition ratios. The classical types use the exact stable trace law of Diaconis–Shahshahani, orthogonal and symplectic identities of Rains, finite

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correction bridges, and uniform trace tails. The exceptional types combine finite character-ring intervals with analytic rectangular tails.

Computer assistance enters only after the mathematical reduction has specified a finite integer, rational, or interval inequality. A source table is not accepted merely because its hash matches. For every low classical ledger and every theorem-consumed exceptional ledger, the clean replay regenerates the claimed moments from the Cartan datum by lemmas 2.4 and 2.5. The exceptional ranges end at degrees $38, 65, 80, 70, 100$ for G_2, F_4, E_6, E_7, E_8, respectively. A separate control then checks those values termwise against the independent ancillary tables of Bourjaily–Plesser–Vergu [14, ancillary file `adjoint_tensor_ranks.tex`]; agreement with that published source is corroboration, not a premise. Likewise, every stable classical moment used in a finite correction is regenerated from the exact three-term recurrence reconstructed by the exact source replay; the archived OEIS row is checked against that regeneration and is not used as a mathematical premise. Exact discrete calculations use GMP integers or rationals, and analytic comparisons use outward-rounded MPFR intervals.

Parts I–II are consolidated in this manuscript. The final-facing reductions and family closures consume a compact finite-correction contract and the accepted certificate ledger below. The detailed computational supplement prints the line-by-line correction derivations and historical ledgers; its numbered correction-prefix proofs and half-stable bridge reduction are incoming nodes for the compact residual closure, while explicitly diagnostic and superseded material adds no hypothesis or incoming edge. The replay contract states which inputs are regenerated, which legacy certificates are audited without a full generator rerun, and which archived files are retained only as diagnostic history.

The choice of the adjoint character is not merely a source of numerical examples. Its powers are the characters of tensor powers of the adjoint module, so every Haar moment is an invariant-tensor multiplicity and can be attacked uniformly through the classification. Part III takes the smallest uniformly closed multiplicative convex cone generated by this character. Ginibre's Proposition 1 promotes the single-generator signed-product inequalities to exactly this generated cone; any claim on a larger cone requires additional input not supplied by that promotion. The resulting theorem is thus a complete statement about the adjoint tensor category, while remaining strictly narrower than a theorem for all central positive-definite functions.

Contribution and relation to prior results. Ginibre's Proposition 1 supplies a promotion mechanism once the required signed products are known on a self-conjugate generating set; it does not establish the repeated-adjoint inequality proved here. The exact stable trace laws of Diaconis–Shahshahani, the power-map identities of Rains, and the character bounds of Garibaldi–Guralnick–Rains provide inputs to individual family arguments, but none of those results compares the two moment diagonals in lemma 2.3 for every exponent and every simple type. Likewise, the adjoint tensor-rank calculations of Bourjaily–Plesser–Vergu provide comparison data for exceptional moments, not the sign theorem.

The contribution of Parts I–II is the uniform classification theorem and the mechanism joining three regimes that do not overlap automatically: stable invariant-tensor moments, finite-rank corrections, and analytic tails. The repeated-adjoint and exactly-two-minus restrictions are essential parts of the statement, not abbreviations for a result about arbitrary characters. In particular, this paper makes no claim for unequal irreducible characters or for all positive-definite class functions. The separate Part III uses Ginibre's promotion only after importing the theorem proved here; it is not used to justify the present theorem or its novelty.

The computer calculations are proof devices rather than the mathematical claim. Each active calculation has a numbered finite contract, and the compact residual B/C contract table below records the exact theorem labels, rank ranges, and moment offsets supplied by the detailed proof.

2. Statement and General Reductions

Definition 2.1 (Adjoint Q_3 functional)**.** For a compact connected Lie group G with simple Lie algebra and normalized Haar probability measure, set

$$Q_3^G(n) = \int_{G\times G} (\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^2 (\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n \, d\,\mathrm{Haar}(g)\, d\,\mathrm{Haar}(h)$$

for $n \in \mathbb{Z}_{\geq 0}$, where χ_{ad} is the adjoint character.

Theorem 2.2 (Two-minus adjoint-character Q3 nonnegativity)**.** *For every compact connected Lie group G with simple Lie algebra and every integer $n \geq 0$,*

$$Q_3^G(n) \geq 0.$$

Lemma 2.3 (Moment formula)**.** *Let*

$$m_k^G = \int_G \chi_{\mathrm{ad}}(g)^k \, d\,\mathrm{Haar}(g) = \mathrm{mult}(\mathbf{1}, \mathrm{ad}^{\otimes k}).$$

Then

$$Q_3^G(n) = 2\sum_{k=0}^{n} \binom{n}{k} \big(m_{k+2}^G m_{n-k}^G - m_{k+1}^G m_{n-k+1}^G\big).$$

Proof. Expand $(\chi_{\mathrm{ad}}(g)+\chi_{\mathrm{ad}}(h))^n$ by the binomial theorem, multiply by $(\chi_{\mathrm{ad}}(g)-\chi_{\mathrm{ad}}(h))^2$, and integrate term by term over $G\times G$. Schur orthogonality identifies $\int_G \chi_{\mathrm{ad}}(g)^k \, d\,\mathrm{Haar}(g)$ with the invariant multiplicity $\mathrm{mult}(\mathbf{1}, \mathrm{ad}^{\otimes k})$. □

Lemma 2.4 (Racah–Speiser adjoint tensor step)**.** *Let V_λ be an irreducible representation of a compact connected semisimple group, let ρ be the half-sum of the positive roots, and let $a(\mu)$ be the multiplicity of the weight μ in the adjoint representation. For every adjoint weight μ, proceed as follows. If $\lambda+\mu+\rho$ lies on a Weyl wall, assign zero. Otherwise let w be the unique Weyl element for which $w(\lambda+\mu+\rho)$ is strictly dominant, put*

$$\nu = w(\lambda+\mu+\rho) - \rho,$$

and assign $\mathrm{sgn}(w)a(\mu)$ to V_ν. After equal dominant weights are collected, the resulting signed sum is the irreducible decomposition of $V_\lambda \otimes \mathrm{ad}$.

Proof. For a weight η write $A_\eta = \sum_{w\in W} \mathrm{sgn}(w) e^{w\eta}$. The Weyl character formula gives $\chi_\lambda = A_{\lambda+\rho}/A_\rho$. Since the adjoint weight multiset is Weyl invariant,

$$A_{\lambda+\rho}\chi_{\mathrm{ad}} = \sum_{\mu} a(\mu) A_{\lambda+\mu+\rho}.$$

An alternant vanishes when its index is singular. If the index is regular, moving it to the strict dominant chamber changes the alternant by $\mathrm{sgn}(w)$, and division by A_ρ gives the character specified in the statement. Collecting equal dominant indices proves the formula. This is the Racah–Speiser procedure; the same shifted-wall convention is described in [14, Appendix B, especially (B.1)–(B.6)]. □

Lemma 2.5 (Character-pairing reconstruction of moments)**.** *Write the exact irreducible decomposition*

$$\chi_{\mathrm{ad}}^j = \sum_{\lambda\in P_+} c_j(\lambda)\chi_\lambda.$$

Then, for every $j \ge 0$,

$$m_{2j}^G = \sum_{\lambda\in P_+} c_j(\lambda)^2, \qquad m_{2j+1}^G = \sum_{\lambda\in P_+} c_j(\lambda)c_{j+1}(\lambda).$$

Consequently all moments through degree M can be reconstructed after iterating lemma 2.4 only through tensor power $\lceil M/2\rceil$.

Proof. The adjoint character is real, so

$$m_{a+b}^G = \int_G \chi_{\mathrm{ad}}(g)^a \overline{\chi_{\mathrm{ad}}(g)^b} \, d\,\mathrm{Haar}(g) = \left\langle \chi_{\mathrm{ad}}^a, \chi_{\mathrm{ad}}^b \right\rangle_{L^2(G)}.$$

Irreducible-character orthonormality makes the last expression $\sum_\lambda c_a(\lambda)c_b(\lambda)$. Taking $(a,b)=(j,j)$ and $(j,j+1)$ proves the two identities. □

Lemma 2.6 (Positivity of adjoint moments)**.** *The adjoint character is real-valued, and*

$$m_0^G = 1, \qquad m_1^G = 0, \qquad m_2^G = 1,$$

while

$$m_k^G = \dim\big((\mathfrak{g}_{\mathbf{C}}^{\otimes k})^G\big) \quad \textit{is a positive integer for every } k \ge 2.$$

In particular, every transition ratio used below has a nonzero positive denominator.

Proof. The adjoint action preserves a positive-definite real inner product B on the compact real form $\mathfrak{g}$, so its complexification is self-dual and its character is real. The compact-semisimple/complex-semisimple correspondence and decomposition into simple root systems in [1, Chapter V, §5, pp. 209–215, especially (5.5) and Theorem (5.8)] show that $\mathfrak{g}_{\mathbf{C}}$ is a complex simple Lie algebra. If a complex subspace of $\mathfrak{g}_{\mathbf{C}}$ is invariant under G, connectedness of G and differentiation make it invariant under $\mathrm{ad}(\mathfrak{g}_{\mathbf{C}})$; it is therefore an ideal. Thus the complexified adjoint representation is irreducible.

Character orthogonality now gives the displayed invariant-space formula and, because the character is real,

$$m_2^G = \int_G |\chi_{\mathrm{ad}}(g)|^2 \, d\,\mathrm{Haar}(g) = 1.$$

Degree zero consists of the scalars. A degree-one invariant is a vector fixed by the adjoint action, hence lies in the center of $\mathfrak{g}_{\mathbf{C}}$; simplicity gives $m_1^G = 0$. The tensor B is a nonzero invariant tensor of degree two. Simplicity and non-abelianness imply that the Cartan tensor

$$\omega(X, Y, Z) = B(X, [Y, Z])$$

is a nonzero invariant tensor of degree three. Every integer $k \geq 2$ has the form $2a + 3b$ with $a, b \geq 0$; tensoring a copies of B and b copies of ω, and using B to identify $\mathfrak{g}$ with its dual, gives a nonzero invariant tensor in degree k. Hence $m_k^G \geq 1$. $\square$

Lemma 2.7 (Central quotient invariance)**.** *If $\pi : \widetilde{G} \to G$ is a finite central cover of compact connected Lie groups with simple Lie algebra, then*

$$Q_3^{\widetilde{G}}(n) = Q_3^G(n)$$

for every $n \geq 0$.

Proof. The adjoint characters satisfy $\chi_{\mathrm{ad}}^{\widetilde{G}} = \chi_{\mathrm{ad}}^G \circ \pi$. The push-forward of normalized Haar measure under π is normalized Haar measure. Apply this in the two variables of the defining integral. $\square$

Lemma 2.8 (Reduction to the simple Lie algebra)**.** *Let G be a compact connected Lie group with simple Lie algebra $\mathfrak{g}$. There is a compact simply connected Lie group G_{sc} with Lie algebra $\mathfrak{g}$ and a finite central covering $G_{\mathrm{sc}} \to G$. Consequently, if G_1 and G_2 are compact connected Lie groups with isomorphic simple Lie algebras, then*

$$Q_3^{G_1}(n) = Q_3^{G_2}(n)$$

for every $n \geq 0$.

Proof. A simple Lie algebra is semisimple. For a compact connected semisimple group, the universal covering group is compact and its kernel is finite and central; moreover, the simply connected compact group is determined by its Lie algebra. These statements are recorded in [1, Chapter V, Remark (7.13) and §8, p. 232]. Thus $G_{\mathrm{sc}} \to G$ has the asserted properties. Isomorphic simple Lie algebras give isomorphic simply connected compact covering groups, so two applications of lemma 2.7 prove the displayed equality. $\square$

Lemma 2.9 (Even exponents)**.** *For every G and every even $n \geq 0$, $Q_3^G(n) \geq 0$.*

Proof. For even n, both $(\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^2$ and $(\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n$ are pointwise nonnegative. $\square$

Definition 2.10 (Chain difference)**.** For odd exponents define

$$D_G(n) = Q_3^G(n+2) - 4Q_3^G(n).$$

Equivalently, with $n = 2m + 1$,

$$D_G(2m+1) = Q_3^G(2m+3) - 4Q_3^G(2m+1).$$

Lemma 2.11 (Chain propagation)**.** *Let $0 \leq m_0 \leq m_1$. Assume $D_G(2m+1) \geq 0$ for $m = m_0, m_0+1, \ldots, m_1$ and $Q_3^G(2m_0+1) \geq 0$. Then*

$$Q_3^G(2m+1) \geq 0$$

for every $m_0 \leq m \leq m_1 + 1$.

Proof. The recurrence

$$Q_3^G(2m+3) = 4Q_3^G(2m+1) + D_G(2m+1)$$

propagates nonnegativity one odd exponent at a time. $\square$

3. Type A: The Adjoint Representation of SU(N)

This section proves the type A branch. Since finite central quotients do not change Q_3 by lemma 2.8, it suffices to use SU(N) as the representative of type A_{N-1}.

Theorem 3.1 (Stable-rank type A branch)**.** *For every $N \geq 2$ and every $n \geq 0$ satisfying $N \geq n+2$,*

$$Q_3^{\mathrm{SU}(N),\mathrm{ad}}(n) \geq 0.$$

Proof. Write $X = |\operatorname{tr} U|^2$ for U Haar-distributed in SU(N). For $0 \leq k \leq N$, scalar invariance identifies the SU(N) integral of X^k with the U(N) integral. Schur expansion and character orthogonality give

$$\mathbf{E}X^k = \sum_{\lambda \vdash k,\ \ell(\lambda) \leq N} (\dim \mathrm{Sp}_\lambda)^2.$$

When $k \leq N$ the length condition is void, and the regular representation of the symmetric group gives $\mathbf{E}X^k = k!$. Since $\chi_{\mathrm{ad}}(U) = X - 1$, this gives

$$m_r^{\mathrm{SU}(N)} = \sum_{j=0}^{r} (-1)^{r-j} \binom{r}{j} j! = !r \qquad (0 \leq r \leq N),$$

where $!r$ is the derangement number. If $N \geq n+2$, all moments appearing in lemma 2.3 are in this stable range.

Let Y have the exponential distribution of mean 1, and set $A = Y - 1$. The exponential generating function $M(z) = \mathbf{E}e^{zA} = e^{-z}(1-z)^{-1}$ satisfies

$$M''(z)M(z) - M'(z)^2 = M(z)^2(\log M)''(z) = \frac{e^{-2z}}{(1-z)^4}.$$

Thus $Q_3^{\mathrm{SU}(N)}(n)/2$ is the nth exponential coefficient of $e^{-2z}(1-z)^{-4}$. If these coefficients are denoted q_n, then $q_0 = 1$, $q_1 = 2$, and differentiating gives

$$q_{n+1} = (n+2)q_n + 2nq_{n-1} \qquad (n \geq 1).$$

Hence $q_n > 0$ for all $n \geq 0$. ∎

Definition 3.2 (Type A transition ratios)**.** For $N \geq 2$, put

$$m_k^{(N)} = \int_{\mathrm{SU}(N)} \chi_{\mathrm{ad}}(U)^k \, dU, \qquad \rho_k^{(N)} = \frac{m_{k+1}^{(N)}}{m_k^{(N)}} \quad (k \geq 3).$$

The transition-ratio inequality at (N,k) is

$$R_{N,k} := m_{k+2}^{(N)} m_k^{(N)} - (m_{k+1}^{(N)})^2 \geq 0,$$

equivalently $\rho_k^{(N)} \leq \rho_{k+1}^{(N)}$.

Lemma 3.3 (Boundary reduction for type A)**.** *Fix $N \geq 4$. If*

$$\rho_3^{(N)} = \frac{9}{2}, \qquad \frac{9}{2} \leq \rho_k^{(N)} \leq \rho_{k+1}^{(N)} \quad (k \geq 3),$$

then $Q_3^{\mathrm{SU}(N)}(n) > 0$ for every odd $n \geq 5$.

Proof. Use the symmetric form

$$\frac{1}{2}Q_3(n) = \frac{1}{2}\sum_{j=0}^{n} \binom{n}{j} \big(m_{j+2}m_{n-j} + m_j m_{n+2-j} - 2m_{j+1}m_{n+1-j} \big).$$

For $3 \leq j \leq n-3$, division by $m_{j+1}m_{n+1-j}$ gives

$$\frac{\rho_{j+1}}{\rho_{n-j}} + \frac{\rho_{n+1-j}}{\rho_j} - 2 \geq 0$$

by monotonicity and AM-GM. The remaining depth-two boundary contribution

$$L_2 = B_0 + nB_1 + \binom{n}{2} B_2$$

uses only $m_0=1, m_1=0, m_2=1, m_3=2, m_4=9$ and the four ratios $R=\rho_{n-2}$, $S=\rho_{n-1}$, $T=\rho_n$, $U=\rho_{n+1}$. Substitution and $U\ge T$ give

$$\frac{L_2}{m_{n-2}} \ge RS\left(T^2+\frac{n^2-5n+2}{2}\right)-2n(n-2)R+\frac{9}{2}n(n-1).$$

Since $R,S,T\ge 9/2$, the right hand side is at least

$$\frac{9(10n^2-66n+765)}{16}>0.$$

Thus the boundary contribution is positive and all interior contributions are nonnegative. □

Lemma 3.4 (Exact formal derivation of the low-rank type A recurrences)**.** *Put*

$$E_k^{(N)}=\mathbf{E}_{\mathrm{U}(N)}|\operatorname{tr}U|^{2k}, \qquad m_k^{(N)}=\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}E_j^{(N)}.$$

Then the following recurrences hold:

N	*recurrence*
3	$(k+3)^2m_{k+1}=k(7k+11)m_k+8k(k+1)m_{k-1}, \quad k\ge 1,$
4	$0=(k^3+22k^2+161k+392)m_{k+4}$ $-(16k^3+230k^2+1054k+1524)m_{k+3}$ $+(10k^3-340k-750)m_{k+2}$ $+(72k^3+522k^2+1242k+972)m_{k+1}$ $+(45k^3+270k^2+495k+270)m_k, \quad k\ge 0,$
5	$0=192(k+1)(k+2)(k+3)(k+6)m_k$ $+16(k+2)(k+3)(22k^2+184k+309)m_{k+1}$ $+(k+3)(129k^3+1245k^2+2570k-1112)m_{k+2}$ $-(k+3)(30k^3+642k^2+4487k+10196)m_{k+3}$ $+(k+8)^2(k+10)^2m_{k+4}, \quad k\ge 0.$

Proof. Gessel's determinant [3, §7, Theorem 16 and the formula following (26)] gives

$$F_N(t):=\sum_{k\ge 0}E_k^{(N)}\frac{t^k}{(k!)^2}=\det\bigl(I_{|i-j|}(2\sqrt{t})\bigr)_{1\le i,j\le N}.$$

Put $y=I_0(2\sqrt{t})$ and $z=y'$. The Bessel recurrence and $ty''+y'-y=0$ give

$$I_1(2\sqrt{t})=\sqrt{t}\,z, \qquad I_{r+1}(2\sqrt{t})=I_{r-1}(2\sqrt{t})-\frac{r}{\sqrt{t}}I_r(2\sqrt{t}), \qquad z'=\frac{y-z}{t}.$$

Expanding the three determinants, with no series truncation, gives

$$\begin{aligned}
F_3&=z(2y^2-yz-2tz^2),\\
F_4&=-\frac{1}{t}\bigl(4t^2z^4-8ty^2z^2+8tyz^3-tz^4+4y^4-8y^3z+4y^2z^2\bigr),\\
F_5&=-\frac{4}{t^2}\bigl(2y^2-yz-(2t+1)z^2\bigr)\\
&\qquad\qquad\cdot\bigl(4y^3-5y^2z+yz^2-4tyz^2+3tz^3\bigr).
\end{aligned}\tag{3.4.1}$$

Here is a finite differential certificate for the coefficient recurrences. On $\mathbb{Q}[t,t^{-1},y,z]$, let ∂ be the derivation

$$\partial t=1, \qquad \partial y=z, \qquad \partial z=(y-z)/t,$$

and put

$$\vartheta=t\partial, \qquad \mathcal{S}=(\vartheta+1)\partial.$$

If $F=\sum_{k\ge 0}a_kt^k/(k!)^2$, then

$$\mathcal{S}F=\sum_{k\ge 0}a_{k+1}\frac{t^k}{(k!)^2}, \qquad q(\vartheta)F=\sum_{k\ge 0}q(k)a_k\frac{t^k}{(k!)^2}.\tag{3.4.2}$$

For the following explicitly displayed polynomials, direct calculation in this Laurent-polynomial differential ring gives

$$\sum_j q_{N,j}(\vartheta)\mathcal{S}^j F_N = 0 \qquad (N = 3, 4, 5). \tag{3.4.3}$$

They are

$$\begin{array}{c|l}
N & (q_{N,0}(x), q_{N,1}(x), \ldots) \\
\hline
3 & \begin{array}{l}(9(x+1)^2,\ -(19x^2+78x+77),\\ \ 11x^2+70x+109,\ -(x+5)^2);\end{array} \\
4 & \begin{array}{l}(-64(x+1)^2(x+2),\\ \ 2(106x^3+763x^2+1791x+1366),\\ \ -(253x^3+2662x^2+9223x+10482),\\ \ 127x^3+1795x^2+8378x+12884,\\ \ -(23x^3+428x^2+2625x+5312),\\ \ (x+8)^2(x+9));\end{array} \\
5 & \begin{array}{l}(225(x+1)^2(x+2)^2,\\ \ -(x+2)^2(934x^2+6226x+10317),\\ \ 1487x^4+22936x^3+130422x^2+324256x+297420,\\ \ -2(554x^4+11290x^3+85109x^2+281496x+344526),\\ \ 367x^4+9494x^3+91097x^2+384340x+601200,\\ \ -(38x^4+1282x^3+15993x^2+87412x+176688),\\ \ (x+10)^2(x+12)^2).\end{array}
\end{array} \tag{3.4.4}$$

Thus (3.4).2–.3 give the $E_k^{(N)}$-recurrences with coefficients $q_{N,j}(k)$.

For completeness, we verify that these are exactly the binomial transforms of the three recurrences in the statement. Let

$$A_N(w) = \sum_{k\ge 0} E_k^{(N)} \frac{w^k}{k!}, \qquad M_N(w) = \sum_{k\ge 0} m_k^{(N)} \frac{w^k}{k!}.$$

The binomial-transform identity is $M_N(w) = e^{-w}A_N(w)$. If a proposed recurrence is written $\sum_r p_{N,r}(k)m_{k+r} = 0$, its exponential-generating-function operator is

$$\sum_r p_{N,r}(wD)D^r, \qquad D = \frac{d}{dw}.$$

Conjugating by e^{-w} replaces D by $D-1$ and wD by $wD-w$. Normal ordering with the single relation

$$Dw = wD + 1$$

and then taking coefficients gives the polynomials in (3.4).4, after index shifts $1, 1, 2$ for $N = 3, 4, 5$, respectively. The finitely many coefficients below those shifts vanish by the hook-length values for $E_k^{(N)}$.

Every algebraic operation just described is replayed without floating point or symbolic-library assumptions by

`character_ring_iter/verify_sun_recurrence_derivation_exact.py`.

It represents a monomial as $t^{a/2}y^bz^c$, expands the determinants over all permutations, normal-orders the Weyl algebra by the displayed commutation relation, applies $\partial z = (y-z)/t$, and requires the final Laurent polynomial to have zero terms. It also regenerates the initial $E_k^{(N)}$ from the hook-length formula and checks the omitted coefficients. The status-zero transcript reports zero reduced-identity terms and zero finite residuals for all three ranks. The source and transcript SHA-256 values are, respectively,

`e08536805bd04122b6cf87ac1ba083bb` `d5076cb2c1620baad6002ae3f42fb100`, `71423c65e6c02d1862e14d3008120816` `e324d15a9642dd7af5143bf530c4a7ec`.

This proves the three formal recurrences for every stated index. ☐

Theorem 3.5 (Low-rank type A ratio theorem)**.** *For $N = 3$ one has*

$$\rho_3^{(3)} = \rho_4^{(3)} = 4, \qquad 4 \le \rho_k^{(3)} \le \rho_{k+1}^{(3)} \quad (k \ge 3).$$

For $N = 4, 5$ one has

$$\rho_3^{(N)} = \frac{9}{2}, \qquad \frac{9}{2} \le \rho_k^{(N)} \le \rho_{k+1}^{(N)} \quad (k \ge 3).$$

Proof. Use the exact formal recurrences of lemma 3.4.

For $N = 3$, set $U_k = 8 - \rho_k$ and

$$G_k = \frac{32k + 100}{(k+4)^2}, \qquad H_k = \frac{32(k + 11/5)}{(k+3)^2}.$$

The first recurrence gives $U_{k+1} = \Phi_{k+1}(U_k)$, where

$$\Phi_k(u) = \frac{k^2 + 37k + 72 - 8k(k+1)/(8-u)}{(k+3)^2}.$$

This map is decreasing for $u < 8$. The exact base values are $U_5 = 111/32$, $G_5 = 260/81$, and $H_5 = 18/5$. The two induction margins are

$$\Phi_{k+1}(H_k) - G_{k+1} = \frac{4(34k^4 + 288k^3 + 683k^2 + 278k - 403)}{(k+4)^2(k+5)^2(5k^2 + 10k + 1)},$$
$$H_{k+1} - \Phi_{k+1}(G_k) = \frac{29k^2 + 91k + 54}{5(k+4)^2(2k^2 + 8k + 7)}.$$

They are positive for $k \ge 5$, so $G_k \le U_k \le H_k$. If r_k^* is the positive root of

$$(k+4)^2 r^2 - (k+1)(7k+18)r - 8(k+1)(k+2) = 0,$$

then, on writing $\epsilon_k^* = 8 - r_k^*$, the discriminant identity

$$(9k^2 + 103k + 238)^2 - 4(k+4)^2(288k + 864) = (9k^2 + 39k + 38)^2 + 16(k-3)(k+2)$$

gives $\epsilon_k^* \le G_k \le U_k$. Hence $\rho_{k+1} \ge \rho_k$ for $k \ge 5$; the exact initial ratios are $\rho_3 = \rho_4 = 4$ and $\rho_5 = 145/32$.

For $N = 4$, set $U_k = 15 - \rho_k$ and

$$G_k = \frac{1125}{10k + 71}, \qquad H_k = \frac{1125}{10k + 66}.$$

Solving the second recurrence for U_{k+3} gives a rational map increasing in U_k, U_{k+1}, U_{k+2} on the box $0 \le U_j \le H_j$ for $j \ge 7$. The exact base margins are

j	$U_j - G_j$	$H_j - U_j$
7	4847/73179	48089/211752
8	120402/1635179	286983/1581034
9	331053/4344746	103677/701636.

Clearing the positive denominators in the lower and upper induction margins gives, respectively,

$$\begin{aligned} P_-(s) &= 2432000s^5 + 100233600s^4 + 1609345435s^3 \\ &\quad + 12643460157s^2 + 48855354258s + 74691137880, \\ P_+(s) &= 128000s^5 + 24662400s^4 + 750722465s^3 \\ &\quad + 8917574583s^2 + 46522820852s + 89208809220, \end{aligned}$$

at $s = k - 7$. The denominators are

$$40(k+7)^2(k+8)(5k-2)(5k+3)(5k+8)(10k+101),$$
$$10(k+7)^2(k+8)(5k+48)(10k-9)(10k+1)(10k+11),$$

so the induction proves $G_k \le U_k \le H_k$ for $k \ge 7$. Since $H_{k+1} < G_k$, the ratios increase. Direct substitution of $m_3 = 2, m_4 = 9, m_5 = 43, m_6 = 245, m_7 = 1557, m_8 = 10829$ checks the preceding indices and the lower bound $\rho_k \ge 9/2$.

For $N = 5$, set $U_k = 24 - \rho_k$ and

$$G_k = \frac{2880}{10k + 109}, \qquad H_k = \frac{288}{k + 10}.$$

The third recurrence similarly gives an increasing three-variable map for $k \geq 6$. Its exact base margins are

j	$U_j - G_j$	$H_j - U_j$
6	92/1859	10/11
7	19597/163248	11345/15504
8	3424/21315	4272/7105.

After clearing the positive denominators, the two induction numerators at $s = k - 6$ are

$$\begin{aligned} R_-(s) &= 3750000s^6 + 174475000s^5 + 3214281000s^4 \\ &\quad + 30095015750s^3 + 153613752915s^2 \\ &\quad + 416461589985s + 482743428300, \\ R_+(s) &= 1900s^5 + 76662s^4 + 1135088s^3 + 7550562s^2 \\ &\quad + 22473804s + 23833584, \end{aligned}$$

with denominators

$$\begin{aligned} 2(k+8)^2(k+10)^2(10k-11)(10k-1)(10k+9)(10k+139), \\ k(k-2)(k-1)(k+8)^2(k+10)^2(k+13). \end{aligned}$$

Thus $G_k \leq U_k \leq H_k$ for $k \geq 6$, and again $H_{k+1} < G_k$. The exact initial ratios $9/2, 44/9, 6, 76/11$ complete the $N = 5$ proof. □

Lemma 3.6 (The $SU(3)$ boundary)**.** *The ratio inequalities in theorem 3.5 imply $Q_3^{\mathrm{SU}(3)}(n) > 0$ for every odd $n \geq 11$.*

Proof. The interior terms in the symmetric moment formula are nonnegative by the same AM–GM argument as in lemma 3.3. For the depth-two boundary put

$$\rho_{n-2} = 4 + s, \qquad \rho_{n-1} = 4 + s + t, \qquad s, t \geq 0.$$

Substitution of the $N = 3$ recurrence from theorem 3.5 gives

$$L_2 = \frac{m_{n-2}}{2(n+3)^2(n+4)^2} P(n, s, t),$$

where

$$\begin{aligned} P &= n^6(s^2 + st + 4s + 4t + 8) + n^5(9s^2 + 9st + 24s + 36t + 56) \\ &\quad + n^4(119s^2 + 119st + 884s + 476t + 2104) \\ &\quad + n^3(479s^2 + 479st + 4256s + 1916t + 10120) \\ &\quad + n^2(716s^2 + 716st + 7184s + 2864t + 17088) \\ &\quad + n(636s^2 + 636st + 6528s + 2544t + 14784) \\ &\quad + 576s^2 + 576st + 4608s + 2304t + 9216. \end{aligned}$$

Every coefficient is nonnegative and the constant term is positive. Thus $L_2 > 0$, while every interior term is nonnegative. □

Theorem 3.7 (The $SU(2)$ row)**.** *For every $n \geq 0$, $Q_3^{\mathrm{SU}(2)}(n) \geq 0$.*

Proof. Parametrize $SU(2)$ conjugacy classes by $\alpha \in [0, \pi]$. The class density is $(2/\pi)\sin^2\alpha\, d\alpha$, and $\chi_{\mathrm{ad}}(\alpha) = 1 + 2\cos(2\alpha)$. For independent angles α, β, put $\theta = \alpha + \beta$ and $\phi = \alpha - \beta$. Folding the resulting diamond uses the identities

$$\begin{aligned} \chi_{\mathrm{ad}}(\alpha) + \chi_{\mathrm{ad}}(\beta) &= 2 + 4\cos\theta\cos\phi, \\ (\chi_{\mathrm{ad}}(\alpha) - \chi_{\mathrm{ad}}(\beta))^2 &= 16\sin^2\theta\sin^2\phi, \\ 4\sin^2\alpha\sin^2\beta = (\cos\theta - \cos\phi)^2, &\qquad d\alpha\, d\beta = \tfrac{1}{2} d\theta\, d\phi. \end{aligned}$$

Thus folding by the C_2 Weyl symmetries sends

$$\frac{1}{2}(\chi_{\mathrm{ad}}(\alpha) - \chi_{\mathrm{ad}}(\beta))^2\, d\mu(\alpha) d\mu(\beta)$$

to the normalized USp(4) Weyl density

$$\frac{8}{\pi^2}(\cos\theta - \cos\phi)^2 \sin^2\theta \sin^2\phi\, d\theta d\phi \quad (0 \leq \theta, \phi \leq \pi)$$

from [1, Chapter IV, (1.11)]. The normalization follows at $n = 0$ from Schur orthogonality. If V is the defining representation of USp(4), then

$$\chi_{\mathrm{ad}}(\alpha) + \chi_{\mathrm{ad}}(\beta) = 2 + 4\cos\theta\cos\phi = \chi_{\Lambda^2 V}(\theta, \phi).$$

Consequently

$$\frac{1}{2} Q_3^{\mathrm{SU}(2)}(n) = \dim\big(((\Lambda^2 V)^{\otimes n})^{\mathrm{USp}(4)}\big) \geq 0.$$

□

Lemma 3.8 (Derangement-ratio gap)**.** *If $D_r = !r$, then for every $r \geq 3$,*

$$\frac{D_{r+2}}{D_{r+1}} - \frac{D_{r+1}}{D_r} \geq \frac{7}{18}.$$

Proof. The recurrence $D_{r+1} = (r+1)D_r + (-1)^{r+1}$ gives

$$\frac{D_{r+2}}{D_{r+1}} - \frac{D_{r+1}}{D_r} = 1 + \frac{(-1)^{r+2}}{D_{r+1}} - \frac{(-1)^{r+1}}{D_r}.$$

For even r this is greater than 1. For odd $r \geq 3$, use $D_r \geq D_3 = 2$ and $D_{r+1} \geq D_4 = 9$ to obtain $1 - 1/9 - 1/2 = 7/18$. □

Lemma 3.9 (Uniform type A transition strip)**.** *For every $N \geq 6$ and every integer k with*

$$N - 1 \leq k \leq N + 33,$$

one has

$$R_{N,k} = m_{k+2}^{(N)} m_k^{(N)} - (m_{k+1}^{(N)})^2 > 0.$$

Proof. Let $D_r = !r$. By Schur–Weyl duality and RSK,

$$a! - E_a^{(N)} = \#\{\pi \in S_a : \mathrm{lds}(\pi) > N\}.$$

If the longest decreasing subsequence has length greater than N, some fixed set of $N + 1$ positions appears in decreasing relative order. The union bound therefore gives

$$0 \leq a! - E_a^{(N)} \leq a! \frac{\binom{a}{N+1}}{(N+1)!}.$$

Fix k in the displayed strip and put $K = k + 2$. Comparing the binomial transforms for $m_r^{(N)}$ and D_r, for $r \leq K$ one obtains

$$\left| m_r^{(N)} - D_r \right| \leq 3r! \frac{\binom{K}{N+1}}{(N+1)!} \leq \tau D_r, \qquad \tau := 9 \frac{\binom{K}{N+1}}{(N+1)!},$$

where $\sum_{j \geq 0} 1/j! < 3$ and $D_r \geq r!/3$ for $r \geq 2$. The latter bound follows directly from $D_r/r! = \sum_{j=0}^{r} (-1)^j/j!$: it is an alternating sum, with the minimum $1/3$ attained at $r = 3$.

By lemma 3.8,

$$D_{k+2} D_k - D_{k+1}^2 \geq \frac{7}{18} D_k D_{k+1}.$$

Writing $m_r^{(N)} = D_r + e_r$, expanding $R_{N,k}$, and using $|e_r| \leq \tau D_r$ together with

$$D_{k+1} \leq (k+2) D_k, \qquad D_{k+2} \leq (k+3) D_{k+1},$$

gives

$$R_{N,k} \geq D_k D_{k+1} \left[\frac{7}{18} - \tau(4k + 10) - \tau^2(2k + 5) \right]. \tag{$*$}$$

For $6 \leq N \leq 18$, the exact C++/GMP hook-length replay evaluates all 35 values $N - 1 \leq k \leq N + 33$ and finds every integer $R_{N,k}$ strictly positive. Now assume $N \geq 19$ and write $k = N + d$, $-1 \leq d \leq 33$. The bracket in $(*)$ is smallest at $d = 33$. In that case

$$\tau_N = 9 \frac{\binom{N+35}{N+1}}{(N+1)!}, \qquad \frac{\tau_{N+1}}{\tau_N} = \frac{N + 36}{(N+2)^2}.$$

Both error terms decrease for $N \geq 19$. Indeed, after putting $s = N - 19$, the cleared numerator for the first ratio comparison is

$$4s^3 + 382s^2 + 10478s + 83928,$$

and for the second it is

$$2s^5 + 277s^4 + 14446s^3 + 362171s^2 + 4408498s + 20862654.$$

All coefficients are positive. At the initial value $N = 19, d = 33$, exact rational arithmetic gives

$$\tau_{19} = \frac{3737279683}{3143467008000},$$

and the bracket in $(*)$ equals

$$\frac{1280170982560916731574699}{9881384830384472064000000} > \frac{1}{10}.$$

Thus $(*)$ is strictly positive for every $N \geq 19$ and every allowed d. The GMP checker independently reconstructs the finite moments, the base rational margin, and the two shifted coefficient lists. $\square$

Lemma 3.10 (Fixed-band determinant criterion)**.** *Let μ be a probability measure on $[-1, A]$, where $A > c > u > b > 1$, and put $m_j = \int x^j \, d\mu(x)$. If*

$$p_B = \mu([b, u]), \qquad p_C = \mu([c, A])$$

satisfy, for an odd integer k,

$$p_B p_C b^k (c-u)^2 > 2(c+1)^2, \tag{3.10.1}$$

then $m_{k+2}m_k - m_{k+1}^2 > 0$.

Proof. For independent X, Y with law μ,

$$m_{k+2}m_k - m_{k+1}^2 = \frac{1}{2}\mathbf{E}\big[X^k Y^k (X-Y)^2\big].$$

The two symmetric rectangles $[b, u] \times [c, A]$ contribute at least $P = p_B p_C b^k c^k (c-u)^2$. The negative contribution for which the positive variable x lies in $[c, A]$ has absolute value at most $H = \int_{[c,A]} x^k (x+1)^2 \, d\mu(x)$. For $x \geq c$, the quotient $(x+1)/(x-u)$ is decreasing. Since $p_C \leq 1$, (3.10).1 gives

$$p_B b^k (x-u)^2 \geq p_B b^k (c-u)^2 \frac{(x+1)^2}{(c+1)^2} > 2(x+1)^2.$$

Integrating first over $[b, u]$ and then over $[c, A]$ shows $P > 2H$. The remaining negative contribution, with positive variable in $(0, c)$, is at most $L = (c+1)^2 c^k$, while (3.10).1 gives $P/2 > L$. All same-sign contributions are nonnegative, proving the claim. $\square$

Lemma 3.11 (Quadratic type-A transition window)**.** *For $N \geq 25$,*

$$R_{N,k} > 0 \qquad \left(N + 16 \leq k \leq \left\lfloor \frac{N^2}{11} \right\rfloor - 2\right).$$

Proof. Put $K = \lfloor N^2/11 \rfloor$, $\theta = 121/176$, and $D_r = !r$. The RSK formula in the proof of lemma 3.9 gives, for $r \leq K$,

$$|m_r^{(N)} - D_r| \leq 3r! \frac{\binom{K}{N+1}}{(N+1)!}.$$

Since $K < (N+1)^2/11$, $(N+1)! \geq ((N+1)/e)^{N+1}$, and $e < 11/4$,

$$\frac{\binom{K}{N+1}}{(N+1)!} \leq \theta^{N+1}.$$

Also $D_r \geq r!/3$ for $r \geq 2$, so $|m_r^{(N)} - D_r| \leq \tau D_r$ with $\tau = 9\theta^{N+1}$. Expanding $R_{N,k}$ about the derangement determinant as in lemma 3.9 yields

$$R_{N,k} \geq D_k D_{k+1} \left[\frac{7}{18} - \tau(4k+10) - \tau^2(2k+5)\right]. \tag{3.11.1}$$

For $k \leq K$ the error terms are bounded by

$$9\theta^{N+1}\left(\frac{4N^2}{11} + 10\right), \qquad 81\theta^{2N+2}\left(\frac{2N^2}{11} + 5\right).$$

They decrease for $N \geq 25$: after clearing denominators, the two successive ratio inequalities reduce respectively to

$$220N^2 - 968N + 5566 > 0, \qquad 32670N^2 - 58564N + 869143 > 0.$$

At $N = 25$ their exact rational sum is less than $129/1000 < 7/18$. Thus the bracket in (3.11).1 is positive. The C++/GMP replay independently checks the two cleared polynomials and the exact $N = 25$ comparison. $\square$

Proposition 3.12 (Uniform type-A endpoint overlaps). *The transition determinant satisfies $R_{N,k} > 0$ in both ranges*

N	*odd* k
$N \geq 24$	$k \geq 71,$
$6 \leq N \leq 23$	$k \geq 53.$

Proof. Let U be Haar-distributed in $\mathrm{SU}(N)$, put $Y = |\operatorname{tr} U|^2$, $X = Y - 1$, and let μ_N be the law of X. Then $X \in [-1, N^2 - 1]$. The $\mathrm{SU}(N)$ and $\mathrm{U}(N)$ moments of Y agree, and

$$\mathbf{E}Y^r = \sum_{\lambda \vdash r,\ \ell(\lambda) \leq N} (f^\lambda)^2 \leq r!, \tag{3.12.1}$$

with equality for $r \leq N$.

For $N \geq 24$, the degree-18 signed-polynomial minorant reconstructed from the manifested verifier source by the C++/GMP replay gives $\mu_N([3/2, 7/2]) \geq \eta_B = 122923229137548665407/(65536 \cdot 1467426013^2)$. Paley–Zygmund applied to Y^{12} gives $\mu_N([4, N^2 - 1]) \geq \eta_C = 88255484124721/992717442773183102976$, and exact rational arithmetic gives $\eta_B \eta_C (3/2)^{71} > 200$. Thus lemma 3.10 applies with $(b, u, c) = (3/2, 7/2, 4)$.

For $6 \leq N \leq 23$, the analogous degree-10 minorant and exact hook moments give $\mu_N([19/10, 47/10]) \geq \delta_B = 44525786465344542780431067563007/43884276324717505323728973379833856$. Paley–Zygmund applied to Y^{15} gives $\mu_N([5, N^2 - 1]) \geq \delta_C = 471756427/178410035553750000$. The replay checks the nonstable ranks, including $\mathbf{E}Y^{15} > 6^{15}$, and verifies $\delta_B \delta_C (19/10)^{52} (3/10)^2 > 72$. Hence lemma 3.10 applies with $(b, u, c) = (19/10, 47/10, 5)$ for every odd $k \geq 53$. $\square$

Proposition 3.13 (Finite type-A overlap rows). *One has $R_{N,k} > 0$ for the following odd transition indices:*

N	k	N	k
24	$57, 59, \ldots, 69$	6	$39, 41, \ldots, 51$
25	$59, 61, \ldots, 69$	7, 8	$41, 43, \ldots, 51$
26	$61, 63, \ldots, 69$	9, 10	$43, 45, \ldots, 51$
27	$65, 67, 69$	11, 12	$45, 47, 49, 51$
		13, 14	$47, 49, 51$
		15, 16	$49, 51$
		17, 18	$51.$

Proof. For $24 \leq N \leq 27$, put $K = k + 2$ and $\tau = 9\binom{K}{N+1}/(N+1)!$. The RSK union bound and determinant expansion in lemma 3.9 give

$$R_{N,k} \geq D_k D_{k+1} \left[\frac{7}{18} - \tau(4k + 10) - \tau^2(2k + 5) \right].$$

Exact rational evaluation over the displayed finite rows gives a bracket greater than $3/10$ in every case.

For $6 \leq N \leq 18$, the exact hook formula in (3.12).1 supplies $\mathbf{E}Y^r$ through $r = 53$; binomial transformation gives the adjoint moments, and direct integer evaluation gives $R_{N,k} > 0$ at every displayed pair. The C++/GMP checker recomputes these partitions and hook dimensions, requires exact division at every step, matches a hard-coded positive row minimum for each N, and rejects any missing transition index. $\square$

Theorem 3.14 (Finite type A ratio theorem). *For every $N \geq 6$,*

$$\rho_3^{(N)} = \frac{9}{2}, \qquad \frac{9}{2} \leq \rho_k^{(N)} \leq \rho_{k+1}^{(N)} \quad (k \geq 3).$$

Proof. The stable part $3 \leq k \leq N - 2$ follows from the derangement moments above. Indeed $m_r^{(N)} = !r$ for $r \leq N$, and lemma 3.8 gives both monotonicity and the lower bound $\rho_k^{(N)} \geq \rho_3^{(N)} = 9/2$.

For even $k = 2q$, Cauchy–Schwarz applied to the real functions χ_{ad}^q and χ_{ad}^{q+1} gives

$$(m_{k+1}^{(N)})^2 \leq m_k^{(N)} m_{k+2}^{(N)}.$$

The denominators are positive by lemma 2.6, so this proves the even transition inequalities.

The uniform strip lemma 3.9 covers $N-1 \le k \le N+33$. The remaining odd transition range is covered by the following proved overlaps:

range	certificate
$N-1 \le k \le N+33$	*lemma* 3.9
k even	Cauchy–Schwarz, as above
$6 \le N \le 18,\ k \le 51$	*proposition* 3.13
$6 \le N \le 23,\ k \ge 53$	*proposition* 3.12
$24 \le N \le 27,\ k < 71$	*lemma* 3.11*and proposition* 3.13
$N \ge 24,\ k \ge 71$	*proposition* 3.12
$N \ge 28,\ k \le 69$	*lemma* 3.11.

These alternatives cover all odd $k \ge N-1$: the first line covers through $N+33$. For $6 \le N \le 18$, the finite rows meet the endpoint range $k \ge 53$; for $19 \le N \le 23$, that endpoint already begins before the first remaining odd index. For $24 \le N \le 27$, the finite and quadratic rows meet $k \ge 71$. If $N \ge 28$ and a remaining odd index is at most 69, then it is at least $N+34 > N+16$, while $\lfloor N^2/11 \rfloor - 2 \ge 69$, so lemma 3.11 applies. Every later odd index is covered by proposition 3.12.

The replay reports the uniform strip, endpoint constants, and every finite row as exact successes. All arithmetic in the finite replay is integer or rational arithmetic; no floating-point comparison is used in a sign decision. The infinite parts are the displayed inequalities in lemmas 3.9 and 3.11 and proposition 3.12. ☐

Corollary 3.15 (Finite-rank type A branch)**.** *For every $N \ge 2$ and every $n \ge 0$ satisfying $N < n+2$,*

$$Q_3^{\mathrm{SU}(N),\mathrm{ad}}(n) \ge 0.$$

Proof. The $N=2$ row is theorem 3.7. For $N=3$, theorem 3.5 and lemma 3.6 cover every odd $n \ge 11$; the hook-length formula gives the exact positive values

n	1	3	5	7	9
$Q_3(n)/2$	2	28	654	21312	902860.

For $N=4,5$, theorem 3.5 and lemma 3.3 give positivity for odd $n \ge 5$. For $N \ge 6$, theorem 3.14 and lemma 3.3 do the same. The remaining odd cases $n=1,3$ follow from the first adjoint moments:

N	m_0	m_1	m_2	m_3	m_5	$Q_3(1)$	$Q_3(3)$
2	1	0	1	1	6	2	8
3	1	0	1	2	32	4	56
4	1	0	1	2	43	4	78
≥ 5	1	0	1	2	44	4	80.

Here $Q_3(1) = 2m_3$ and $Q_3(3) = 2(m_5 - 2m_3)$. Even n is lemma 2.9. ☐

Theorem 3.16 (Type A closure)**.** *For every $N \ge 2$ and every $n \ge 0$,*

$$Q_3^{\mathrm{SU}(N),\mathrm{ad}}(n) \ge 0.$$

Proof. If $N \ge n+2$, use theorem 3.1. If $N < n+2$, use corollary 3.15. ☐

4. The Non-A Classical Families

The non-A classical Lie algebras are represented by the standard B_b, C_b, and D_b families. The proof uses exact adjoint moments, Rains stabilization, and a final finite certificate. We use the irreducible-root-system rank conventions

$$B_b\ (b \ge 2), \qquad C_b\ (b \ge 2), \qquad D_b\ (b \ge 4).$$

The low-rank coincidences $B_1 = C_1 = A_1$ and $D_3 = A_3$ are already covered by theorem 3.16; $D_2 = A_1 \times A_1$ is not simple. The coincidence $B_2 = C_2$ causes harmless overlap between the two finite classical rows.

Definition 4.1 (Classical constants and first hits)**.** Let

$$C_G = \max_{g \in G}(-\chi_{\mathrm{ad}}(g)).$$

Define

$$s(G) = \begin{cases} 2b-1, & G = B_b, C_b, \\ \text{largest odd integer } \le b-3, & G = D_b, \end{cases}$$

the largest odd exponent covered by the stable moment comparison. After the row-gated bridge through $m = 29$ the first possible unbridged odd exponent is

$$N_{\text{hit}}^{(29)}(G) = \max(63, s(G) + 2).$$

Equivalently,

$$\begin{array}{ll} B_b, C_b: & N_{\text{hit}}^{(29)} = \max(63, 2b + 1), \\ D_b: & N_{\text{hit}}^{(29)} = \max(63, b - 1) \text{ for even } b, \quad \max(63, b - 2) \text{ for odd } b. \end{array}$$

Lemma 4.2 (Classical negative endpoint)**.** *For the standard B_b, C_b, D_b representatives,*

$$C_{B_b} = b, \qquad C_{C_b} = b, \qquad C_{D_b} = \begin{cases} b, & b \text{ even}, \\ b - 2, & b \text{ odd}. \end{cases}$$

Proof. Write the defining eigenvalue pairs as z_j, z_j^{-1} and put $x_j = (z_j + z_j^{-1})/2 \in [-1, 1]$ and $S = \sum_j x_j$, $Q = \sum_j x_j^2$. For B_b the defining trace is $T_1 = 1 + 2S$ and $T_2 = 1 + 4Q - 2b$. Since the adjoint representation is Λ^2 of the defining representation,

$$\chi_{\text{ad}} = \frac{T_1^2 - T_2}{2} = b + 2S + 2(S^2 - Q).$$

This expression is affine in each x_j separately, so its minimum occurs at a vertex. At a vertex $Q = b$ and $S = b-2k$, hence $\chi_{\text{ad}} = 2S^2 + 2S - b$; the minimum is $-b$.

For D_b, $T_1 = 2S$ and $T_2 = 4Q - 2b$, again with $\chi_{\text{ad}} = (T_1^2 - T_2)/2$. Thus

$$\chi_{\text{ad}} = b + 2(S^2 - Q).$$

At a vertex this is $2S^2 - b$. If b is even, $S = 0$ is allowed and the minimum is $-b$; if b is odd, the nearest allowed values are $S = \pm 1$ and the minimum is $2 - b$.

For C_b, the adjoint representation is Sym^2 of the defining representation, so

$$\chi_{\text{ad}} = \frac{T_1^2 + T_2}{2} = 2S^2 + 2Q - b.$$

This is bounded below by $-b$, and equality is attained at $x_1 = \cdots = x_b = 0$. □

Lemma 4.3 (Stable classical edge)**.** *For the classical families the stable comparison proves the following Chain differences:*

$$D_G(2r + 1) = Q_3^G(2r + 3) - 4Q_3^G(2r + 1) > 0$$

whenever

G	*stable range*
B_b, C_b	$b \geq r + 2$
D_b	$b \geq 2r + 6.$

Equivalently, the stable edge covers all odd exponents at most $s(G)$, where $s(G)$ is defined in definition 4.1.

Proof. Write $T_j(U) = \mathrm{Tr}(U^j)$ in the defining representation. The adjoint characters are

$$\chi_{\text{ad}}^{\text{SO}} = (T_1^2 - T_2)/2, \qquad \chi_{\text{ad}}^{\text{Sp}} = (T_1^2 + T_2)/2.$$

Diaconis–Shahshahani [4, Theorem 4, p. 56, and Theorem 6, p. 59] give the exact stable joint trace moments

$$(T_1, T_2)_{\text{SO}} \longmapsto (Z_1, 1 + \sqrt{2} Z_2), \qquad (T_1, T_2)_{\text{Sp}} \longmapsto (Z_1, -1 + \sqrt{2} Z_2),$$

where Z_1, Z_2 are independent standard normals. Hence the stable orthogonal and symplectic adjoint variables are

$$X_{\text{SO}} = \frac{Z_1^2 - 1 - \sqrt{2} Z_2}{2}, \qquad X_{\text{Sp}} = \frac{Z_1^2 - 1 + \sqrt{2} Z_2}{2},$$

which have the same distribution. Their common moment generating function is

$$Z(t) = \frac{\exp(t^2/4 - t/2)}{\sqrt{1 - t}}.$$

It remains to justify the stated finite-rank ranges. For B_b, the adjoint character is the Schur function $s_{(1,1)}$. Expand $s_{(1,1)}^s = \sum_\lambda c_\lambda s_\lambda$. Rains' orthogonal Schur integral criterion [5, §3, paragraph preceding Lemma 3.1] says that only partitions with even row lengths contribute to the $O(2b + 1)$ integral. Such a partition has size $2s$ and at most s rows; thus $2b + 1 \geq s$ removes the length cutoff. The integrand is unchanged by $U \mapsto -U$, so the $SO(2b + 1)$ and $O(2b + 1)$ integrals agree.

For C_b, the adjoint character is $s_{(2)}$. Every partition in the Pieri expansion of $s_{(2)}^s$ has at most s rows. Rains' symplectic criterion in the same source paragraph retains precisely the partitions in which every row length has even multiplicity. Their number of rows is even and at most $2\lfloor s/2 \rfloor$, so $b \geq \lfloor s/2 \rfloor$ removes the symplectic length cutoff.

For D_b, expand χ_{ad}^s into monomials $T_1^{2j} T_2^{s-j}$, each of weighted trace degree $2s$. If $b \geq s+1$, then $2s < 2b$, so Diaconis–Shahshahani's exact orthogonal trace-moment identity applies. In this strict degree range the $SO(2b)$ integral equals the $O(2b)$ integral: the difference is the determinant-isotypic integral, and the determinant representation first occurs in matrix-entry degree $2b$. This proves the finite-rank stabilization for every adjoint moment m_s in the three stated ranges.

For independent copies of the common stable variable, the exponential generating function of Q_3 is

$$F(t) = \mathbf{E}[(X-Y)^2 e^{t(X+Y)}] = \big((1-t)^{-3} + (1-t)^{-1}\big) e^{-t+t^2/2}.$$

Hence

$$F''(t) - 4F(t) = P(t)G(t),$$

where

$$G(t) = e^{-t+t^2/2}, \qquad P(t) = 12(1-t)^{-5} - 7(1-t)^{-3} - 4(1-t)^{-1} + (1-t).$$

Write $P(t) = \sum p_k t^k$ and $G(t) = \sum (-1)^k h_k t^k$. The coefficients satisfy

$$p_{k+1} > p_k, \qquad h_k \geq h_{k+1} > 0 \qquad (k \geq 0),$$

because

$$p_k = 12\binom{k+4}{4} - 7\binom{k+2}{2} - 4 + [k=0] - [k=1],$$

and

$$(k+1)h_{k+1} = h_k + h_{k-1}, \qquad h_{-1} = 0, \quad h_0 = 1.$$

Indeed, $p_0 = 2$, while $p_1 - p_0 = 32$, $p_2 - p_1 = 100$, and for $k \geq 2$,

$$p_{k+1} - p_k = 12\binom{k+4}{3} - 7(k+2) = (k+2)\big(2(k+4)(k+3) - 7\big) > 0.$$

The recurrence gives $h_1 = 1$ and positivity of every h_k by induction. For $k \geq 1$, its shifted form

$$kh_k = h_{k-1} + h_{k-2} \geq h_{k-1}$$

then implies

$$h_{k+1} = \frac{h_k + h_{k-1}}{k+1} \leq h_k.$$

For $N = 2r+1$,

$$D_{\mathrm{st}}(N) = N! \sum_{j=0}^{(N-1)/2} \big(p_{2j+1} h_{N-2j-1} - p_{2j} h_{N-2j}\big) > 0.$$

Here every summand is strictly positive: with $a = N - 2j - 1$, the coefficient inequalities give

$$p_{2j+1} h_a - p_{2j} h_{a+1} = (p_{2j+1} - p_{2j}) h_a + p_{2j}(h_a - h_{a+1}) > 0.$$

The Chain difference at $2r+1$ uses only $m_0^G, \ldots, m_{2r+5}^G$. For B_b, C_b, the condition $b \geq r+2$ implies $b \geq \lfloor (2r+5)/2 \rfloor$; for D_b, the condition $b \geq 2r+6$ is exactly $b \geq (2r+5)+1$. The preceding stabilization therefore identifies every required finite-rank moment with the stable one. Substitution gives the displayed inequality. ☐

Definition 4.4 (Classical row gate)**.** For an odd Chain target $2m+3$ and a finite row G with $C_G < 296$, the row is active in the exact middle bridge if

$$0 \leq m \leq 29, \qquad s(G) < 2m+3 \leq 61.$$

The stable edge lemma 4.3 covers odd exponents $\leq s(G)$, while the post-bridge hit begins at $N_{\mathrm{hit}}^{(29)}(G) = \max(63, s(G)+2)$. Thus the row gate is exactly the finite interval between stable coverage and the post-$m = 29$ tail.

Lemma 4.5 (Pushforward tail criterion)**.** *Let $X = \chi_{\mathrm{ad}}(g)$ and $Y = \chi_{\mathrm{ad}}(h)$ for independent Haar variables, and let ψ_G be the positive finite measure on $\mathbb{R}$ defined by*

$$\int f(u)\, d\psi_G(u) = \mathbf{E}\big[(X-Y)^2 f(X+Y)\big].$$

Then $\psi_G(\mathbb{R}) = 2$ and

$$Q_3^G(n) = \int u^n\, d\psi_G(u).$$

For $c > 2C_G$ and $P_G(c) = \psi_G([c, 2\dim G])$,

$$Q_3^G(n) \geq c^n P_G(c) - 2(2C_G)^n$$

for every odd n. More generally, for $0 < a < 2C_G < c$ and $N_G(a) = \psi_G([-2C_G, -a])$,

$$Q_3^G(n) \geq c^n P_G(c) - (2C_G)^n N_G(a) - 2a^n.$$

Therefore the inequality

$$P_G(c) \geq N_G(a)\left(\frac{2C_G}{c}\right)^n + 2\left(\frac{a}{c}\right)^n$$

implies $Q_3^G(n) \geq 0$, and if it holds at an odd n_0 it holds for all later odd exponents.

Proof. The first identity is the definition of ψ_G with $f(u) = u^n$. By lemma 2.6, $\mathbf{E}X = 0$ and $\mathbf{E}X^2 = 1$, hence $\psi_G(\mathbb{R}) = \mathbf{E}(X-Y)^2 = 2$. For odd n, split the integral into $u \geq c$, $0 \leq u < c$, and $u < 0$. The middle piece is nonnegative. Since $\chi_{\mathrm{ad}} \geq -C_G$, the negative support is contained in $[-2C_G, 0]$. On that interval $u^n \geq -(2C_G)^n$, and its ψ_G-mass is at most $\psi_G(\mathbb{R}) = 2$, giving the first lower bound. For the second, on $[-2C_G, -a]$ one has $u^n \geq -(2C_G)^n$, while on $[-a, 0]$ one has $u^n \geq -a^n$. Therefore

$$\int_{-2C_G}^{0} u^n\, d\psi_G(u) \geq -(2C_G)^n N_G(a) - a^n \psi_G([-a,0]) \geq -(2C_G)^n N_G(a) - 2a^n.$$

After division by c^n, the two terms in the sufficient condition are nonnegative multiples of $(2C_G/c)^n$ and $(a/c)^n$. Both bases lie in $(0,1)$, proving the monotonicity assertion. ☐

Lemma 4.6 (Weyl caps and elementary pushforward conversion)**.** *Let $G = D_b = \mathrm{SO}(2b)$, let $r = b$, $d = b(2b-1)$, and let $d_1, \dots, d_r$ be the degrees of its Weyl group W. Use the type-D_b roots $e_i \pm e_j$, all of squared length 2, and write*

$$\sum_{\alpha>0} \alpha(\theta)^2 = \kappa|\theta|^2.$$

Suppose $R > 0$ is small enough that the ball $\kappa|\theta|^2 \leq R$ lies in one torus eigenangle chart, and suppose

$$\prod_{\alpha>0}\left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2}\right)^2 \geq \eta_R > 0 \tag{4.6.1}$$

throughout that ball. If $X = \chi_{\mathrm{ad}}(g)$ for Haar g, then

$$\Pr\{X \geq d - R\} \geq \frac{\eta_R}{|W|}\frac{\prod_{i=1}^r d_i!}{(2\pi)^{r/2}\Gamma(d/2+1)}\left(\frac{R}{2\kappa}\right)^{d/2}. \tag{4.6.2}$$

Write $C = C_G$, put $x_0 = d - R$, and let P_G be the pushforward mass of lemma 4.5. If $x_0 - C > 2C$ and $x_0 \geq 1$, then

$$P_G(x_0 - C) \geq \Pr\{X \geq x_0\}\frac{(x_0-1)^2}{C+1}. \tag{4.6.3}$$

If $t > 1$ and $x_0 - t > 2C$, then

$$P_G(x_0 - t) \geq \Pr\{X \geq x_0\}(x_0 - t)^2(1 - t^{-2}). \tag{4.6.4}$$

Proof. For $g = \exp(\theta)$ in the chosen torus chart,

$$d - \chi_{\mathrm{ad}}(g) = 2\sum_{\alpha>0}(1 - \cos\alpha(\theta)) \leq \sum_{\alpha>0}\alpha(\theta)^2 = \kappa|\theta|^2.$$

Weyl integration, (4.6).1, and the Macdonald–Mehta integral therefore reduce the cap mass to

$$\frac{\eta_R}{|W|(2\pi)^r}\int_{\kappa|\theta|^2 \leq R}\prod_{\alpha>0}\alpha(\theta)^2\, d\theta.$$

The root product is homogeneous of degree $d-r$. Taking the radial part of the Macdonald–Mehta Gaussian integral gives

$$\int_{\kappa|\theta|^2\le R}\prod_{\alpha>0}\alpha(\theta)^2\,d\theta=\frac{(2\pi)^{r/2}\prod_i d_i!}{\Gamma(d/2+1)}\left(\frac{R}{2\kappa}\right)^{d/2},$$

which proves (4.6).2.

Let Y be an independent copy of X. Since $Y\ge -C$ and $\mathbf{E}Y=0$, if $p=\Pr\{Y>1\}$ then $0=\mathbf{E}Y\ge p-C(1-p)$; hence $\Pr\{Y\le 1\}\ge 1/(C+1)$. On $\{X\ge x_0, Y\le 1\}$ one has $X+Y\ge x_0-C$ and $(X-Y)^2\ge (x_0-1)^2$. Independence proves (4.6).3. Also $\mathbf{E}Y^2=1$ by lemma 2.6, so Chebyshev gives $\Pr\{|Y|\le t\}\ge 1-t^{-2}$. On $\{X\ge x_0,|Y|\le t\}$, both $X+Y$ and $X-Y$ are at least x_0-t. This proves (4.6).4. ☐

Lemma 4.7 (Root-normalized type-B Weyl caps). *Let $G=B_b=\mathrm{SO}(2b+1)$, let $r=b$, $d=b(2b+1)$, and let $d_i=2i$ for $1\le i\le b$. Use the actual type-B_b roots*

$$e_i\pm e_j\quad (i<j),\qquad e_i\quad (1\le i\le b),$$

and put $\kappa=2b-1$. Suppose the ball $\kappa|\theta|^2\le R$ lies in one torus eigenangle chart and

$$\prod_{\alpha>0}\left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2}\right)^2\ge \eta_R>0\qquad (\kappa|\theta|^2\le R). \tag{4.7.1}$$

If $X=\chi_{\mathrm{ad}}(g)$ for Haar $g\in G$, then

$$\Pr\{X\ge d-R\}\ge \frac{2^{-b}\eta_R}{|W|}\frac{\prod_{i=1}^b d_i!}{(2\pi)^{b/2}\Gamma(d/2+1)}\left(\frac{R}{2\kappa}\right)^{d/2}. \tag{4.7.2}$$

With $C=C_G$, $x_0=d-R$, and P_G as in lemma 4.5, the conversion bounds (4.6).3 and (4.6).4 also hold.

Proof. For the displayed positive roots,

$$\sum_{\alpha>0}\alpha(\theta)^2=\sum_{i<j}\big((\theta_i+\theta_j)^2+(\theta_i-\theta_j)^2\big)+\sum_i\theta_i^2=(2b-1)|\theta|^2.$$

The Macdonald–Mehta normalization used with the degrees $2,4,\dots,2b$ replaces each short root e_i by the squared-length-2 normal $\sqrt{2}e_i$, while leaving the roots $e_i\pm e_j$ unchanged. Hence the square of the actual Weyl discriminant is 2^{-b} times the Macdonald–Mehta discriminant square. Repeating the radial calculation in the proof of lemma 4.6 with this factor and with $\kappa=2b-1$ gives (4.7).2. The proofs of (4.6).3 and (4.6).4 use only the lower support bound for the adjoint character, $\mathbf{E}X=0$, $\mathbf{E}X^2=1$, and independence, so the same calculation applies to B_b. ☐

Lemma 4.8 (One-sided moment pushforward criterion). *In the setting of lemma 4.5, put*

$$K_j^G=\int_{\mathbb{R}}u^j\,d\psi_G(u)=Q_3^G(j).$$

Let k,m be nonnegative integers, let $0<a<2C_G<c$, and suppose $2k+m\le n$. Define

$$H_{k,m}^G(a)=\sum_{j=0}^{2m}(-1)^j\binom{2m}{j}a^{-j}K_{2k+j}^G.$$

Then

$$\int_{-a}^0|u|^n\,d\psi_G(u)\le\frac{a^{n-2k}}{4^m}H_{k,m}^G(a).$$

Consequently,

$$P_G(c)\ge N_G(a)\left(\frac{2C_G}{c}\right)^n+\frac{a^{n-2k}}{4^mc^n}H_{k,m}^G(a) \tag{4.8.1}$$

implies $Q_3^G(n)\ge 0$. If the quantities on the right are fixed and $a,2C_G<c$, validity at one odd exponent implies validity at every later odd exponent.

Proof. For $-a\le u\le 0$, write $x=-u/a\in[0,1]$. The arithmetic–geometric mean inequality gives

$$\frac{(1+x)^2}{4}\ge x.$$

Since $n-2k\ge m$, it follows that

$$|u|^n=a^nx^n\le\frac{a^{n-2k}}{4^m}u^{2k}(1-u/a)^{2m}.$$

The polynomial on the right is nonnegative on all of $\mathbb{R}$. Integration and binomial expansion give the first assertion. Splitting the negative part of the integral defining $Q_3^G(n)$ at $-a$ gives

$$Q_3^G(n) \geq c^n P_G(c) - (2C_G)^n N_G(a) - \frac{a^{n-2k}}{4^m} H_{k,m}^G(a),$$

which proves the criterion. The two ratios in (4.8).1 are strictly below one, proving the last assertion. □

Lemma 4.9 (Exact push moments from adjoint moments)**.** *Let X, Y be independent copies of the adjoint character of a compact group, and write $m_j = \mathbf{E}[X^j]$. Then the pushforward moments in lemma 4.8 satisfy, for every $q \geq 0$,*

$$K_q^G = 2\sum_{i=0}^{q} \binom{q}{i} \big(m_{i+2}m_{q-i} - m_{i+1}m_{q-i+1}\big). \tag{4.9.1}$$

Consequently $K_0^G, \ldots, K_q^G$ are determined exactly by $m_0, \ldots, m_{q+2}$.

Proof. Expand $(X-Y)^2(X+Y)^q$ and use independence. The two pure-square sums are equal after replacing i by $q-i$; collecting them with the mixed term gives (4.9).1. □

Lemma 4.10 (Stable finite-rank push moments)**.** *Define the integers*

$$K_j = j![t^j]\left(\big((1-t)^{-3} + (1-t)^{-1}\big)e^{-t+t^2/2}\right).$$

For $G = B_b$ or C_b and every $0 \leq j \leq 2b-2$, one has

$$K_j^G = K_j.$$

For $G = D_b$ the same identity holds for every $0 \leq j \leq b-3$. In particular, if $2k+2m \leq 2b-2$ in types B, C, or if $2k+2m \leq b-3$ in type D, then

$$H_{k,m}^G(a) = \sum_{j=0}^{2m} (-1)^j \binom{2m}{j} a^{-j} K_{2k+j}.$$

Proof. The proof of lemma 4.3 computes the exponential generating function

$$\mathbf{E}[(X-Y)^2 e^{t(X+Y)}] = \big((1-t)^{-3} + (1-t)^{-1}\big)e^{-t+t^2/2}$$

for the stable orthogonal/symplectic adjoint variable. The same proof invokes Rains' Schur-integral stabilization to identify the finite B_b, C_b adjoint moments through degree $2r+5$ whenever $b \geq r+2$. Take $r = b-2$. Since K_j^G uses adjoint moments only through degree $j+2 \leq 2b$, the finite and stable values agree in types B, C. In type D_b, the exact decomposition of lemma 4.45 has $A_q(b) = B_q(b) = 0$ for $q < b$. Thus the finite adjoint moments used by K_j^G agree with their stable values whenever $j+2 < b$, equivalently $j \leq b-3$. The final formula is the binomial expansion in lemma 4.8. □

Lemma 4.11 (Trace-tail reduction)**.** *Let U be Haar-distributed in the defining representation of B_b, C_b, D_b, and put*

$$T_1(U) = \operatorname{Tr}(U), \qquad \tau_G(U) = \begin{cases} \operatorname{Tr}(U^2), & G = B_b, D_b, \\ -\operatorname{Tr}(U^2), & G = C_b. \end{cases}$$

For $1 < a < 2C_G < c$, define

$$p_1(G; c, a) = \Pr\{|T_1(U)| \geq \sqrt{2c+3a}\}, \qquad p_2(G; a) = \Pr\{\tau_G(U) \geq a\}.$$

Then

$$P_G(c) \geq c^2(1-a^{-2})(p_1(G; c, a) - p_2(G; a)), \qquad N_G(a) \leq 2(C_G^2+1)p_2(G; a).$$

Proof. The defining-character identities are

$$\chi_{\mathrm{ad}}(U) = \frac{T_1(U)^2 - T_2(U)}{2} \quad (B_b, D_b), \qquad \chi_{\mathrm{ad}}(U) = \frac{T_1(U)^2 + T_2(U)}{2} \quad (C_b).$$

On the event $|T_1(U)| \geq \sqrt{2c+3a}$ and $\tau_G(U) \leq a$ one has $\chi_{\mathrm{ad}}(U) \geq c+a$. If V is an independent Haar element with $|\chi_{\mathrm{ad}}(V)| \leq a$, then $\chi_{\mathrm{ad}}(U) + \chi_{\mathrm{ad}}(V) \geq c$ and $|\chi_{\mathrm{ad}}(U) - \chi_{\mathrm{ad}}(V)| \geq c$. Chebyshev and Schur orthogonality give $\Pr\{|\chi_{\mathrm{ad}}(V)| \leq a\} \geq 1-a^{-2}$, proving the lower bound for $P_G(c)$. For the negative tail, $\chi_{\mathrm{ad}}(U) + \chi_{\mathrm{ad}}(V) \leq -a$

implies one of the two characters is at most $-a/2$. Put $X = \chi_{\mathrm{ad}}(U)$, $Y = \chi_{\mathrm{ad}}(V)$, and $A = \{X \leq -a/2\}$. Symmetry, followed by independence and lemma 2.6, gives

$$\begin{aligned} N_G(a) &\leq 2\,\mathbf{E}\big[(X-Y)^2\mathbf{1}_A\big] \\ &= 2\left(\mathbf{E}[X^2\mathbf{1}_A] - 2\mathbf{E}[X\mathbf{1}_A]\mathbf{E}[Y] + \Pr(A)\mathbf{E}[Y^2]\right) \\ &\leq 2(C_G^2+1)\Pr(A). \end{aligned}$$

The displayed adjoint formulas imply $\{\chi_{\mathrm{ad}}(U) \leq -a/2\} \subseteq \{\tau_G(U) \geq a\}$. □

Lemma 4.12 (Decoupled trace-tail reduction and excess-square negative tail)**.** *Use the notation of lemma 4.11. Let*

$$1 < a < 2C_G < c, \qquad d > 1, \qquad q > 1,$$

and put

$$p_1(G; c, d, q) = \Pr\{|T_1(U)| \geq \sqrt{2c + 2d + q}\}, \qquad p_2(G; q) = \Pr\{\tau_G(U) \geq q\}.$$

Then

$$P_G(c) \geq c^2(1 - d^{-2})\big(p_1(G; c, d, q) - p_2(G; q)\big). \tag{4.12.1}$$

Moreover,

$$N_G(a) \leq 2\mathbf{E}\big[(\tau_G(U) - a)_+^2\big]\,. \tag{4.12.2}$$

For every $v > 0$, this implies

$$N_G(a) \leq \frac{8e^{-2}}{v^2}e^{-va}\mathbf{E}e^{v\tau_G(U)}. \tag{4.12.3}$$

Proof. On the event $|T_1(U)| \geq \sqrt{2c+2d+q}$ and $\tau_G(U) \leq q$, the two defining-character identities in lemma 4.11 give $\chi_{\mathrm{ad}}(U) \geq c + d$. For an independent Haar element V, the event $|\chi_{\mathrm{ad}}(V)| \leq d$ therefore gives both $\chi_{\mathrm{ad}}(U) + \chi_{\mathrm{ad}}(V) \geq c$ and $|\chi_{\mathrm{ad}}(U) - \chi_{\mathrm{ad}}(V)| \geq c$. Schur orthogonality and Chebyshev's inequality give probability at least $1 - d^{-2}$ for the latter event, proving (4.12).1.

Write $X = \chi_{\mathrm{ad}}(U)$ and $Y = \chi_{\mathrm{ad}}(V)$. Partition the event $X + Y \leq -a$ into $X \leq Y$ and $Y < X$. The two contributions are equal, and the diagonal has zero $(X-Y)^2$ weight. On the first part,

$$0 \leq Y - X \leq -a - 2X.$$

For all three classical families the defining-character identity is $X = (T_1(U)^2 - \tau_G(U))/2$, so $-2X \leq \tau_G(U)$. Hence

$$(X-Y)^2 \leq (\tau_G(U) - a)^2, \qquad \tau_G(U) \geq a,$$

which proves (4.12).2. Finally, for $z \geq a$, the maximum of $(z-a)^2e^{-vz}$ is $4e^{-2}v^{-2}e^{-va}$, attained at $z = a + 2/v$. Applying this pointwise and then (4.12).2 proves (4.12).3. □

Lemma 4.13 (Rains-square high-rank trace certificate)**.** *Set*

$$r = \frac{2000001}{1000000}, \qquad \eta = \frac{359}{2000}, \qquad A = \frac{4571}{2000}, \qquad L = 2r + 3\eta.$$

For every B_b, C_b, D_b row with $C_G \geq 296$, the hypotheses of lemma 4.11 hold at $a = \eta C_G$ and $c = rC_G$, and the comparison in lemma 4.5 holds at the stable edge.

Proof. The positive T_1 tail is supplied by the one-trace density estimate of Courteaut–Johansson–Lambert, Theorem 1.4, combined with the Mills lower bound, giving

$$\Pr\{|T_1| \geq \sqrt{LC_G}\} \geq \exp(-AC_G)$$

throughout the checked range. The determinant and parity shifts in the orthogonal cases change T_1 by at most a deterministic unit, and the one-trace L^1 error is absorbed by the replayed inequality $E_1(C_G) \leq \exp(-AC_G)$.

For the τ_G tail, use the power-map decomposition at power 2. Rains' Theorem 1.5, equations (22)–(23), and Theorem 1.7 give the following centered case split. Here $X_N^U := 2\,\mathrm{Re}\,\mathrm{Tr}\,V$ for Haar $V \in U(N)$; thus it is the trace of Rains' real-projection eigenvalue multiset and is centered at mean 0, and $X_N^{\pm}$ is the centered trace in the nonfixed N-pair part of the orthogonal determinant component $O^{\pm}$. The deterministic fixed-trace contribution deleted by Rains' convention is

$$\kappa(O^+(2N)) = \kappa(O^-(2N)) = 0, \qquad \kappa(O^+(2N+1)) = 1, \qquad \kappa(O^-(2N+1)) = -1.$$

The full square-trace means are $\mathbf{E}\,\mathrm{Tr}(U^2) = 1$ for B_b, D_b and $\mathbf{E}\,\mathrm{Tr}(U^2) = -1$ for C_b, by the adjoint character identities in lemma 4.11 and Schur orthogonality. Thus centering τ_G at $\mathbf{E}\tau_G = 1$ gives:

G	$\tau_G - 1$	component pair-count lower bound
B_b	X_b^U	$b = C_G$,
D_b	$X^+_{\lceil b/2\rceil} + X^-_{\lceil (b-1)/2\rceil}$	$\lfloor b/2\rfloor \ge \lfloor C_G/2\rfloor$,
C_b	$-X^-_{\lceil (b-1)/2\rceil} - X^+_{\lceil b/2\rceil}$	$\lfloor b/2\rfloor = \lfloor C_G/2\rfloor$.

The deterministic shifts in this table are componentwise as follows. For B_b, equation (23) with $p = 2$ gives $O^+(2b+1)^2 \sim \mathrm{Re}\,U(b)$ after the source-side fixed $+1$ eigenvalue is omitted. Restoring that source contribution gives

$$\mathrm{Tr}(U^2) = 1 + X_b^U.$$

For D_b, equation (22) with $p = 2$ gives, with $n_0 = \lceil b/2\rceil$ and $n_1 = \lceil (b-1)/2\rceil$,

$$O^+(2b)^2 \sim O^+(2n_0) \oplus O^-(2n_1 + 1),$$

where the source component has fixed contribution 0, the $O^+(2n_0)$ component has fixed contribution 0, and the $O^-(2n_1+1)$ component has a deleted fixed contribution -1. Since $X^-_{n_1}$ denotes only the centered nonfixed trace, the centered full source trace is

$$\mathrm{Tr}(U^2) - 1 \sim X^+_{\lceil b/2\rceil} + X^-_{\lceil (b-1)/2\rceil}.$$

For C_b, Rains' equation (36) identifies $\mathrm{Sp}(2b)$ with $O^-(2b+2)$ after omitting fixed ± 1 eigenvalues, whose trace contribution is $+1 - 1 = 0$. Applying equation (22) with $n = b + 1$ gives

$$O^-(2b+2)^2 \sim O^-(2\lceil (b+1)/2\rceil) \oplus O^+(2\lceil b/2\rceil + 1).$$

The first component again has deleted fixed contribution 0, while the second has deleted fixed contribution $+1$. With the centered nonfixed traces denoted by $X^-_{\lceil (b-1)/2\rceil}$ and $X^+_{\lceil b/2\rceil}$, and with $\mathbf{E}\,\mathrm{Tr}(U^2) = -1$, the convention $\tau_G = -\mathrm{Tr}(U^2)$ gives

$$-\mathrm{Tr}(U^2) - 1 \sim -X^-_{\lceil (b-1)/2\rceil} - X^+_{\lceil b/2\rceil}.$$

Thus every non-unitary component has at least $N(C_G) = \lfloor C_G/2\rfloor$ nontrivial conjugate pairs. In the B_b row, Courteaut–Johansson–Lambert, Theorem 1.1, applies to the unitary trace and projection cannot increase total variation; in the C_b, D_b rows, Courteaut–Johansson–Lambert, Theorem 1.4, applies componentwise. The replay checks that the B_b unitary-projection error is bounded by the same two-component envelope used below. For $N(C) = \lfloor C/2\rfloor$, Theorem 1.4 gives the component bound

$$E_1(N) = \frac{5(\log(2N))^{1/4}}{\sqrt{\Gamma(2N+1)}}, \qquad N(C) = \lfloor C/2\rfloor.$$

The L^1 distance of the convolution from the corresponding Gaussian convolution is at most the sum of the two component errors. The Gaussian convolution has variance 2, so the one-sided Gaussian tail gives

$$\Pr\{\tau_G \ge \eta C_G\} \le \exp\left(-\frac{(\eta C_G - 1)^2}{4}\right) + 2E_1(N(C_G)).$$

For completeness, we print the scalar certificate rather than leaving it as an unnamed replay. Put

$$\widehat{E}(C) = 5(\log(2C))^{1/4}\exp\{-C(\log(2C) - 1)\}, \qquad E_U(C) = \frac{19C^{1/4}(\log C)^{1/2}}{\Gamma(C+2)},$$

$$T(C) = \exp\left(-\frac{(\eta C - 1)^2}{4}\right) + 2\widehat{E}(\lfloor C/2\rfloor), qquadd(C) = 1 - \frac{1}{\eta^2 C^2}.$$

The elementary bound $(2C)! \ge (2C/e)^{2C}$ gives $E_1(C) \le \widehat{E}(C)$. With the rational constants r, η, A, L displayed in the statement, define

$$\begin{aligned} G(C) &= (A - L/2)C + \tfrac{1}{2}\log(LC) - \log(LC+1) - \tfrac{3}{2}\log 2, \\ R_0(C) &= e^{AC}T(C), \\ R_1(C) &= \frac{r(1 + C^{-2})}{2d(C)} e^{AC} T(C) \left(\frac{2}{r}\right)^C, \\ R_2(C) &= \frac{4r}{\eta^3 C^2 d(C)} \exp\bigl(-\bigl(\log(r/\eta) - A\bigr)C\bigr). \end{aligned} \tag{4.13.1}$$

The finite scalar certificate consists exactly of

$$G(C) \geq 0, \quad \widehat{E}(C)e^{AC} \leq 1, \quad E_U(C) \leq \widehat{E}(\lfloor C/2 \rfloor), \quad R_i(C) \leq \frac{1}{2} \ (i = 0, 1, 2). \tag{4.13.2}$$

The first two comparisons control the Gaussian lower tail and its total variation error, the third absorbs the type-B unitary projection error, and the last three are precisely the positive-tail/negative-tail ratios left by lemmas 4.5 and 4.11. The directed-MPFR replay evaluates (4.13).2 for every integer $296 \leq C \leq 10000$. At $C = 296$ it also checks, with outward rounding, the sign conditions

$$A - L/2 - (2C)^{-1} > 0, quad A - \log(2C) + (4C\log(2C))^{-1} < 0,$$
$$A - \eta(\eta C - 1)/2 < 0, quad A - \eta(\eta C - 1)/2 + \log(2/r) < 0, quad A - \log(r/\eta) < 0. \tag{4.13.3}$$

These are the derivative upper bounds used for the five monotone factors in (4.13).2; their left sides only improve with C. Consequently the endpoint check propagates beyond 10000. This lists every scalar predicate decided by the high-rank replay. ☐

Proposition 4.14 (Stable and high-rank classical tail)**.** *For the classical families B_b, C_b, D_b, all rows with the classical size parameter $C_G \geq 296$ satisfy*

$$Q_3^G(n) \geq 0$$

for every odd exponent $n \geq s(G) + 2$.

Proof. Rains moment stabilization identifies the adjoint moment prefix with the stable orthogonal/symplectic moment sequence through the required degree, and the stable Chain differences are exact positive integers. The finite-rank correction is controlled by lemma 4.13: after lemma 4.11 reduces the correction to the two defining-trace probabilities, the exact one-variable replay verifies the pushforward comparison in lemma 4.5 at the stable edge for every $C_G \geq 296$. The monotonicity clause of lemma 4.5 then gives all later odd exponents. ☐

Proposition 4.15 (Source-generated B/C row-gated bridge)**.** *For $G = B_b$ or C_b, $2 \leq b \leq 30$, and every row-gated index $b - 1 \leq m \leq 29$, one has*

$$D_G(2m + 1) > 0.$$

All 870 inequalities are regenerated from the Pieri rule and checked with exact integer arithmetic.

Proof. Write

$$e_2^j = \sum_{\lambda \vdash 2j} a_j(\lambda) s_\lambda, \qquad a_j(\lambda) \in \mathbb{Z}_{\geq 0}. \tag{4.15.1}$$

The orthogonal and symplectic Schur-integral criteria used in proposition 4.62 give

$$m_j^{B_b} = \sum_{\substack{\lambda:\ \lambda_i \text{ even} \\ \ell(\lambda) \leq 2b+1}} a_j(\lambda), \qquad m_j^{C_b} = \sum_{\substack{\lambda:\ \lambda_i \text{ even} \\ \lambda_1 \leq 2b}} a_j(\lambda). \tag{4.15.2}$$

Indeed, the first identity is the finite odd-orthogonal criterion. For the second, apply the standard involution ω: it sends h_2^j to e_2^j and s_α to $s_{\alpha'}$. A partition α has its row lengths in equal pairs precisely when every row length of α' is even, while $\ell(\alpha) \leq 2b$ is equivalent to $(\alpha')_1 \leq 2b$. This proves (4.15).2.

Pieri's rule gives the exact recurrence

$$a_{j+1}(\mu) = \sum_{\lambda:\ \mu/\lambda \text{ is a vertical two-strip}} a_j(\lambda), \qquad a_0(\varnothing) = 1. \tag{4.15.3}$$

For a band with largest active rank r, the verifier retains exactly the union

$$\ell(\lambda) \leq 2r + 1 \quad \text{or} \quad \lambda_1 \leq 2r. \tag{4.15.4}$$

A discarded partition has both height and width beyond these bounds. Adding boxes cannot decrease either quantity, so none of its descendants can enter either sum in (4.15).2. Thus this truncation is exact. A retained partition is stored in ordinary coordinates when its height is bounded and in conjugate coordinates otherwise; in the latter coordinates (4.15).3 becomes horizontal two-strip Pieri. This is a bijective change of representation, not a further truncation.

The three simultaneous bands and their exact moment windows are

band	b	finite moments generated through
low	$2, 3, 4$	61
middle	$5, 6, 7, 8$	$47, 45, 43, 43$
high	$9, 10, \ldots, 19$	41.

(4.15.5)

For the termwise source-ledger audit only, the same reconstruction is extended to degrees $51, 53, 57$ in B_{11}, B_{12}, B_{13} and to degrees $43, 51, 45$ in C_{13}, C_{14}, C_{17}, respectively. These extensions do not enlarge the row-gated bridge box. The recurrence is evaluated modulo the first five, four, or three primes, respectively, from

$$\begin{aligned}(2305843009213694963, 2305843009214695031, 2305843009215694967,\\ 2305843009216695041, 2305843009217694971).\end{aligned} \tag{4.15.6}$$

The exact inversions in each CRT reconstruction verify that the moduli are pairwise coprime; primality is not required. The relevant products are

$$\begin{aligned}
P_3 &= 12259964326943078111827907482335947634252293072155482851 > s_{41},\\
P_4 &= 28269553036527760477584768395351886838659733871027006344612486973772241891 > s_{47},\\
P_5 &= 65185151242986397660135212167169058667750503816445442673844289788983521680612515978066230161 > s_{61}.
\end{aligned} \tag{4.15.7}$$

Each sum in (4.15).2 lies in $[0, s_j]$ by nonnegativity and the stable Schur sum. Hence its modular residues have a unique representative in that interval, and the GMP Chinese-remainder reconstruction is the exact finite moment. The code substitutes the result back into every prime and also verifies stabilization before $j = 2b + 2$.

Put $m_j^G = s_j + \delta_j^G$. Outside the generated windows, proposition 4.62 and nonnegativity give the proved box

$$-s_j \leq \delta_j^G \leq 0. \tag{4.15.8}$$

For each row the verifier expands, over $\mathbb{Z}$,

$$D_G(2m+1) = D_{\mathrm{st}}(2m+1) + \sum_j L_{m,j}\delta_j^G + \sum_{i \leq j} Q_{m,i,j}\delta_i^G\delta_j^G. \tag{4.15.9}$$

An exact correction is substituted. For an interval correction, each linear term is minimized at the appropriate endpoint and each quadratic term at all four endpoint products; for a diagonal term the value at zero is included. These are the extrema because every interval in (4.15).8 is one-sided. Thus the resulting integer is a rigorous lower bound for (4.15).9.

There are

$$2\sum_{b=2}^{30}(31 - b) = 870$$

row-gated checks. The replay generates 832 correction values, evaluates 416 steps exactly and 454 by the stated interval substitution, and compares all 182 consumed archived correction claims in 14 low-rank rows termwise with the determinant reconstruction. It reports no failure. Its least lower bound is the exact value 1046 at $B_2, m = 1$. The source `character_ring_iter/verify_bc_row_gated_bridge_bounded_littlewood_gmp.cpp` and accepted transcript `certificates/classical_bridge/bc_row_gated_bounded_littlewood_certificate.log` have SHA-256 hashes

$$\begin{aligned}&\texttt{834f0005c2ef8146736a75106d327c6ca}\\ &\texttt{985ba266ebb26cdb4f35a5c5304c1dc},\\ &\texttt{62fb4525ab8d130c39c8950ed11514fd}\\ &\texttt{2ea98970d2f065907fec961d3905735f}.\end{aligned} \tag{4.15.10}$$

This proves every asserted strict inequality. ☐

Proposition 4.16 (Correction-aware classical row-gated bridge)**.** *The classical rows in the finite bridge through $m = 29$ satisfy*

$$D_G(2m+1) \geq 0$$

for every Chain step $m = 0, \ldots, 29$ whose target exponent $2m + 3$ is covered by the row gate.

Proof. The B_b, C_b rows are exactly proposition 4.15. For type D, the ranks $4 \leq b \leq 24$ are proposition 4.54, the ranks $25 \leq b \leq 52$ lie in the correction-aware interval range proved in proposition 4.51, and the ranks $b \geq 53$ are proposition 4.48. These ranges exhaust every classical row admitted by definition 4.4; all inequalities are strict, hence in particular nonnegative. ☐

Proposition 4.17 (First-hit trace rows)**.** *The 57 top finite rows at the first post-bridge hit satisfy*

$$Q_3^G(n) \geq 0$$

for every later odd exponent.

Proof. At the first hit

$$N_{\text{hit}}^{(29)}(G) = \max(63, s(G) + 2),$$

the 384-bit directed-rounding C++/MPFR certificate checks the one-trace lower bound against the negative correction terms for exactly 57 rows:

$$B : 276, 278, \ldots, 295, \quad C : 276, 278, \ldots, 295, \quad D : 276, 278, 280, \ldots, 295, 297.$$

Every interval lower endpoint is positive. The replay output is:

rows interval-certified	57,
minimum row	D_{281},
minimum log-margin interval	[0.6644444976107690011484081126, 0.6644444976107690011484081126].

The checker uses outward rounding for every rational operation, logarithm, exponential, and logarithmic subtraction. Its archived output has SHA-256

`ca3a938c9d45d65eb892f6ba637ed8c3`
`a7b1d32a571a5be52e2f5efb80477176`.

The monotonicity clause of lemma 4.5 then carries the inequality to all later odd exponents. ☐

Proposition 4.18 (Direct post-$m = 29$ low-rank tails)**.** *The following low-rank post-$m = 29$ rows satisfy*

$$Q_3^G(n) \geq 0$$

for every odd $n \geq N_{\text{hit}}^{(29)}(G)$:

family	*ranks*
B	$2 \leq b \leq 13$
C	$2 \leq b \leq 19$
D	$4 \leq b \leq 24.$

Proof. The exact box integrals for B_2, C_2, B_3, C_3 are replayed by the flags

`-b2-c2-post-m29-tail-certificate`,
`-b3-c3-post-m29-tail-certificate`.

For the rank-three Weyl source, signed permutations preserve both the adjoint character and the Weyl density. The replay therefore stores one coefficient for each dominant absolute-value triple. A transition from an orbit of size o_λ contributes its exact total mass $o_\lambda c_\lambda$ to the target orbits; division by each target orbit size is checked to be integral. This is the full Laurent recurrence modulo the order-48 Weyl action, and the unreduced and orbit recurrences are checked termwise through moments $m_0, \ldots, m_{12}$ before the reduced recurrence is continued to m_{63}. For B_4, B_5, B_6, B_8, the fixed rational cap replays apply lemma 4.7; in particular they include the factor 2^{-b} absent from an unnormalized non-simply-laced Macdonald–Mehta product. Their cap data are

G	R	x_0	independent-variable event	c
B_4	9	27	$Y \leq 1$	23
B_5	9	46	$Y \leq 1$	41
B_6	1053/50	2847/50	$\lvert Y\rvert \leq 3/2$	1386/25
B_8	1609/25	1791/25	$\lvert Y\rvert \leq 3/2$	3507/50.

The corrected exact-rational margins are positive in all four rows.

The rows

$$B_7, B_9, B_{10}, B_{11}, B_{12}, B_{13}, \qquad C_4, \ldots, C_{19}$$

use the exact determinant tails of proposition 4.41. Since $b \leq 19$ here, definition 4.1 gives $N_{\text{hit}}^{(29)}(G) = 63$. For B_{11}, B_{12}, B_{13} and C_{13}, C_{14}, C_{17}, the first-hit certificate proposition 4.44 supplies $Q_3^G(63) \geq 0$ and proposition 4.42 supplies every odd exponent before the exact-MGF onset. For C_{16}, the same first-hit certificate supplies the exponent 63 and the exact-MGF tail starts at 65. Every other row in the display has exact-MGF onset 63. Thus these certificates cover every odd exponent at least 63 in all B/C rows listed in the proposition.

For $D_4, \ldots, D_9$, the fixed rational ball-cap and product-sine replays

`-d4-post-m29-tail-certificate`, `-d5-post-m29-tail-certificate`,
`-d6-b6-post-m29-tail-certificate`, `-d7-b7-c7-post-m29-tail-certificate`,
`-d8-post-m29-tail-certificate`, `-b8-c9-d9-post-m29-tail-certificate`

instantiate lemma 4.6. Their type-D cap and central-variable data are

b	x_0	independent-variable event	c
4	26	$Y \le 1$	22
5	41	$Y \le 1$	38
6	949/20	$\lvert Y\rvert \le 7/5$	921/20
7	73	$\lvert Y\rvert \le 8/5$	357/5
8	84	$\lvert Y\rvert \le 8/5$	412/5
9	767/10	$\lvert Y\rvert \le 8/5$	751/10.

For D_4, D_5, the replay uses the per-root bound $\big(\sin(\alpha/2)/(\alpha/2)\big)^2 \ge 1/4$. For D_6, D_7 it uses

$$\prod_{\alpha>0}\left(\frac{\sin(\alpha/2)}{\alpha/2}\right)^2 \ge \left(1-\frac{R}{24}\right)^2,$$

which follows from $\sin z/z \ge 1-z^2/6$ and $\prod_i(1-u_i) \ge 1-\sum_i u_i$. For D_8, the Euler product and $\sum_{\alpha>0}(\alpha/2)^2 \le 9$ give the lower bound $(\sin 3/3)^2$; the replay inserts the rational inequality $\sin 3 \ge 15/106-(15/106)^3/6$. For D_9, put $L = |\alpha|^2 = 2$. Concavity of $\log(1-u)$ and $\alpha(\theta)^2 \le LR/\kappa$ give

$$\prod_{\alpha>0}\left(1-\frac{\alpha(\theta)^2}{24}\right)^2 \ge \left(1-\frac{LR}{24\kappa}\right)^{2\kappa/L}.$$

The exact rational replays substitute these bounds in (4.6).2–(4.6).4 and prove the pushforward-tail inequality from odd exponent 63. For D_{10} and $D_{12},\dots,D_{24}$, the exact finite-rank certificates in proposition 4.55 also start at odd exponent 63. The row D_{11} instead has an exact correction-aware Chain bridge through odd exponent 75 and an exact-MGF tail from exponent 75 onward. Finally, proposition 4.54 gives every row-gated Chain difference through $m = 29$, so the stable edge and lemma 2.11 reach odd exponent 61 in every row. These three cases cover every odd exponent at least 63. □

Proposition 4.19 (Post-$m=29$ high-edge Chernoff trace tails)**.** *For $G = B_b$ or $G = C_b$ with $218 \le b \le 275$, or $b = 277$, one has*

$$Q_3^G(n) \ge 0$$

for every odd $n \ge N_{\mathrm{hit}}^{(29)}(G)$.

Proof. This is Proposition 24.17E in the proof ledger. The active replay uses 384-bit directed rounding in `character_ring_iter/trace_square_cutoff_mpfr.cpp`. Its source-supported CJL/Rains Chernoff estimate uses

$$(r,\eta) \in \{(2001/1000, 9/20), (10001/5000, 56/125), (99/50, 56/125)\}.$$

It checks the one-trace source range, the Chernoff admissibility window, and the trace-pushforward inequality of lemma 4.5 at $N_{\mathrm{hit}}^{(29)}(G)$, with outward rounding for every rational operation, logarithm, exponential, and logarithmic subtraction. The local log

`certificates/classical_trace/first_hit_trace_replay.log`

exits with status 0 and has SHA-256 hash

`ca3a938c9d45d65eb892f6ba637ed8c3`
`a7b1d32a571a5be52e2f5efb80477176`.

All 118 interval lower endpoints are positive. The smallest is attained at B_{219} and C_{219}, and is enclosed by

$$[0.307429969543813258390342958 8, 0.3074299695438132583903429588].$$

□

Lemma 4.20 (Type B unitary square-trace tail)**.** *Let O be Haar-distributed in the defining representation of B_b, and put $\tau_B(O) = \mathrm{Tr}(O^2)$. For every real $a > 1$,*

$$\Pr\{\tau_B(O) \ge a\} \le 4\exp\left(-\frac{(a-1)^2}{4}\right).$$

Proof. Use the standard representative $O \in \mathrm{SO}(2b+1)$. Rains [6, Theorem 1.5, equation (23)], specialized to power $p = 2$, identifies the nonfixed eigenvalues of O^2 with those of $\operatorname{Re} U(b)$. Thus, if V is Haar-distributed in $U(b)$, restoring the fixed $+1$ eigenvalue gives the distributional identity

$$\tau_B(O) \overset{d}{=} 1 + 2\operatorname{Re}\operatorname{Tr} V.$$

Courteaut–Johansson–Lambert [11, Lemma 2.2, with $m = 1$] gives, for every $L > 0$,

$$\Pr\big\{(\operatorname{Re}\operatorname{Tr} V, \operatorname{Im}\operatorname{Tr} V) \notin [-L/2, L/2]^2\big\} < 4e^{-L^2/4}.$$

For every $0 < L < a - 1$, the event $\{2\operatorname{Re}\operatorname{Tr} V \geq a - 1\}$ is contained in the displayed complement of the square. Hence its probability is at most $4e^{-L^2/4}$. Letting $L \uparrow a - 1$ proves the claim. □

Proposition 4.21 (Post-$m = 29$ middle B tails from the unitary square trace)**.** *For every* $124 \leq b \leq 217$ *and every odd* $n \geq 2b + 1$*,*

$$Q_3^{B_b}(n) \geq 0.$$

Proof. Set

$$r = \frac{2001}{1000}, \qquad \eta = \frac{7}{20}, \qquad c = rb, \qquad a = \eta b, \qquad L = 2r + 3\eta = \frac{1263}{250}.$$

Then $1 < a < 2b < c$. In the notation of lemma 4.11, put

$$p_1 = \Pr\{|T_1(O)| \geq \sqrt{Lb}\}, \qquad p_2 = \Pr\{\tau_B(O) \geq \eta b\}.$$

The defining trace has mean zero. Courteaut–Johansson–Lambert [11, Theorem 1.4], which applies because $b \geq 124$, and the two-sided Mills lower bound with $\pi < 4$ give

$$p_1 \geq G_b - E_b,$$

where

$$G_b = \frac{e^{-Lb/2}\sqrt{Lb}}{\sqrt{2}(Lb+1)}, \qquad E_b = \frac{5(\log(2b))^{1/4}}{\sqrt{(2b)!}}.$$

By lemma 4.20,

$$p_2 \leq U_b := 4\exp\left(-\frac{(\eta b - 1)^2}{4}\right).$$

The directed-rounding MPFR replay

`certificates/post_m29/ post_m29_b_unitary_square_trace_mpfr_certificate.log`

checks, for every integer $124 \leq b \leq 217$, the four inequalities

$$E_b \leq \frac{1}{4}G_b, \qquad U_b \leq \frac{1}{4}G_b,$$

$$(rb)^2\big(1 - (\eta b)^{-2}\big)\frac{G_b}{2} \geq 2\left(2(b^2+1)U_b(2/r)^{2b+1}\right),$$

and

$$(rb)^2\big(1 - (\eta b)^{-2}\big)\frac{G_b}{2} \geq 2\left(2(\eta/r)^{2b+1}\right).$$

It uses 384-bit MPFR intervals, rounds every lower bound downward and every upper bound upward, exits with status 0, and reports `rows_checked=94` and `failures=0`. The four smallest natural-log margins are all attained at B_{124} and are respectively

$$241.28842853377649\ldots, \qquad 129.87520977348063\ldots,$$
$$130.69322950465274\ldots, \qquad 126.27950899088545\ldots.$$

The log has SHA-256

`7780bfe3253bf9e030a9a544dcd7327f`
`cd905aed8a2890ded191dc38efae8230`,

and the C++ source has SHA-256

`d0d503ffc4bd6cb0485a97a4cef2178a`
`930f3f00c38ffd4486cb66aa01652aa7`.

The first two inequalities imply $p_1 - p_2 \geq G_b/2$. Hence lemma 4.11 gives

$$P_{B_b}(c) \geq (rb)^2\big(1 - (\eta b)^{-2}\big)\frac{G_b}{2}, \qquad N_{B_b}(a) \leq 2(b^2+1)U_b.$$

The final two checked inequalities imply

$$P_{B_b}(c) \geq N_{B_b}(a)(2/r)^{2b+1} + 2(\eta/r)^{2b+1}.$$

Thus lemma 4.5 applies at the odd exponent $2b+1$ and, by its monotonicity clause, at every later odd exponent. □

Definition 4.22 (CJL Fredholm remainder)**.** For an integer $d \geq 1$ and $0 \leq v < d+1$, set

$$\mathfrak{b}_d(v) = \frac{\exp\big(v^2/(d+1)\big)}{1 - v^2/(d+1)^2} \frac{v^d}{d!}.$$

Definition 4.23 (CJL trace-norm envelope)**.** For an integer $d \geq 1$ and $0 \leq v < d+1$, set

$$\mathfrak{a}_d(v) = \frac{\exp(v^2/(d+1))v^d}{d!} \frac{1}{1 - v/(d+1)} \frac{1}{\sqrt{1 - v^2/(d+1)^2}}.$$

Lemma 4.24 (Extended small-Fredholm trace MGF bounds)**.** *Let Z be a centered full defining trace in one of the four orthogonal Jacobi components of Courteaut–Johansson–Lambert, or in the corresponding symplectic component, and let d be the total matrix dimension attached to that component. If*

$$0 \leq v < d+1, \qquad \mathfrak{a}_d(v) < 1,$$

then $\mathfrak{b}_d(v) < 1$ and

$$e^{v^2/2}(1 - \mathfrak{b}_d(v)) \leq \mathbf{E}e^{\pm vZ} \leq \frac{e^{v^2/2}}{1 - \mathfrak{b}_d(v)}. \tag{4.24.1}$$

Proof. Courteaut–Johansson–Lambert [11, Proposition 3.3] writes the centered transform as

$$\mathbf{E}e^{zZ} = e^{z^2/2} \det(I + QK_zQ).$$

The proof of their Lemma 3.6 starts from the trace-norm inequality in their equation (3.4), inserts their Bessel bound (2.16), and sums two geometric series. If the least Bessel index is denoted by the total dimension d, the calculation before their equation (3.5) gives, without using the later sufficient restriction $|z| < 2e^{-5/4}n$,

$$\|QK_zQ\|_{\mathcal{J}_1} \leq \frac{\exp(|z|^2/(d+1))|z|^d}{d!} \frac{1}{1 - |z|/(d+1)} \frac{1}{\sqrt{1 - |z|^2/(d+1)^2}} = \mathfrak{a}_d(|z|)$$

whenever $|z| < d+1$. The four kernels listed in Proposition 3.3 have least indices $2n, 2n+1, 2n+1, 2n+2$, respectively, exactly their corresponding total dimensions, so the same displayed calculation applies to every case.

The proof of their Lemma 3.7, in particular equation (3.6), also gives directly

$$\sum_{j \geq 1} \frac{1}{j} \left|\mathrm{Tr}(QK_zQ)^j\right| \leq -\log(1 - \mathfrak{b}_d(|z|)).$$

Here $\mathfrak{b}_d(v) < \mathfrak{a}_d(v) < 1$. Thus the trace-norm hypothesis justifies Plemelj's series, and the last display bounds the absolute value of the logarithm of the determinant. For real $z = \pm v$, the determinant is positive because it equals $e^{-v^2/2}\mathbf{E}e^{\pm vZ}$. Exponentiation proves (4.24).1. □

Lemma 4.25 (Layered exponential-tilt lower tail)**.** *Let Z satisfy (4.24).1 at every argument used below, with the same dimension d. Fix $s, l > 0$, put $t = s - l$, and choose $\lambda_- > 0$ and pairs (u_i, λ_i), $i = 0, \ldots, L-1$, such that*

$$0 < u_0 < \cdots < u_{L-1}.$$

Write $D(v) = 1 - \mathfrak{b}_d(v)$ and define

$$A_- = \frac{\exp(\lambda_-^2/2 - \lambda_- l)}{D(s)D(s - \lambda_-)}, \qquad A_i^+ = \frac{\exp(\lambda_i^2/2 - \lambda_i u_i)}{D(s)D(s + \lambda_i)},$$
$$g_i = 1 - A_- - A_i^+.$$

Suppose

$$0 < g_0 \leq g_1 \leq \cdots \leq g_{L-1}.$$

Then, with $w_i = e^{-su_i}$,

$$\Pr\{Z \geq t\} \geq V_d(s,l) := D(s)e^{-s^2/2}\left(w_{L-1}g_{L-1} + \sum_{i=0}^{L-2}(w_i - w_{i+1})g_i\right). \tag{4.25.1}$$

If the same MGF bounds hold after replacing Z by $-Z$, then

$$\Pr\{|Z| \geq t\} \geq 2V_d(s,l). \tag{4.25.2}$$

Proof. Tilt the law by $d\mathbf{P}_s = M(s)^{-1}e^{sZ}\,d\mathbf{P}$, where $M(s) = \mathbf{E}e^{sZ}$. Exponential Markov at the independently chosen parameters λ_- and λ_i, followed by lemma 4.24, gives

$$\mathbf{P}_s\{Z - s \le -l\} \le A_-, \qquad \mathbf{P}_s\{Z - s \ge u_i\} \le A_i^+.$$

Consequently

$$\mathbf{P}_s\{t < Z \le s + u_i\} \ge g_i.$$

For a decreasing function, the least integral compatible with these nested cumulative-mass lower bounds places each required mass increment at the right endpoint. Hence

$$\mathbf{E}_s[e^{-s(Z-s)}\mathbf{1}_{\{Z>t\}}] \ge w_{L-1}g_{L-1} + \sum_{i=0}^{L-2}(w_i - w_{i+1})g_i.$$

Multiplying by $M(s)e^{-s^2}$ and using $M(s) \ge e^{s^2/2}D(s)$ proves (4.25).1. Applying the same argument to $-Z$ and adding the two disjoint tails proves (4.25).2. $\square$

Lemma 4.26 (Arbitrary-parameter $B/C/D$ square-trace MGF bounds)**.** *For B_b, let $\tau_B = \mathrm{Tr}(O^2)$ and, for $x > 1$, $v > 0$, put*

$$U_B(x,v) = \exp\bigl(-v(x-1) + v^2\bigr).$$

Then

$$e^{-vx}\mathbf{E}e^{v\tau_B} \le U_B(x,v). \tag{4.26.1}$$

For C_b, use the integers d_0, d_1 of lemma 4.32 and suppose $\mathfrak{a}_{d_i}(v) < 1$ for $i = 0, 1$. Put

$$U_C(x,v) = \frac{\exp(-v(x-1) + v^2)}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Then

$$e^{-vx}\mathbf{E}e^{v\tau_C} \le U_C(x,v). \tag{4.26.2}$$

For D_b, put

$$m_0 = \left\lceil \frac{b}{2} \right\rceil, \qquad m_1 = \left\lceil \frac{b-1}{2} \right\rceil, \qquad d_0 = 2m_0, \qquad d_1 = 2m_1 + 1,$$

and suppose $\mathfrak{a}_{d_i}(v) < 1$ for $i = 0, 1$. With $\tau_D = \mathrm{Tr}(O^2)$, put

$$U_D(x,v) = \frac{\exp(-v(x-1) + v^2)}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Then

$$e^{-vx}\mathbf{E}e^{v\tau_D} \le U_D(x,v). \tag{4.26.3}$$

Proof. For B_b, the Rains identity used in lemma 4.20 gives $\tau_B \overset{d}{=} 1 + 2\,\mathrm{Re}\,\mathrm{Tr}\,V$ with V Haar in $U(b)$. Courteaut–Johansson–Lambert [11, Lemma 2.12], applied to the real function $f(\theta) = 2v\cos\theta$, gives

$$\mathbf{E}e^{2v\,\mathrm{Re}\,\mathrm{Tr}\,V} \le e^{v^2}.$$

This proves (4.26).1. For C_b, the independent Rains components Y_0, Y_1 in the proof of lemma 4.32 satisfy $\tau_C - 1 \overset{d}{=} -(Y_0 + Y_1)$. Apply the upper bound in lemma 4.24 to each centered component at $-v$, multiply the two bounds, and include the deterministic $+1$. This gives (4.26).2.

For D_b, Rains' power-two decomposition [6, Theorem 1.5, equation (22)], with the fixed eigenvalues restored as in the proof of lemma 4.13, gives independent centered full orthogonal traces Y_0, Y_1 of total dimensions d_0, d_1 such that

$$\tau_D - 1 \overset{d}{=} Y_0 + Y_1.$$

Apply the upper bound of lemma 4.24 at v to both components, multiply, and include the deterministic $+1$. This proves (4.26).3. $\square$

Lemma 4.27 (Type B defining-trace MGF sandwich)**.** *Let O be Haar-distributed in $\mathrm{SO}(2b+1)$, put $T_b = \mathrm{Tr}\,O$, and, for $v \ge 0$, define*

$$\beta_b(v) = \mathfrak{b}_{2b+1}(v).$$

If $0 \le v < 2e^{-5/4}b$, then $\beta_b(v) < 1$ and

$$e^{v^2/2}\bigl(1 - \beta_b(v)\bigr) \le \mathbf{E}e^{vT_b} \le \frac{e^{v^2/2}}{1 - \beta_b(v)}.$$

Proof. In the notation of Courteaut–Johansson–Lambert, the b nonfixed eigenvalue pairs of $O^+(2b+1)$ have Jacobi parameters $(1/2,-1/2)$. Their trace variable in Section 3 omits the deterministic $+1$ eigenvalue and has mean -1. Thus the linear factor in [11, Proposition 3.3] cancels when that eigenvalue is restored. Specializing the proposition at $\xi=-iv$ gives

$$\mathbf{E}e^{vT_b}=e^{v^2/2}\det\bigl(I+Q_bK_v^{+-}Q_b\bigr).$$

For $v<2e^{-5/4}b$, Lemmas 3.6–3.7 of the same source, with total matrix size $d=2b+1$, give

$$\bigl|\log\det\bigl(I+Q_bK_v^{+-}Q_b\bigr)\bigr|\le-\log\bigl(1-\beta_b(v)\bigr).$$

The determinant is positive because it is $e^{-v^2/2}\mathbf{E}e^{vT_b}$. Exponentiating the last display proves both bounds. □

Lemma 4.28 (Exponentially tilted type B defining-trace tail)**.** *In the setting of lemma 4.27, let $t,h>0$, put $s=t+h$, and suppose $t+2h<2e^{-5/4}b$. Define*

$$\rho_b(t,h)=1-\frac{e^{-h^2/2}}{1-\beta_b(s)}\left(\frac{1}{1-\beta_b(t)}+\frac{1}{1-\beta_b(t+2h)}\right).$$

If $\rho_b(t,h)>0$, then

$$\Pr\{T_b\ge t\}\ge V_b(t,h):=\bigl(1-\beta_b(s)\bigr)\rho_b(t,h)\exp\left(-\frac{s^2}{2}-sh\right).$$

Proof. Write $M(v)=\mathbf{E}e^{vT_b}$ and tilt Haar probability by

$$d\mathbf{P}_s=M(s)^{-1}e^{sT_b}\,d\mathbf{P}.$$

Exponential Markov and lemma 4.27 give

$$\mathbf{P}_s\{T_b-s\ge h\}\le\frac{e^{-h^2/2}}{(1-\beta_b(s))(1-\beta_b(s+h))}$$

and

$$\mathbf{P}_s\{T_b-s\le -h\}\le\frac{e^{-h^2/2}}{(1-\beta_b(s))(1-\beta_b(s-h))}.$$

Consequently $\mathbf{P}_s\{s-h<T_b<s+h\}\ge\rho_b(t,h)$. Since $s-h=t$,

$$\begin{aligned}\Pr\{T_b\ge t\}&\ge M(s)\,\mathbf{E}_s\bigl[e^{-sT_b}\mathbf{1}_{\{s-h<T_b<s+h\}}\bigr]\\&\ge M(s)e^{-s(s+h)}\rho_b(t,h)\\&\ge(1-\beta_b(s))\rho_b(t,h)e^{-s^2/2-sh},\end{aligned}$$

as claimed. □

Proposition 4.29 (Post-$m=29$ middle B tails from an MGF tilt)**.** *For every $62\le b\le 123$ and every odd $n\ge 2b+29$,*

$$Q_3^{B_b}(n)\ge 0.$$

Proof. Set

$$r=\frac{20001}{10000},\qquad \alpha=\frac{979}{250},\qquad h=\frac{601}{500},\qquad c_b=rb,\qquad a_b=\alpha\sqrt{b},$$

and put

$$t_b=\sqrt{2c_b+3a_b},\qquad s_b=t_b+h,\qquad V_b=V_b(t_b,h),$$

where the final quantity is defined in lemma 4.28. One has $1<a_b<2b<c_b$. In the notation of lemma 4.11,

$$p_1\ge\Pr\{T_b\ge t_b\}\ge V_b,\qquad p_2\le U_b:=4\exp\left(-\frac{(a_b-1)^2}{4}\right),$$

where the second inequality is lemma 4.20.

The 384-bit directed-rounding MPFR replay

`certificates/post_m29/ post_m29_bc_tilted_mgf_mpfr_certificate.log`

checks, for every integer $62\le b\le 123$, that

$$t_b+2h<2e^{-5/4}b,\qquad \rho_b(t_b,h)>0,\qquad V_b\ge 2U_b,$$

and that

$$c_b^2(1-a_b^{-2})\frac{V_b}{2}\ge 2\left(2(b^2+1)U_b(2/r)^{2b+29}\right),$$

$$c_b^2(1-a_b^{-2})\frac{V_b}{2} \ge 2\left(2(a_b/c_b)^{2b+29}\right).$$

Every lower quantity is rounded downward and every upper quantity upward. The combined B/C log exits with status 0, reports `B_rows_checked=62` and `failures=0`, and places all five B-side minima at B_{62}. The minimum domain and tilted-mass lower bounds are respectively

$$14.669512561242725\ldots, \qquad 0.028831039797142247\ldots,$$

and the three minimum natural-log comparison margins are

$$0.11377835996790686\ldots, \qquad 0.12021572562760810\ldots, \qquad 0.15645659337912115\ldots.$$

The log has SHA-256

`ca6860204466e4a6b38d9c5597511cd1`
`1cfebb4b1b65cab88db6ee45f9477111`,

and the C++ source has SHA-256

`0f5ad1037bcf738ab4ddd9c5b12f151a`
`adc1bff2b5c4726c60622f608ef9e0e7`.

The first comparison gives $p_1 - p_2 \ge V_b/2$. Hence lemma 4.11 gives

$$P_{B_b}(c_b) \ge c_b^2(1-a_b^{-2})\frac{V_b}{2}, \qquad N_{B_b}(a_b) \le 2(b^2+1)U_b.$$

The final two checked comparisons imply the hypothesis of lemma 4.5 at the odd exponent $2b+29$. Its monotonicity clause proves every later odd exponent. $\square$

Lemma 4.30 (Type C defining-trace MGF sandwich)**.** *Let U be Haar-distributed in* $\mathrm{Sp}(2b)$*, put $S_b = \operatorname{Tr} U$, and define*

$$\gamma_b(v) = \mathfrak{b}_{2b}(v).$$

If $0 \le v < 2e^{-5/4}b$, then $\gamma_b(v) < 1$ and

$$e^{v^2/2}\big(1-\gamma_b(v)\big) \le \mathbf{E}e^{vS_b} \le \frac{e^{v^2/2}}{1-\gamma_b(v)}.$$

Proof. For $\mathrm{Sp}(2b)$, the Jacobi parameters in [11, Section 3, equation (3.1)] are $(1/2,1/2)$, b nontrivial conjugate pairs, and total matrix dimension $d = 2b$. There are no deterministic eigenvalues and the linear factor in [11, Proposition 3.3] vanishes. Substitution of $\xi = -iv$ in that proposition therefore gives

$$\mathbf{E}e^{vS_b} = e^{v^2/2}\det\big(I + Q_bK_v^{++}Q_b\big).$$

For $v < 2e^{-5/4}b$, Lemmas 3.6–3.7 of the same source, with $d = 2b$, give

$$\left|\log\det\big(I + Q_bK_v^{++}Q_b\big)\right| \le -\log\big(1-\mathfrak{b}_{2b}(v)\big).$$

The determinant is positive because it equals $e^{-v^2/2}\mathbf{E}e^{vS_b}$. Exponentiation proves both bounds. $\square$

Lemma 4.31 (Exponentially tilted type C defining-trace tail)**.** *In the setting of lemma 4.30, let $t,h > 0$, put $s = t+h$, and suppose $t+2h < 2e^{-5/4}b$. Define*

$$\rho_b^C(t,h) = 1 - \frac{e^{-h^2/2}}{1-\gamma_b(s)}\left(\frac{1}{1-\gamma_b(t)} + \frac{1}{1-\gamma_b(t+2h)}\right).$$

If $\rho_b^C(t,h) > 0$, then

$$\Pr\{S_b \ge t\} \ge W_b(t,h) := \big(1-\gamma_b(s)\big)\rho_b^C(t,h)\exp\!\left(-\frac{s^2}{2} - sh\right).$$

Proof. The proof of lemma 4.28 uses only the two displayed MGF bounds in lemma 4.27. Apply its exponential tilt at $s = t+h$ with S_b in place of T_b and with γ_b in place of β_b. The two exponential-Markov estimates become

$$\mathbf{P}_s\{S_b - s \ge h\} \le \frac{e^{-h^2/2}}{(1-\gamma_b(s))(1-\gamma_b(s+h))},$$

$$\mathbf{P}_s\{S_b - s \le -h\} \le \frac{e^{-h^2/2}}{(1-\gamma_b(s))(1-\gamma_b(s-h))}.$$

Their sum leaves interval mass at least $\rho_b^C(t,h)$. On $s-h < S_b < s+h$, one has $e^{-sS_b} \ge e^{-s(s+h)}$; the lower MGF bound of lemma 4.30 then gives exactly the displayed $W_b(t,h)$. $\square$

Lemma 4.32 (Type C square-trace MGF tail)**.** *Let U be Haar-distributed in $\mathrm{Sp}(2b)$, where $b \geq 2$, and put $\tau_C(U) = -\operatorname{Tr}(U^2)$. Define*

$$m_0 = \left\lceil \frac{b+1}{2} \right\rceil, \quad m_1 = \left\lceil \frac{b}{2} \right\rceil, \quad n_0 = m_0 - 1, \quad n_1 = m_1,$$

$$d_0 = 2m_0, \qquad d_1 = 2m_1 + 1.$$

For $a > 1$, put $v = (a-1)/2$. If

$$v < 2e^{-5/4} \min(n_0, n_1),$$

then $\mathfrak{b}_{d_i}(v) < 1$ for $i = 0, 1$ and

$$\Pr\{\tau_C(U) \geq a\} \leq \frac{\exp\bigl(-(a-1)^2/4\bigr)}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Proof. Rains [6, Theorem 1.5 and the subsequent remark, equation (36)] identifies the nonfixed eigenvalues of $\mathrm{Sp}(2b)$ with those of $O^-(2b+2)$. The omitted eigenvalues are $+1, -1$, so after squaring their trace contribution is 2. Equation (22) of the same theorem at $p = 2$ decomposes the remaining eigenvalue distribution into independent components

$$O^-(2m_0) \oplus O^+(2m_1 + 1).$$

Let Y_0, Y_1 be the full defining traces of these two components. The first component has fixed eigenvalues $+1, -1$, of total trace 0; the second has a fixed $+1$. Restoring these fixed eigenvalues in Rains' identity therefore gives

$$\operatorname{Tr}(U^2) \stackrel{d}{=} Y_0 + Y_1 - 1, \qquad \tau_C(U) - 1 \stackrel{d}{=} -(Y_0 + Y_1).$$

By Courteaut–Johansson–Lambert [11, Section 3, equation (3.1)], the first component has Jacobi parameters $(1/2, 1/2)$ and n_0 nontrivial pairs, while the second has parameters $(1/2, -1/2)$ and n_1 nontrivial pairs. Proposition 3.3 shows that the nonfixed trace means are respectively 0 and -1; after the fixed traces 0 and $+1$ are restored, both Y_0, Y_1 are centered. Lemmas 3.6–3.7, at the argument $-v$, now give

$$\mathbf{E}e^{-vY_i} \leq \frac{e^{v^2/2}}{1 - \mathfrak{b}_{d_i}(v)} \qquad (i = 0, 1).$$

Independence and exponential Markov yield

$$\Pr\{-(Y_0 + Y_1) \geq a - 1\} \leq \frac{e^{-v(a-1)+v^2}}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Since $v = (a-1)/2$, the exponent is $-(a-1)^2/4$. ☐

Proposition 4.33 (Post-$m = 29$ middle C tails from MGF tilts)**.** *For every $62 \leq b \leq 217$ and every odd $n \geq 2b + 29$,*

$$Q_3^{C_b}(n) \geq 0.$$

Proof. Set

$$r = \frac{20001}{10000}, \qquad \alpha = \frac{979}{250}, \qquad h = \frac{601}{500}, \qquad c_b = rb, \qquad a_b = \alpha\sqrt{b},$$

and put

$$t_b = \sqrt{2c_b + 3a_b}, \qquad s_b = t_b + h, \qquad W_b = W_b(t_b, h),$$

where the final quantity is defined in lemma 4.31. For the square trace, use the integers m_i, n_i, d_i of lemma 4.32, put $v_b = (a_b - 1)/2$, and define

$$U_b^C = \frac{\exp\bigl(-(a_b-1)^2/4\bigr)}{(1 - \mathfrak{b}_{d_0}(v_b))(1 - \mathfrak{b}_{d_1}(v_b))}.$$

One has $1 < a_b < 2b < c_b$. In the notation of lemma 4.11,

$$p_1 \geq \Pr\{S_b \geq t_b\} \geq W_b, \qquad p_2 \leq U_b^C,$$

by lemmas 4.31 and 4.32.

The 384-bit directed-rounding MPFR replay

`certificates/post_m29/ post_m29_bc_tilted_mgf_mpfr_certificate.log`

checks, for every integer $62 \leq b \leq 217$, that

$$t_b + 2h < 2e^{-5/4}b, \qquad v_b < 2e^{-5/4}\min(n_0, n_1), \qquad \rho_b^C(t_b, h) > 0, \qquad W_b \geq 2U_b^C,$$

and that

$$c_b^2(1-a_b^{-2})\frac{W_b}{2} \ge 2\left(2(b^2+1)U_b^C(2/r)^{2b+29}\right),$$
$$c_b^2(1-a_b^{-2})\frac{W_b}{2} \ge 2\left(2(a_b/c_b)^{2b+29}\right).$$

Every lower quantity is rounded downward and every upper quantity upward. The replay exits with status 0, reports `C_rows_checked=156` and `failures=0`, and gives the minimum defining-trace domain, square-trace domain, and tilted-mass lower bounds

$$14.669512561242725\ldots, \qquad 2.7221542041383810\ldots, \qquad 0.028831039797142247\ldots.$$

The three minimum natural-log comparison margins are

$$1.5000727210859520\ldots, \qquad 1.5065100867456532\ldots, \qquad 0.15645659337912115\ldots.$$

The log has SHA-256

`ca6860204466e4a6b38d9c5597511cd1`
`1cfebb4b1b65cab88db6ee45f9477111`,

and the C++ source has SHA-256

`0f5ad1037bcf738ab4ddd9c5b12f151a`
`adc1bff2b5c4726c60622f608ef9e0e7`.

The comparison $W_b \ge 2U_b^C$ gives $p_1 - p_2 \ge W_b/2$. Hence lemma 4.11 gives

$$P_{C_b}(c_b) \ge c_b^2(1-a_b^{-2})\frac{W_b}{2}, \qquad N_{C_b}(a_b) \le 2(b^2+1)U_b^C.$$

The final two checked comparisons imply the hypothesis of lemma 4.5 at the odd exponent $2b+29$. Its monotonicity clause proves every later odd exponent. ☐

Lemma 4.34 (Exact odd-orthogonal defining-trace MGFs)**.** *Let O be Haar-distributed in $\mathrm{SO}(2b+1)$, put $Z_b = \operatorname{Tr} O$, and, for $v > 0$ and $\epsilon \in \{+1,-1\}$, set $M_{b,\epsilon}(v) = \mathbf{E}e^{\epsilon v Z_b}$. Then*

$$M_{b,+}(v) = e^{v}\det\left(I_{|j-k|}(2v) - I_{j+k+1}(2v)\right)_{0\le j,k<b}, \tag{4.34.1}$$

$$M_{b,-}(v) = e^{-v}\det\left(I_{|j-k|}(2v) + I_{j+k+1}(2v)\right)_{0\le j,k<b}, \tag{4.34.2}$$

where I_m is the modified Bessel function. Both displayed matrices are positive definite.

Proof. For $O^+(2b+1)$, Courteaut–Johansson–Lambert [11, Section 3, equation (3.1)] identify the b nontrivial eigenvalue pairs with Jacobi parameters $(1/2,-1/2)$. Their exact Toeplitz–Hankel identity [11, Lemma 3.1, the E_n^{+-} formula], applied to $\psi(x) = e^{2vx}$, has Fourier coefficients $\widehat{\phi}_m = I_m(2v)$. Restoring the deterministic $+1$ eigenvalue gives (4.34).1.

For the argument $-v$, use $I_m(-2v) = (-1)^m I_m(2v)$. Conjugating the resulting matrix by $\operatorname{diag}(1,-1,1,-1,\ldots)$ changes it to the plus-Hankel matrix in (4.34).2 and does not change its determinant. The two matrices are moment Gram matrices for the strictly positive Jacobi weights occurring in Lemma 3.1 of the cited source, so they are positive definite. ☐

Lemma 4.35 (Exact even-orthogonal defining-trace MGFs)**.** *For $b \ge 2$, let $Z_b^+ = \operatorname{Tr} O$ with O Haar-distributed in $\mathrm{SO}(2b) = O^+(2b)$. For $r \ge 1$, let $Z_r^- = \operatorname{Tr} O$ with O Haar-distributed in the determinant-minus component $O^-(2r+2)$. For $v > 0$,*

$$\mathbf{E}e^{vZ_b^+} = \frac{1}{2}\det\left(I_{|j-k|}(2v) + I_{j+k}(2v)\right)_{0\le j,k<b}, \tag{4.35.1}$$

and

$$\mathbf{E}e^{vZ_r^-} = \det\left(I_{|j-k|}(2v) - I_{j+k+2}(2v)\right)_{0\le j,k<r}. \tag{4.35.2}$$

Both displayed matrices are positive definite.

Proof. Courteaut–Johansson–Lambert [11, Section 3, equation (3.1)] identify $O^+(2b)$ with the Jacobi parameters $(-1/2,-1/2)$ and $O^-(2r+2)$ with the parameters $(1/2,1/2)$. Apply their exact Toeplitz–Hankel identities [11, Lemma 3.1, the E_b^{--} and E_r^{++} formulas] to $\psi(x) = e^{2vx}$, whose Fourier coefficients are $I_j(2v)$.

For completeness, the factor $1/2$ in (4.35).1 is forced by the normalized $(-1/2,-1/2)$ Jacobi measure. In the orthonormal cosine basis $1, \sqrt{2}T_1, \sqrt{2}T_2, \ldots$, Andreief's identity gives a Gram determinant. The unscaled Toeplitz–Hankel matrix in the display is obtained by multiplying the first Gram row and column by $\sqrt{2}$, so its determinant is twice the normalized expectation. In particular it equals 2 at $v = 0$. The $(1/2,1/2)$ formula is already normalized at $v = 0$. Finally, both matrices are Gram matrices, up to this positive diagonal rescaling, for their strictly positive Jacobi weights. Hence every principal pivot is positive. $\square$

Lemma 4.36 (Exact low-rank trace supports)**.** *For $O \in \mathrm{SO}(2b+1)$,*

$$-(2b-1) \le \operatorname{Tr} O \le 2b+1. \tag{4.36.1}$$

For $O \in O^-(2r+2)$,

$$|\operatorname{Tr} O| \le 2r. \tag{4.36.2}$$

For $U \in \mathrm{Sp}(2b)$ and $\tau_C = -\operatorname{Tr}(U^2)$,

$$|\operatorname{Tr} U| \le 2b, \qquad |\tau_C| \le 2b. \tag{4.36.3}$$

Proof. The nonreal eigenvalues occur in conjugate unit-circle pairs and each pair contributes a number in $[-2, 2]$ to the trace. An element of $\mathrm{SO}(2b+1)$ has a forced $+1$ eigenvalue, giving the upper endpoint $2b+1$ and the lower endpoint $1-2b$. An element of $O^-(2r+2)$ has forced eigenvalues $+1, -1$, whose trace contributions cancel, leaving r conjugate pairs. A symplectic matrix has b such pairs and no forced real eigenvalue; squaring preserves the unit-circle pair structure. These observations give all three displays. $\square$

Lemma 4.37 (Exact type-C defining-trace MGF)**.** *Let U be Haar-distributed in $\mathrm{Sp}(2b)$, $b \ge 1$, and put $S_b = \operatorname{Tr} U$. For every $v > 0$,*

$$\mathbf{E}e^{vS_b} = \mathbf{E}e^{-vS_b} = \det\left(I_{|j-k|}(2v) - I_{j+k+2}(2v)\right)_{0\le j,k<b}. \tag{4.37.1}$$

The displayed matrix is positive definite.

Proof. The remark following Rains' Theorem 1.5 [6, equation (36)] identifies the nonfixed eigenvalues of $\mathrm{Sp}(2b)$ with those of $O^-(2b+2)$. The latter component has fixed eigenvalues $+1, -1$, of total trace zero, so the full traces have the same distribution. Equation (4.35).2 with $r = b$ gives the determinant in (4.37).1 and its positive definiteness. Multiplication of U by the central element $-I$ preserves Haar measure and changes S_b to $-S_b$, proving the equality of the two MGFs. $\square$

Lemma 4.38 (Exact type-C square-trace MGF)**.** *Let U be Haar-distributed in $\mathrm{Sp}(2b)$, $b \ge 2$, and put $\tau_C = -\operatorname{Tr}(U^2)$. Define*

$$m_0 = \left\lceil \frac{b+1}{2} \right\rceil, \qquad m_1 = \left\lceil \frac{b}{2} \right\rceil,$$

and

$$H_m(v) = \det\left(I_{|j-k|}(2v) - I_{j+k+2}(2v)\right)_{0\le j,k<m-1}.$$

With $M_{m_1,-}$ as in (4.34).2, one has, for every $v > 0$,

$$\mathbf{E}e^{v\tau_C} = e^v H_{m_0}(v) M_{m_1,-}(v). \tag{4.38.1}$$

Both determinant matrices on the right are positive definite.

Proof. Use Rains' identification $\mathrm{Sp}(2b) \sim O^-(2b+2)$ after omitting the fixed eigenvalues $+1, -1$ [6, remark following Theorem 1.5, equation (36)]. Theorem 1.5 at $p = 2$, specifically equation (22), decomposes the nonfixed eigenvalues after squaring into independent components

$$O^-(2m_0) \oplus O^+(2m_1+1).$$

If Y_0, Y_1 are their full traces, then the first component's fixed eigenvalues have total trace zero and the second component has a fixed $+1$. Restoring these eigenvalues gives

$$\tau_C - 1 \overset{d}{=} -(Y_0 + Y_1).$$

The law of Y_0 is symmetric because multiplication by $-I$ preserves the determinant-minus component in even dimension. Thus $\mathbf{E}e^{-vY_0} = H_{m_0}(v)$ by (4.35).2. By definition, $\mathbf{E}e^{-vY_1} = M_{m_1,-}(v)$. Independence proves (4.38).1. Positive definiteness is supplied by lemmas 4.34 and 4.35. $\square$

Lemma 4.39 (Exact type-D square-trace MGF)**.** *Let O be Haar-distributed in $\mathrm{SO}(2b)$ and put $\tau_D = \operatorname{Tr}(O^2)$. Define*

$$m_0 = \left\lceil \frac{b}{2} \right\rceil, \qquad m_1 = \left\lceil \frac{b-1}{2} \right\rceil.$$

If $G_{m,+}(v) = \mathbf{E}e^{vZ_m^+}$ denotes the even-orthogonal MGF in (4.35).1 and $M_{m,-}(v)$ denotes the odd-orthogonal MGF in (4.34).2, then, for $v > 0$,

$$\mathbf{E}e^{v\tau_D} = e^v G_{m_0,+}(v) M_{m_1,-}(v). \tag{4.39.1}$$

Proof. Rains' power-two decomposition [6, Theorem 1.5, equation (22)], with the fixed eigenvalues restored, gives independent full traces Y_0, Y_1 on $O^+(2m_0)$ and $O^-(2m_1+1)$. The latter component has a forced eigenvalue -1, which is omitted in Rains' convention, and therefore $\tau_D - 1 \overset{d}{=} Y_0 + Y_1$; this is the type-$D$ identity used in the proof of lemma 4.26. Independence and lemmas 4.34 and 4.35 give (4.39).1. ☐

Lemma 4.40 (Layered tilt with exact two-sided MGFs)**.** *Let Z be a real random variable and write $M_\epsilon(v) = \mathbf{E}e^{\epsilon v Z}$ for $\epsilon \in \{+1,-1\}$. Fix $s>0$ and $t>0$, and choose $\lambda_- > 0$ and pairs (x_i,λ_i), $i = 0,\dots,L-1$, with $t < x_0 < \cdots < x_{L-1}$ and $\lambda_i > 0$. Assume the displayed MGFs are finite. For each sign define*

$$A_{\epsilon,-} = e^{\lambda_- t}\frac{M_\epsilon(s-\lambda_-)}{M_\epsilon(s)}, \qquad A^+_{\epsilon,i} = e^{-\lambda_i x_i}\frac{M_\epsilon(s+\lambda_i)}{M_\epsilon(s)},$$

$$g_{\epsilon,i} = 1 - A_{\epsilon,-} - A^+_{\epsilon,i}, \qquad w_i = e^{-s x_i}.$$

If $\epsilon Z \le x_i$ almost surely, one may instead set $A^+_{\epsilon,i} = 0$. If $0 < g_{\epsilon,0} \le \cdots \le g_{\epsilon,L-1}$, then

$$\Pr\{\epsilon Z \ge t\} \ge V_\epsilon := M_\epsilon(s)\left(w_{L-1}g_{\epsilon,L-1} + \sum_{i=0}^{L-2}(w_i - w_{i+1})g_{\epsilon,i}\right). \tag{4.40.1}$$

Consequently $\Pr\{|Z| \ge t\} \ge V_+ + V_-$.

Proof. For fixed ϵ, tilt the law by $d\mathbf{P}_{\epsilon,s} = M_\epsilon(s)^{-1}e^{\epsilon s Z}\,d\mathbf{P}$. Exponential Markov gives

$$\mathbf{P}_{\epsilon,s}\{\epsilon Z < t\} \le A_{\epsilon,-}, \qquad \mathbf{P}_{\epsilon,s}\{\epsilon Z > x_i\} \le A^+_{\epsilon,i}.$$

The second inequality is replaced by the deterministic zero bound when $\epsilon Z \le x_i$ almost surely. Thus the tilted mass of $t \le \epsilon Z \le x_i$ is at least $g_{\epsilon,i}$. The same summation-by-parts argument as in lemma 4.25 bounds the integral of the decreasing weight $e^{-s\epsilon Z}$ by the bracket in (4.40).1. Since

$$\Pr\{\epsilon Z \ge t\} = M_\epsilon(s)\mathbf{E}_{\epsilon,s}[e^{-s\epsilon Z}\mathbf{1}_{\{\epsilon Z \ge t\}}],$$

multiplication by $M_\epsilon(s)$ proves the displayed inequality. The positive and negative tail events are disjoint for $t>0$, so their lower bounds add. ☐

Proposition 4.41 (Low-rank B/C tails from exact determinant MGFs)**.** *For the rows and onsets in the following table,*

$$Q_3^G(n) \ge 0 \qquad \text{for every odd } n \ge N_G.$$

G	B_7	B_9	B_{10}	B_{11}	B_{12}	B_{13}
N_G	63	63	63	75	81	83

G	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
N_G	63	63	63	63	63	63	63	63

G	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}
N_G	63	71	75	63	65	73	63	63

Proof. The manifested verifier source contains the exact rational parameter row for every displayed onset. The 384-bit directed-rounding C++/MPFR replay evaluates the exact Toeplitz–Hankel defining-trace MGFs, the square-trace Chernoff bounds, and the stable push moments, and checks the sufficient inequality of lemma 4.8. It checks all 22 rows with no failure; the minimum final natural-log margin is $0.0743728374600429420826635512 2 > 0$ at C_{15}. It also checks $2k+2m \le 2b-2$, $2b/c < 1$, and $a/c < 1$ in every row. Thus the criterion holds at the listed onset and its monotonicity clause gives every later odd exponent. The accepted log and current C++ source are authenticated by the replay ledger. ☐

Proposition 4.42 (Finite Chain bridges to the low-rank exact-MGF onsets)**.** *The Chain differences are nonnegative in the following ranges:*

G	$\{m : D_G(2m+1) \ge 0\}$
B_{11}	$31 \le m \le 35$
B_{12}	$31 \le m \le 38$
B_{13}	$31 \le m \le 39$
C_{13}	$31 \le m \le 33$
C_{14}	$31 \le m \le 35$
C_{17}	$31 \le m \le 34.$

(4.42.1)

Consequently, if $Q_3^G(63) \ge 0$ in one of these six rows, then $Q_3^G(n) \ge 0$ for every odd $63 \le n < N_G$, where N_G is the onset in proposition 4.41.

Proof. Expanding each Chain difference in $m_j^G = m_j^{\mathrm{st}} + \delta_j^G$ gives an exact linear–quadratic polynomial in the corrections. The exact-integer replay consumes the authenticated consecutive exact-correction logs and uses $-m_j^{\mathrm{st}} \le \delta_j^G \le 0$ only beyond the supplied prefix, as given by the Pieri omitted-summand formula and moment nonnegativity. It checks all 34 displayed differences. Its smallest exact integer lower bound is positive and occurs at $C_{13}, m = 31$. Hence lemma 2.11 carries the value at odd exponent 63 to each exact-MGF onset. $\square$

Proposition 4.43 (Residual B/C tails from layered MGFs and one-sided moments)**.** *For every B_b with $14 \le b \le 61$ and every C_b with $20 \le b \le 61$, one has*

$$Q_3^G(n) \ge 0$$

for every odd $n \ge 2b + 29$.

Proof. The following table gives four exact rational parameter sets. In each row put

$$c = rb, \quad q = \alpha_q\sqrt{b}, \quad a = \alpha_a\sqrt{b}, \quad t = \sqrt{2c + 2d + q}, \quad s = t + l,$$

$$u_i = u_0 + i\Delta \quad (0 \le i \le 7), \qquad \lambda_- = l, \qquad \lambda_i = f_+ u_i,$$

$$v_q = f_q(q-1)/2, \qquad v_a = f_a(a-1)/2.$$

G	r	α_q	α_a	d	l	u_0	Δ	f_+	f_q	f_a	(k, m)
B_{14}–B_{15}	$\frac{16}{5}$	$\frac{529}{100}$	$\frac{213}{50}$	$\frac{143}{100}$	$\frac{109}{100}$	$\frac{13}{10}$	$\frac{31}{1000}$	1	1	$\frac{509}{500}$	$(5, b-6)$
B_{16}	$\frac{200001}{10^5}$	$\frac{45546}{10^4}$	$\frac{4284}{10^3}$	$\frac{1277}{10^3}$	$\frac{856}{10^3}$	$\frac{1847}{10^3}$	$\frac{34}{10^3}$	$\frac{567}{10^3}$	1	$\frac{101331}{10^5}$	$(5, 10)$
B_{17}	$\frac{20446}{10^4}$	$\frac{44739}{10^4}$	$\frac{42902}{10^4}$	$\frac{1341}{10^3}$	$\frac{972}{10^3}$	$\frac{15366}{10^4}$	$\frac{332}{10^4}$	$\frac{7291}{10^4}$	1	$\frac{101331}{10^5}$	$(5, 10)$
B_{18}–B_{61}	$\frac{20674}{10^4}$	$\frac{441721}{10^5}$	$\frac{429465}{10^5}$	$\frac{13501}{10^4}$	$\frac{103311}{10^5}$	$\frac{139822}{10^5}$	$\frac{31205}{10^6}$	1	1	$\frac{101331}{10^5}$	$(5, 8)$
C_{20}–C_{61}	$\frac{226172}{10^5}$	$\frac{460576}{10^5}$	$\frac{438971}{10^5}$	$\frac{138552}{10^5}$	$\frac{105713}{10^5}$	$\frac{135104}{10^5}$	$\frac{30177}{10^6}$	1	$\frac{75841}{10^5}$	$\frac{79538}{10^5}$	$(5, 8)$

For B_{14}, B_{15}, let $V_{b,+}, V_{b,-}$ be the exact-MGF lower bounds in lemma 4.40, using lemma 4.34, $L = 8$, $\lambda_- = l$, and $x_i = s + u_i$, $\lambda_i = u_i$, and put $L_b = V_{b,+} + V_{b,-}$. In the other four rows, let V_b be the lower bound in lemma 4.25 with the displayed parameters and put $L_b = 2V_b$. For $G = B_b, C_b$, respectively, write U_G for the corresponding function in lemma 4.26. Define

$$P_b^* = c^2(1 - d^{-2})\big(L_b - U_G(q, v_q)\big), \qquad N_b^* = \frac{8e^{-2}}{v_a^2} U_G(a, v_a),$$

and, using the stable push moments of lemma 4.10,

$$H_{k,m}(a) = \sum_{j=0}^{2m} (-1)^j \binom{2m}{j} a^{-j} K_{2k+j}.$$

The 384-bit directed-rounding MPFR replay

`certificates/post_m29/ post_m29_bc_layered_mgf_mpfr_certificate.log`

checks every intermediate domain and sign condition and, at $n_b = 2b + 29$, the final comparison

$$P_b^* \ge N_b^* \left(\frac{2b}{c}\right)^{n_b} + \frac{a^{n_b - 2k}}{4^m c^{n_b}} H_{k,m}(a). \tag{4.43.1}$$

For B_{14}, B_{15} it also evaluates

$$I_\nu(2v) = \sum_{r=0}^{180} \frac{v^{2r+\nu}}{r!(r+\nu)!} + R_{\nu,v}, \qquad 0 < R_{\nu,v} \le \frac{v^{362+\nu}}{181!(181+\nu)!} \frac{1}{1 - v^2/(181(181+\nu))},$$

after checking that the final ratio is below one for every argument and $0 \le \nu \le 29$. It evaluates both exact Toeplitz–Hankel determinants by interval Cholesky decomposition and checks every pivot strictly positive. It constructs the integers K_j exactly in GMP from their generating function, converts them to outward-rounded MPFR intervals, and uses outward rounding for every subsequent arithmetic operation. The combined replay checks 333 rows: 48 type-B, 42 type-C, and the 243 type-D rows used later in proposition 4.49. It exits with status 0 and reports `failures=0`. Its B/C lower margins include

condition	minimum directed lower bound
$c-2b,\ 2b-a,\ a-1,\ d-1$	0.00016
$1-\mathfrak{a}_d$	0.24176531795234819...
exact Cholesky pivot	0.85461901487040600...
g_0	0.0028157129657487152...
$\min_i(g_{i+1}-g_i)$	0.008012263371171611...
$\log P_b^* - \log(\text{right side of } (4.43).1)$	1.358823017167476129...

The two exact-determinant rows have final natural-log margins

$$6.045003937030598629\ldots \quad \text{and} \quad 7.590922674135796189\ldots.$$

Among the B/C rows the final minimum is attained at B_{16}. The combined-log minimum is the positive D_{220} margin quoted in proposition 4.49. The log has SHA-256

`3e4900e5e2de2ec1a7fb04830ac59dea`
`7369f8eebb1932ec08ac9ef73ba11654`,

and the C++ source has SHA-256

`71cc2293c4ad5ba50d216c8f939cd1ec`
`604ce7b86d6a19e9be1fcb27adc421fe`.

By lemmas 4.25 and 4.40, as applicable, the defining-trace event in lemma 4.12 has probability at least L_b. Equations (4.26).1–(4.26).2 bound the positive square-trace loss by $U_G(q, v_q)$. Therefore (4.12).1 gives $P_G(c) \ge P_b^*$. Equations (4.12).2–(4.12).3 give $N_G(a) \le N_b^*$. The degree conditions for lemma 4.10 are $2k+2m = 2b-2$ for B_{14}, B_{15}, $2k+2m = 30 = 2(16)-2$ for B_{16}, B_{17}, and $2k+2m = 26 \le 2b-2$ in the uniform rows. Thus (4.43).1 and lemma 4.8 prove the claim at n_b. Both ratios $2b/c$ and a/c are strictly below one, so the monotonicity clause of that lemma proves every later odd exponent. □

Proposition 4.44 (Post-$m=29$ first-hit lower bound)**.** *For the remaining post-$m=29$ finite residue rows*

$$B_{11}, \ldots, B_{275}, B_{277}, \qquad C_{13}, \ldots, C_{275}, C_{277}, \qquad D_{25}, \ldots, D_{275}, D_{277}, D_{279},$$

one has

$$D_G\big(N_{\text{hit}}^{(29)}(G) - 2\big) \ge 0$$

where

$$N_{\text{hit}}^{(29)}(G) = \max(63, s(G)+2).$$

Consequently $Q_3^G(N_{\text{hit}}^{(29)}(G)) \ge 0$ for each of these rows.

Proof. The 783 rows split into five arithmetic buckets:

bucket	rows	minimum positive margin row
BC exact $m=30$ window	14	B_{11}
BC stable-tail $m=30$	26	B_{19}
BC stable-tail sloped	490	B_{32}
D exact two-sided interval	28	D_{25}
D A-free frontier	225	D_{53}

For a row whose first hit is $n = 2m+3$, let

$$\Delta_{\text{st}}(m) = D_{\text{st}}(2m+1)$$

be the stable Chain difference, and write the finite-row moment correction as $m_j^G = m_j^{\text{st}} + \delta_j^G$. Expanding definition 2.10 in the variables δ_j^G gives

$$D_G(2m+1) = \Delta_{\text{st}}(m) + \sum_j L_{m,j}\delta_j^G + \sum_{i\le j} Q_{m,i,j}\delta_i^G\delta_j^G$$

with integer coefficients $L_{m,j}$ and $Q_{m,i,j}$ determined from lemma 2.3. The B/C checker computes these coefficients exactly and proves a positive integer lower bound in each of its 530 rows. Those rows use the proved nonpositive boundary correction, so the positive linear tail is compared with the negative quadratic terms. The accepted replay is restricted to B/C and does not parse a determinant-only type-D row. It reports

rows checked	530,
stable moments generated	$m_0, \ldots, m_{557}$,
scope	one Chain step at $N^{(29)}_{\mathrm{hit}}(G)$,
exit status	0.

Its three minimum margins have the following decimal sizes:

bucket	minimum row	$\log_{10}$(minimum margin)
BC exact $m = 30$ window	B_{11}	85.79
BC stable-tail $m = 30$	B_{19}	85.79
BC stable-tail sloped	B_{32}	93.85

The replay log has SHA-256

```
21642c87b90affa8e618bb9f97bf6ad3
46289380832690546488c6dcc7a7300e
```
.

For $D_{25}, \ldots, D_{52}$, the exact two-sided interval replay of proposition 4.51 checks every Chain step from the stable edge through its direct-tail onset, hence in particular the displayed first-hit step. Its 707 exact checks have minimum lower endpoint $469598628053992849245107 2 > 0$. For every displayed D_b with $b \geq 53$, the short-frontier replay of proposition 4.49 checks the required moment ceiling $N_D(b) + 2 \leq 2b$ and every Chain step from $m = 30$ to the direct-tail onset using lemma 4.46.3. Its worst positive logarithmic lower bound is $167.515421966088354\ldots$ at $D_{53}, m = 50$. The two status-0 logs have SHA-256 hashes

```
e47196c8097fb2a087b4673d47a5e34c
aae1cd1a316058759e17b13708e2d888
```
,

```
876d3f752d03b8f35183f0b28af7c15e
9e28d104396fc65c215e5d854d8ae31f
```
.

Every one of the five buckets therefore has a strictly positive lower margin. Thus $D_G(N^{(29)}_{\mathrm{hit}}(G) - 2) \geq 0$ in every displayed row. The preceding odd exponent is either covered by the stable edge or by the row-gated bridge, so lemma 2.11 gives $Q^G_3(N^{(29)}_{\mathrm{hit}}(G)) \geq 0$ in every displayed row. The B/C first-hit replay supplies only this one Chain step and is not used as an all-later tail. The all-later claims are proved separately below by the interval, layered-MGF, tilted-MGF, and correction-bridge propositions. $\square$

Lemma 4.45 (Exact finite-D moment decomposition)**.** *For $D_b = \mathrm{SO}(2b)$ write*

$$e_2^j = \sum_{\lambda \vdash 2j} c_j(\lambda) s_\lambda,$$

where the nonnegative integers $c_j(\lambda)$ are given by the iterated vertical-two-strip Pieri rule. Put

$$A_j(b) = \sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b}} c_j(2\nu_1, 2\nu_2, \ldots)$$

and

$$B_j(b) = \begin{cases} 0, & j < b, \\ \displaystyle\sum_{\substack{\nu \vdash j-b \\ \ell(\nu) \leq 2b}} c_j(1 + 2\nu_1, \ldots, 1 + 2\nu_{\ell(\nu)}, 1^{2b-\ell(\nu)}), & j \geq b. \end{cases}$$

If $s_j = m_j^{\mathrm{st}}$, then the exact correction is

$$m_j^{D_b} - s_j = B_j(b) - A_j(b).$$

In particular, the determinant-isotypic sum $B_j(b)$ alone is not the finite-rank correction once $A_j(b)$ is nonzero.

Proof. The adjoint character of $\mathrm{SO}(2b)$ is e_2 in the defining eigenvalues. Normalized Haar measure on $\mathrm{SO}(2b)$ is obtained from normalized Haar measure on $O(2b)$ by multiplication by $1 + \det$. Expand e_2^j by Pieri's rule. Rains' orthogonal Schur-integral criterion [5, §3] says that the untwisted $O(2b)$ integral retains exactly the Schur summands with even row lengths and at most $2b$ rows. The stable sum retains all even-row summands, so the omitted part is exactly $A_j(b)$.

For the determinant-twisted integral, use $\det s_\lambda = s_{\lambda+(1^{2b})}$ on $O(2b)$. The same criterion is nonzero precisely when λ has exactly $2b$ rows, all odd. Such a partition has the unique displayed form with $\nu \vdash j-b$; none exists for $j<b$. Its total contribution is $B_j(b)$. Adding the untwisted and determinant-twisted integrals gives $m_j^{D_b} = s_j - A_j(b) + B_j(b)$. □

Lemma 4.46 (Type-D determinant component and the A-free envelope)**.** *Let $b \geq 4$, and use the notation of lemma 4.45. Put*

$$p_j(b) = s_j - A_j(b).$$

Then

$$p_j(b) - B_j(b) = \int_{\mathrm{Sp}(2b-2)} \chi_{\omega_2}(V)^j \, dV, \tag{4.46.1}$$

where ω_2 is the second fundamental representation of $\mathrm{Sp}(2b-2)$. Consequently, for every $j \geq 0$,

$$0 \leq B_j(b) \leq p_j(b) \leq s_j. \tag{4.46.2}$$

Moreover $A_j(b) = 0$ for $0 \leq j \leq 2b$. Hence throughout this window the actual type-D correction satisfies

$$0 \leq m_j^{D_b} - s_j = B_j(b) \leq s_j. \tag{4.46.3}$$

Proof. Normalized Haar measure on $O(2b)$ is the half-sum of normalized Haar measure on its determinant components. Since

$$p_j(b) = \int_{O(2b)} e_2(U)^j \, dU, \qquad B_j(b) = \int_{O(2b)} \det(U) e_2(U)^j \, dU,$$

the normalized $O^-(2b)$ component moment is $p_j(b) - B_j(b)$.

Rains [6, remark following Theorem 1.5, equation (36)] identifies the nonfixed eigenvalue distribution of $O^-(2b)$ with that of $\mathrm{Sp}(2b-2)$. Restoring the fixed eigenvalues $+1, -1$, let V denote the defining $2b-2$ dimensional symplectic representation. On eigenvalue multisets,

$$e_2(V \oplus \mathbf{1} \oplus (-\mathbf{1})) = e_2(V) - 1.$$

The symplectic decomposition $\Lambda^2 V \cong \mathbf{1} \oplus V_{\omega_2}$ therefore turns the last character into χ_{ω_2}. This proves (4.46).1. Its right side is the multiplicity of the trivial representation in $V_{\omega_2}^{\otimes j}$, so it is a nonnegative integer. Since $A_j(b), B_j(b) \geq 0$, this proves (4.46).2.

Finally, a partition ν with more than $2b$ nonzero parts has $|\nu| \geq 2b+1$. The defining sum for $A_j(b)$ is therefore empty when $j \leq 2b$. Substitution in lemma 4.45 proves (4.46).3. □

Lemma 4.47 (Two-sided type-D correction box)**.** *Use the notation of lemma 4.45 and put $\delta_j(b) = m_j^{D_b} - s_j$ and $U_j = s_j$. Then*

$$\delta_j(b) = 0 \quad (0 \leq j < b), \tag{4.47.1}$$

$$0 \leq \delta_j(b) \leq U_j \quad (b \leq j \leq 2b), \tag{4.47.2}$$

and

$$-s_j \leq \delta_j(b) \leq U_j \quad (j > 2b). \tag{4.47.3}$$

Proof. For $j < b$, both sums $A_j(b), B_j(b)$ vanish. For $j \leq 2b$, the sum $A_j(b)$ vanishes, so lemma 4.46 gives $\delta_j(b) = B_j(b) \geq 0$. For all j, the same lemma gives $0 \leq B_j(b) \leq s_j$ and the definition gives $0 \leq A_j(b) \leq s_j$; hence $\delta_j(b) \geq -s_j$. The same inequalities give $\delta_j(b) = B_j(b) - A_j(b) \leq B_j(b) \leq s_j = U_j$ for every parity. Together with the preceding vanishing and nonnegativity statements, this proves all three displays. □

Proposition 4.48 (Unconditional A-free type-D row-gated bridge)**.** *For every $b \geq 53$, the type-D_b rows in the finite bridge through $m = 29$ satisfy*

$$D_{D_b}(2m+1) \geq 0$$

at every row-gated Chain step.

Proof. At a row-gated step $m \leq 29$, the two diagonals in the exact moment formula use moment indices only through $2m+5 \leq 63$. Since $63 \leq 2b$ for $b \geq 53$, lemma 4.47 supplies

$$\delta_j^{D_b} = 0 \quad (j < b), \qquad 0 \leq \delta_j^{D_b} \leq s_j \quad (b \leq j \leq 63).$$

The exact GMP/OpenMP verifier expands each Chain difference as in (4.15).9 and minimizes every linear and quadratic monomial over these intervals. For $53 \leq b \leq 63$ it checks the 36 row-gated pairs

$$\left\lfloor \frac{b-4}{2} \right\rfloor \leq m \leq 29.$$

Every lower bound is positive; the least is

$$1295951275047754385311107869606245254765288973412864475150697991241728 > 0$$

at $D_{53}, m = 24$. For $b \geq 64$ the row-gated range through $m = 29$ is empty. The verifier source and deterministic machine-C output have SHA-256 hashes

`1e574e0c53e624df30e9c69bc24280b`
`8e6c56a099923428f0a27036b74a7454b`,

`a84df851316bf376f00933ff958b7ad0d`
`ac444acfa8f45d05d4221f17ddadd12`.

Thus every nonvacuous row in the proposition has a source-generated exact integer certificate. □

Proposition 4.49 (Unconditional short-onset type-D tails)**.** *For every* $53 \leq b \leq 295$ *and every odd integer*

$$n \geq N_{\mathrm{hit}}^{(29)}(D_b),$$

one has

$$Q_3^{D_b}(n) \geq 0.$$

Proof. For each rank, let $N_D(b)$ be the odd onset in

`certificates/post_m29/ post_m29_bc_layered_mgf_mpfr_certificate.log`.

The directed-rounding verifier uses

$$c_b = \frac{20674}{10000}b, \quad q_b = \frac{441721}{100000}\sqrt{b}, \quad a_b = \frac{429465}{100000}\sqrt{b}, \quad d = \frac{13501}{10000},$$
$$l = \frac{103311}{100000}, \quad u_i = \frac{139822}{100000} + i\frac{31205}{1000000} \quad (0 \leq i < 8), \quad k = 5, \quad m = 8.$$

The lower and upper defining-trace tilt fractions and the positive square-trace tilt fraction are 1; the negative square-trace fraction is 101331/100000. The full defining trace of $\mathrm{SO}(2b)$ is centered, and lemmas 4.24 and 4.25 give the checked two-sided defining-trace lower bound. The type-D clause (4.26).3 gives both checked square-trace upper bounds. Since $2k + 2m = 26 \leq b - 3$, the type-D clause of lemma 4.10 supplies the exact polynomial $H_{5,8}^{D_b}(a_b)$.

For each b, the verifier starts at $N_{\mathrm{hit}}^{(29)}(D_b)$ and increases the odd exponent until the strict comparison

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2b}{c_b}\right)^{N_D(b)} + \frac{a_b^{N_D(b)-10}}{4^8 c_b^{N_D(b)}} H_{5,8}^{D_b}(a_b) \tag{4.49.1}$$

holds. For odd b, $C_{D_b} = b-2$, so the first ratio in lemma 4.8 is no larger than the conservative $2b/c_b$ used in (4.49).1. All geometry, trace-norm, determinant, nested-mass, polynomial, and final comparisons are outward-rounded at 384-bit MPFR precision. The machine-C run checks all 243 ranks, reports

`SUMMARY rows_checked=333 B_rows_checked=48 C_rows_checked=42 D_rows_checked=243 failures=0`,

and has minimum final natural-log margin

$$0.0078804615513483267599166693134 > 0$$

at D_{220}. Its log and C++ source have respective SHA-256 hashes

`3e4900e5e2de2ec1a7fb04830ac59dea`
`7369f8eebb1932ec08ac9ef73ba11654`,

`71cc2293c4ad5ba50d216c8f939cd1ec`
`604ce7b86d6a19e9be1fcb27adc421fe`.

The verifier also checks $N_D(b) + 2 \leq 2b$ in every row, with minimum slack 1. Thus lemma 4.8 proves the assertion for every odd $n \geq N_D(b)$.

It remains to reach these onsets. The exact GMP/OpenMP replay

`certificates/post_m29/ post_m29_d53_295_short_frontier_gmp_machine_c_certificate.log`

reads each onset exactly once, rejects an even or missing onset, and independently checks $N_D(b) + 2 \leq 2b$. It expands every Chain difference in the stable moments and the correction variables and uses lemma 4.56 and corollary 4.57 together with lemma 4.46.3. It verifies every active step $m = 30, \ldots, 163$ for all $D_{53}, \ldots, D_{295}$, reports `failures=0`, and records minimum A-free slack 1. The worst positive frontier is $m = 50$, odd target 103, frontier D_{53}, with logarithmic lower bound

$$167.515421966088354.$$

The log and source hashes are

$$\begin{matrix}\texttt{876d3f752d03b8f35183f0b28af7c15e}\\\texttt{9e28d104396fc65c215e5d854d8ae31f}\end{matrix},\quad \begin{matrix}\texttt{444cf4dbe59d5000a3315f31054ef07e}\\\texttt{c1cf710edf91a7c685f973644d65ad74}\end{matrix}.$$

The stable edge and proposition 4.48 give $Q_3^{D_b}(61) > 0$. The replayed Chain differences and lemma 2.11 reach $N_D(b)$, after which the monotone direct tail just proved supplies every later odd exponent. This proves the proposition. □

Proposition 4.50 (Unconditional D_{19} closure)**.** *For every odd integer $n \geq s(D_{19}) + 2 = 17$, one has*

$$Q_3^{D_{19}}(n) \geq 0.$$

Proof. By lemma 4.45, the status-0 machine-C C++/GMP logs give the adjoint moments through degree 34; expanding $(X-Y)^2(X+Y)^q$ determines exactly the push moments through degree 32. With $(k, m) = (6, 10)$, the directed 384-bit MPFR verifier applies lemmas 4.8, 4.35, 4.39 and 4.40 and proves the monotone tail from odd $n = 63$, with strict log margin $6.7370668175795237151\ldots$. For the bridge, lemma 4.46 makes the determinant rows full corrections through $j = 38$; the source replay supplies the full $B_{39} - A_{39}$ value. With the remaining boxes of lemma 4.47, the exact-GMP verifier checks all 24 steps $m = 7, \ldots, 30$, with minimum lower bound $36778114931645730 > 0$. The identity ledger and `d19_exact_mgf_machine_c.sha256` authenticate the two status-0 outputs and every exact input. Finally, lemmas 2.11 and 4.3 supplies the prefix through 15 and propagates these steps to 63. □

Proposition 4.51 (Unconditional D_{25}–D_{52} closure)**.** *For every $25 \leq b \leq 52$ and every odd integer $n \geq s(D_b)+2$, one has*

$$Q_3^{D_b}(n) \geq 0.$$

Proof. First consider the monotone direct tails. Use the rational parameters and tilt fractions in the proof of proposition 4.49; only the MGF supplier and the following polynomial pairs (k_b, m_b) change:

$$(5,6),\ (4,7),\ (5,7),\ (5,7) \quad \text{at } b = 25, 26, 27, 28, \qquad (k_b, m_b) = (5,8) \quad (29 \leq b \leq 52). \tag{4.51.1}$$

Thus $2k_b + 2m_b \leq b - 3$ in every row, and lemma 4.10 supplies the exact polynomial $H^{D_b}_{k_b,m_b}(a_b)$.

For $25 \leq b \leq 28$, the defining-trace lower bound is obtained from lemmas 4.35 and 4.40; the trace law is symmetric because $-I \in \mathrm{SO}(2b)$. For $29 \leq b \leq 52$, it is obtained from lemmas 4.24 and 4.25. In every row, lemma 4.39 supplies both square-trace upper bounds. The directed 384-bit MPFR verifier evaluates each Bessel series through term 180, adds the positive geometric remainder bound displayed in the proof of proposition 4.43, and uses interval Cholesky decomposition for every exact determinant. It verifies all geometry, pivot, Fredholm, nested-mass, polynomial, and final comparison inequalities.

Let T_b be the resulting odd onset. The exact list is recorded row by row in the machine-C log cited below. At T_b the verifier proves the strict inequality

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2b}{c_b}\right)^{T_b} + \frac{a_b^{T_b-2k_b}}{4^{m_b}c_b^{T_b}} H^{D_b}_{k_b,m_b}(a_b). \tag{4.51.3}$$

The conservative factor $2b/c_b$ dominates the actual endpoint factor also when b is odd. Hence lemma 4.8 proves every odd exponent at least T_b. The machine-C log

`certificates/post_m29/ post_m29_d25_52_exact_mgf_mpfr_machine_c_certificate.log`

checks 28 rows with no failures. Its minimum final natural-log margin is

$$0.013751090874926124112662444772 > 0$$

at D_{39}. The exact defining- and square-determinant pivot minima are, respectively, $0.8522815995\ldots$ and $0.8481405784\ldots$. The log and source hashes are respectively

$$\begin{matrix}\texttt{f87ab4bce0b189878ef4c3528ce34687}\\\texttt{1721bff05cb00ccb553a6c4e84e1f049}\end{matrix},\quad \begin{matrix}\texttt{71cc2293c4ad5ba50d216c8f939cd1ec}\\\texttt{604ce7b86d6a19e9be1fcb27adc421fe}\end{matrix}. \tag{4.51.4}$$

It remains to connect the stable edge to T_b. Expand each Chain difference in the correction variables as in lemma 4.56. The exact C++/GMP/OpenMP verifier substitutes the boxes of lemma 4.47; for every bilinear term it takes the minimum over the four interval endpoints, and for a diagonal term it also accounts for an interval containing zero. It checks every Chain index

$$\left\lfloor \frac{b-4}{2} \right\rfloor \leq m \leq \frac{T_b - 3}{2} \tag{4.51.5}$$

for every $25 \leq b \leq 52$. These are exactly the steps whose target exponents run from $s(D_b)+2$ through T_b. The replay performs 707 exact integer checks and reports no failure. Its smallest exact lower bound is

$$4695986280539928492451072 > 0$$

at $D_{25}, m = 10$. The machine-C log

`certificates/post_m29/ post_m29_d25_52_correction_interval_gmp_machine_c_certificate.log`

and its C++ source have respective SHA-256 hashes

$$\begin{matrix}\texttt{e47196c8097fb2a087b4673d47a5e34c} \\ \texttt{aae1cd1a316058759e17b13708e2d888}\end{matrix}, \quad \begin{matrix}\texttt{9a6cf4da5b866d3b8fc2ce6d1d1a442f} \\ \texttt{d3b419702fda2296da83d2fb788379e4}\end{matrix}. \tag{4.51.6}$$

The stable edge, these positive Chain differences, and lemma 2.11 reach T_b; the direct tail then supplies every later odd exponent. ☐

Definition 4.52 (Legacy finite type-D envelope)**.** For $4 \leq b \leq 52$, let T_b be the first odd exponent of the monotone pushforward tail in the accepted row certificate cited in proposition 4.18. Let DEnv denote the following finite statement: for $4 \leq b \leq 52$ and every $b \leq j \leq T_b + 2$, the quantities of lemma 4.45 satisfy

$$0 \leq B_j(b) - A_j(b) \leq s_j.$$

It contains only finitely many integer inequalities and no asymptotic assertion.

Proposition 4.53 (The legacy envelope is false)**.** *The statement* DEnv *of definition 4.52 is false. Specifically, at* $b = 4$ *and* $j = 18$*,*

$$s_{18} = 658906772228668, \qquad A_{18}(4) = 369382858189470,$$
$$B_{18}(4) = 288601773705246, \qquad B_{18}(4) - A_{18}(4) = -80781084484224 < 0.$$

Proof. The exact C++/GMP Pieri replay on machine C reports

$$m_{18}^{D_4} = 578125687744444 = s_{18} - 369382858189470 + 288601773705246.$$

The archived log

`certificates/post_m29/ d4_j18_denv_counterexample_gmp_machine_c.log`

has SHA-256

$$\begin{matrix}\texttt{463e3c9632bb92936620fa65c03d3ff5} \\ \texttt{abdb86940624f1621ffa976156d7d02f}\end{matrix},$$

and the C++ source has SHA-256

$$\begin{matrix}\texttt{bc22efcafafe2d40b5f5430cb8f9cb04} \\ \texttt{6bc6b9b188da684e61dc6fdfaa27077e}\end{matrix}.$$

All displayed identities are exact integer identities. The negative correction contradicts the lower inequality in DEnv. ☐

Proposition 4.54 (Exact D_4–D_{24} row-gated bridge)**.** *For every* $4 \leq b \leq 24$*, the type-*D_b *row satisfies*

$$D_{D_b}(2m+1) \geq 0$$

at every row-gated Chain step $0 \leq m \leq 29$*.*

Proof. The exact signed adjoint-moment sources are generated from the two Pieri sums in lemma 4.45. They supply the following full moment prefixes:

b	4	5	6, 7	8, . . . , 17	18	19	20, . . . , 24
exact through m_j	59	44	40	38	38	34	30.

(4.54.1)

At D_4, five distinct deterministic-primality-checked 63-bit moduli have product greater than 28^{59}. The Chinese remainder reconstruction is therefore the unique integer in the character bound $0 \leq m_j^{D_4} \leq 28^j$ for every $0 \leq j \leq 59$. It agrees with two independent archived reconstruction summaries through m_{59} and with the full Pieri source at all 37 overlapping indices $m_4, \ldots, m_{40}$.

For every index after the prefix in (4.54).1, the verifier uses the proved intervals of lemma 4.47. It expands each Chain difference exactly as an integer polynomial in the corrections; a bilinear term is minimized over all four interval endpoints, and a diagonal term also tests zero when its interval contains zero. The fixed-mode GMP/OpenMP replay

`certificates/post_m29/ post_m29_d4_24_exact_prefix_interval_gmp_certificate.log` checks all 530 row-gated steps and reports `failures=0`. Its smallest exact lower endpoint is $34 > 0$, at $D_5, m = 0$. The separate D4 CRT replay reports five primes, 60 reconstructed moments, 37 full-source cross-checks, and no failure. Every signed source log records exactly one zero status, machine-C host `nb1cb2f`, and source SHA-256

`bc22efcafafe2d40b5f5430cb8f9cb04`
`6bc6b9b188da684e61dc6fdfaa27077e`.

Before a correction is used, the C++ loader verifies both $O_j^{\mathrm{even}} + O_j^{\det} = \delta_j$ and $s_j + \delta_j = m_j$. The parent accepted-artifact manifest authenticates both replay outputs and all source logs. Thus every exact or interval-substituted integer lower endpoint used in the stated rows is positive. ☐

Proposition 4.55 (Exact D_{10}–D_{24} terminal tails)**.** *For every $10 \le b \le 24$ and every odd integer $n \ge 63$,*

$$Q_3^{D_b}(n) \ge 0.$$

Proof. First consider D_{10}. Put $d_{10} = \dim D_{10}$ and use the following rational cap and squared-Chebyshev data:

b	d_b	C_{D_b}	R_b	κ_b	M_b	ℓ_b	S_b	a_b	J_b
10	190	10	107	18	12400	9	4650	14	38.

(4.55.1)

Here $\sum_{\alpha>0} \alpha(\theta)^2 = (2b-2)|\theta|^2$, which gives the displayed values of κ_b. More explicitly, if $p_{2j} = \sum_i \theta_i^{2j}$, expansion over the positive roots $e_i \pm e_j$ gives

$$\sum_{\alpha>0} \alpha(\theta)^4 = 6p_2^2 + (2b-8)p_4 \le \kappa_b p_2^2$$

and

$$\sum_{\alpha>0} \alpha(\theta)^6 = 30p_2p_4 + (2b-32)p_6 \le \kappa_b p_2^3.$$

For the second inequality, after putting $y_i = \theta_i^2/p_2$, the difference between the right and left sides, divided by p_2^3, is

$$\sum_i y_i(1-y_i)\big((2b-2) + (2b-32)y_i\big) \ge 0$$

for $b = 10$. On the cap $Q(\theta) = \kappa_b|\theta|^2 \le R_b$, these formulas also give $|\alpha(\theta)|^2 \le 2R_b/\kappa_b$.

The cap width is $w = 6/5$ and the positive threshold is $c_{10} = d_{10} - R_{10} - w = 409/5$. Weyl integration in exponential coordinates [1, Chapter IV, (1.11)], the Macdonald–Mehta normalization [12, Theorem 3.1(i)], and

$$\prod_{\alpha>0} \left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2}\right)^2 \ge M_b^{-1}$$

give an exact rational lower bound for $P_{D_b}(c_b)$. The checker proves the sine-product inequality by the rational exponential majorant

$$\frac{R_b}{12} + \frac{R_b^2}{1440\kappa_b} + \frac{R_b^3}{90720\kappa_b^2(1 - 2R_b/(39\kappa_b))},$$

using 90 Taylor terms. All remaining cap, quartic-tail, monotonicity, Weyl-order, and factorial comparisons are exact integer or rational inequalities.

For completeness, the displayed sine exponent follows directly from the Euler product for $\sin z/z$. The quadratic, quartic, and sextic terms are bounded by

$$\frac{R_b}{12}, \qquad \frac{R_b^2}{1440\kappa_b}, \qquad \frac{R_b^3}{90720\kappa_b^2},$$

respectively. Since $\pi^2 > 39/4$, each remaining Euler-product term is at most $2R_b/(39\kappa_b)$ times its predecessor; the geometric sum is the third term in the stated majorant. Also $R_b < \kappa_b\pi^2$, so the cap is contained in one eigenangle chart. Writing $(d_i) = (2, 4, \ldots, 2b-2, b)$, its resulting probability lower bound is

$$\frac{1}{|W(D_b)|M_b} \frac{\prod_i d_i!}{(2\pi)^{b/2}\Gamma(d_b/2+1)} \left(\frac{R_b}{2\kappa_b}\right)^{d_b/2}. \tag{4.55.2}$$

The verifier replaces the powers of π by an explicit rational lower bound. Namely, $\pi < 22/7$ gives

$$\frac{1}{(2\pi)^5\Gamma(96)} > \frac{1}{(44/7)^5\, 95!} \tag{4.55.2a}$$

for D_{10}. The checker uses this bound only in the direction decreasing the cap probability.

Let Y be an independent adjoint character. By lemma 4.45, its first four moments are stable; since $\mathbf{E}Y = 0$, $\mathbf{E}Y^2 = \mathbf{E}Y^3 = 1$, and $\mathbf{E}Y^4 = 6$, for $r = 1/2$, $s = 15/4$ one has

$$\mathbf{E}(Y-r)^2(Y-s)^2 = 6 - 2(r+s) + r^2 + s^2 + 4rs + r^2s^2 =: V_4.$$

If $y < -w$ and $x \geq d_b - R_b \geq r$, then

$$\frac{x-y}{x+w} \leq \frac{(r-y)(s-y)}{(r+w)(s+w)}.$$

Thus the lost $(X-Y)^2$-weight is at most $V_4(x+w)^2/((r+w)^2(s+w)^2)$. The verifier checks that the resulting quadratic function of x is positive and increasing from $x = d_b - R_b$ onward. This proves that the cap contribution is a lower bound for $P_{D_b}(c_b)$, rather than merely for unweighted Haar mass.

For a real variable u, define

$$R_b^{\mathrm{Ch}}(u) = T_{\ell_b}\left(1 - \frac{4d_b}{S_b}u + \frac{2}{S_b}u^2\right)^2.$$

For $u \leq -a_b$, the quantity $u(u-2d_b)$ is at least $a_b(a_b+2d_b)$; since $T_{\ell_b}(z)^2$ increases for $z \geq 1$, this polynomial is at least $R_b^{\mathrm{Ch}}(-a_b) > 0$. Hence

$$N_{D_b}(a_b) \leq \frac{\int R_b^{\mathrm{Ch}}(u)\, d\psi_{D_b}(u)}{R_b^{\mathrm{Ch}}(-a_b)}. \tag{4.55.3}$$

The numerator has degree $4\ell_b$ and is an exact linear combination of $K_0^{D_b}, \ldots, K_{4\ell_b}^{D_b}$. By lemma 4.9, these are reconstructed from the full adjoint moments through $m_{J_b}^{D_b}$. The machine-C C++/GMP sources provide the gap-free range $m_{10}, \ldots, m_{38}$ and verify, row by row,

$$m_j^{D_b} = s_j + O_j^{\mathrm{even}}(D_b) + O_j^{\det}(D_b). \tag{4.55.4}$$

The exact-rational C++/GMP replay then proves

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2C_{D_b}}{c_b}\right)^{63} + 2\left(\frac{a_b}{c_b}\right)^{63}. \tag{4.55.5}$$

Its output is

`certificates/post_m29/ post_m29_d10_exact_tail_gmp_certificate.log`.

For the exact-MGF rows $11 \leq b \leq 24$, use the common parameters

$$\begin{gathered} c_b = \frac{20674}{10000}b, \quad a_b = \frac{429465}{100000}\sqrt{b}, \quad d_0 = \frac{13501}{10000}, \quad l = \frac{103311}{100000}, \\ u_i = \frac{139822}{100000} + i\frac{31205}{1000000} \quad (0 \leq i < 8), \qquad f_- = f_i = f_{\mathrm{sq}} = 1, \qquad f_{\mathrm{neg}} = \frac{101331}{100000}, \end{gathered} \tag{4.55.6}$$

Here the four f's are respectively the lower and upper defining-trace tilt fractions and the positive and negative square-trace tilt fractions. Use the following rank-dependent polynomial and square-trace parameters:

b	(k_b, m_b)	$q_b/\sqrt{b}$
11	$(3,7)$	24/5
$12, 13, 14$	$(4,6)$	19/4
$15, 16$	$(4,6)$	9/2
17	$(5,5)$	441721/100000
18	$(4,7)$	441721/100000
19	$(5,6)$	441721/100000
$20, 21$	$(5,7)$	441721/100000
22	$(5,8)$	441721/100000
23	$(5,8)$	17/4
24	$(5,9)$	441721/100000.

(4.55.7)

The source loader first requires one status-zero machine-C provenance line, the pinned C++/GMP generator hash, and both exact identities in (4.55).4 for every consumed moment. It then uses lemma 4.9 to reconstruct the finite-rank $K_j^{D_b}$ needed by $H_{k_b,m_b}^{D_b}(a_b)$.

Put $n_{11} = 75$ and $n_b = 63$ for $12 \leq b \leq 24$. The exact even-orthogonal defining-trace determinant of lemma 4.35, the exact square-trace determinant of lemma 4.39, and the layered tilt of lemma 4.40 supply the positive trace tail. The same square-trace determinant and (4.12).3 supply $N_{D_b}(a_b)$. With directed 384-bit MPFR intervals, the verifier proves at $n = n_b$

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2b}{c_b}\right)^{n_b} + \frac{a_b^{n_b-2k_b}}{4^{m_b}c_b^{n_b}} H^{D_b}_{k_b,m_b}(a_b). \tag{4.55.8}$$

The factor $2b/c_b$ conservatively dominates $2C_{D_b}/c_b$. For D_{11}, the machine-C replay reports defining- and square-MGF pivot lower bounds $0.8426844977700063107\ldots$ and $0.9814906225170877019\ldots$, and final natural-log margin $0.09113279249064900255 9\ldots > 0$. Its output is

`certificates/post_m29/ post_m29_d11_exact_mgf_mpfr_machine_c_certificate.log`.

The separate exact C++/GMP/OpenMP bridge consumes the gap-free machine-C corrections through $m_{46}^{D_{11}}$, checks all 34 Chain steps from the stable edge through odd exponent 75, and reports no failure. Its minimum exact lower endpoint is $11976538 > 0$. Its output is

`certificates/post_m29/ post_m29_d11_onset75_bridge_gmp_machine_c_certificate.log`.

By the stable edge and lemma 2.11, the bridge proves the required odd exponents through 75; the exact-MGF comparison and lemma 4.8 prove every odd exponent from 75 onward.

For $D_{12}, \ldots, D_{24}$, the replay checks all 13 rows with no failure. Its minimum defining- and square-MGF Cholesky pivots are respectively $0.8380062150\ldots$ and $0.8377469953\ldots$, and its minimum final natural-log margin is

$$0.09200855750026939896 85\ldots > 0$$

at D_{24}. The output is

`certificates/post_m29/ post_m29_d12_24_exact_mgf_mpfr_certificate.log`.

Finally, lemma 4.5 propagates (4.55).5 for D_{10}, while lemma 4.8 propagates (4.55).8 for $D_{11}, \ldots, D_{24}$. ☐

Lemma 4.56 (D stable-envelope frontier monotonicity)**.** *Fix $m \geq 31$. Let $s_j = m_j^{\mathrm{st}}$ be the stable D-row moments, let $\Delta_{\mathrm{st}}(m)$ be the stable Chain difference, and let $L_{m,j}$ and $Q_{m,i,j}$ be the integer coefficients in the expansion*

$$D_G(2m+1) = \Delta_{\mathrm{st}}(m) + \sum_j L_{m,j}\delta_j^G + \sum_{i \leq j} Q_{m,i,j}\delta_i^G\delta_j^G,$$

where $m_j^G = s_j + \delta_j^G$. For an integer rank boundary r, define

$$F_m(r) = \Delta_{\mathrm{st}}(m) + \sum_{\substack{j \geq r \\ L_{m,j} < 0}} L_{m,j}s_j + \sum_{\substack{i \leq j,\ i,j \geq r \\ Q_{m,i,j} < 0}} Q_{m,i,j}s_is_j.$$

Then $F_m(r') \geq F_m(r)$ whenever $r' \geq r$. Moreover, if a type D_b row has determinant corrections satisfying

$$\delta_j^{D_b} = 0 \quad (j < b), \qquad 0 \leq \delta_j^{D_b} \leq s_j \quad (j \geq b),$$

then

$$D_{D_b}(2m+1) \geq F_m(b).$$

Proof. The difference $F_m(r') - F_m(r)$ is the negative of the sum of those displayed linear and quadratic summands whose indices lie in the set removed when the boundary is raised from r to r'. Every removed displayed summand is nonpositive because $s_j \geq 0$ and only negative coefficients are displayed; hence the difference is nonnegative.

For the row bound, expand $D_{D_b}(2m+1)$ in the corrections $\delta_j^{D_b}$. Terms with positive coefficient are nonnegative and may be discarded in a lower bound. If a negative linear coefficient occurs, then $0 \leq \delta_j^{D_b} \leq s_j$ gives $L_{m,j}\delta_j^{D_b} \geq L_{m,j}s_j$. If a negative quadratic coefficient occurs, then the product vanishes unless both indices are at least b, and $0 \leq \delta_i^{D_b}\delta_j^{D_b} \leq s_is_j$ gives $Q_{m,i,j}\delta_i^{D_b}\delta_j^{D_b} \geq Q_{m,i,j}s_is_j$. Adding these lower bounds gives $D_{D_b}(2m+1) \geq F_m(b)$. ☐

Corollary 4.57 (D frontier bridge reduction)**.** *Let $\mathcal{I}$ be a finite interval of type D ranks for which the stable-envelope hypotheses of lemma 4.56 hold. For a fixed m, let r_m be the smallest rank in $\mathcal{I}$ whose finite bridge still includes the Chain step $2m+1$. If $F_m(r_m) > 0$, then*

$$D_{D_b}(2m+1) > 0$$

for every active rank $b \in \mathcal{I}$ at that Chain step.

Proof. Every active rank satisfies $b \geq r_m$. By lemma 4.56, $D_{D_b}(2m+1) \geq F_m(b) \geq F_m(r_m) > 0$. ☐

Proposition 4.58 (Post-$m = 29$ D-interval finite tails)**.** *For $G = D_b$ with $98 \le b \le 275$ or $b = 277, 279$, one has*

$$Q_3^G(n) \ge 0$$

for every odd $n \ge N_{\mathrm{hit}}^{(29)}(G)$.

Proof. Every displayed rank lies in $53 \le b \le 295$, so this is the corresponding subrange of proposition 4.49. The former $R = 38b$ onset and long frontier artifacts are retained only as superseded provenance; no theorem now consumes their false stable-envelope hypothesis. ☐

Proposition 4.59 (Post-$m = 29$ B/C interval direct tails)**.** *For $G = B_b$ with $14 \le b \le 217$ or $G = C_b$ with $20 \le b \le 217$, let $N_{\mathrm{dir}}^{BC}(G)$ be the odd onset recorded for the row in*

`certificates/post_m29/bc_interval_formula_mpfr_certificate.log`.

Then

$$Q_3^G(n) \ge 0$$

for every odd $n \ge N_{\mathrm{dir}}^{BC}(G)$.

Proof. The accepted C++/Optimus verifier

`certificates/post_m29/bc_cheb80_negative_tail_gmp_certificate.log`

first supplies the common negative-tail input used by the interval direct-tail template. It uses exact GMP rational arithmetic with the degree-8 Chebyshev majorant, scale 12000, and cutoff 80. The log exits with status 0, reports `failures=0`, and has SHA-256

`80fce36378ec2192dca2c0b616c083b4`
`6168cf5134b66e73775156b995ea303b`.

Its worst row is B_{217}, with exact upper mass log $-15.1757329227092583 < -15$.

The directed-rounding MPFR verifier

`certificates/post_m29/bc_interval_formula_mpfr_certificate.log`

then checks the hypothesis of lemma 4.5 at the logged row onset for each of the 402 rows. It exits with status 0, reports `failures=0`, and has SHA-256

`53530cfe3d4b1947a77bf45c5c5aaa56`
`a052ec1cde93c0704b68e004f22f2cb5`.

The largest logged onset is $N_{\mathrm{dir}}^{BC}(B_{217}) = 30897$. The smallest directed lower log margin at an onset, attained at B_{20}, is

$$0.0061301728071582916875791 > 0.$$

By the monotonicity clause of lemma 4.5, the same inequality holds for every later odd exponent in the row. ☐

Corollary 4.60 (Uniform residual B/C direct-tail onset bound)**.** *For every row $G = B_b$ with $14 \le b \le 217$ or $G = C_b$ with $20 \le b \le 217$,*

$$N_{\mathrm{dir}}^{BC}(G) \le 30897.$$

Proof. This is the largest-onset clause in the directed-rounding MPFR verifier used in proposition 4.59: the largest logged onset is $N_{\mathrm{dir}}^{BC}(B_{217}) = 30897$. ☐

Corollary 4.61 (Initial residual B direct-tail onset values)**.** *The directed-rounding MPFR verifier used in proposition 4.59 records*

$$N_{\mathrm{dir}}^{BC}(B_{14}) = 925, \qquad N_{\mathrm{dir}}^{BC}(B_{15}) = 415, \qquad N_{\mathrm{dir}}^{BC}(B_{16}) = 329, \qquad N_{\mathrm{dir}}^{BC}(B_{17}) = 303.$$

Proof. These are the row entries for B_{14}, B_{15}, B_{16}, and B_{17} in the status-0 directed-rounding MPFR verifier log

`certificates/post_m29/bc_interval_formula_mpfr_certificate.log`,

whose SHA-256 hash and failure-free summary are cited in proposition 4.59. ☐

Proposition 4.62 (B/C Pieri correction sign)**.** *For $G = B_b$ or $G = C_b$, put $s_j = m_j^{\mathrm{st}}$ and $m_j^G = s_j + \delta_j^G$ in the first nonstable moment range $j \geq 2b+2$. Then*

$$\delta_j^G \leq 0.$$

More precisely, writing

$$e_2^j = \sum_\lambda c_j(\lambda) s_\lambda, \qquad h_2^j = \sum_\lambda d_j(\lambda) s_\lambda,$$

the coefficients $c_j(\lambda)$ and $d_j(\lambda)$ are nonnegative integers, and

$$\delta_j^{B_b} = - \sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b+1}} c_j(2\nu_1, 2\nu_2, \ldots),$$

while

$$\delta_j^{C_b} = - \sum_{\substack{\mu \vdash j \\ \ell(\mu) > b}} d_j(\mu_1, \mu_1, \ldots, \mu_{\ell(\mu)}, \mu_{\ell(\mu)}).$$

Proof. For B_b, the adjoint character is $e_2 = s_{(1,1)}$ in the defining eigenvalues. Pieri's rule expands e_2^j by adding vertical two-strips, so each $c_j(\lambda)$ is a nonnegative integer. Rains' orthogonal Schur-integral criterion [5, §3] says that the stable orthogonal integral keeps exactly the Schur summands whose row lengths are even, while the finite $\mathrm{SO}(2b+1)$ integral keeps exactly those among them with at most $2b+1$ nonzero rows. The omitted stable summands are precisely the partitions $(2\nu_1, 2\nu_2, \ldots)$ with $\ell(\nu) > 2b+1$, giving the displayed negative sum.

For C_b, the adjoint character is $h_2 = s_{(2)}$. Pieri's rule expands h_2^j by adding horizontal two-strips, so each $d_j(\lambda)$ is nonnegative. Rains' symplectic Schur-integral criterion [5, §3] says that the stable symplectic integral keeps exactly the Schur summands whose row lengths occur with even multiplicity, while the finite $\mathrm{Sp}(2b)$ integral keeps exactly those among them with at most $2b$ nonzero rows. The omitted summands are exactly $(\mu_1, \mu_1, \ldots, \mu_{\ell(\mu)}, \mu_{\ell(\mu)})$ with $\ell(\mu) > b$, giving the second negative sum. ☐

Proposition 4.63 (Local B/C half-stable bridge)**.** *Fix one of the rows*

$$B_{14}, \ldots, B_{123}, \qquad C_{20}, \ldots, C_{217},$$

and an integer m for which

$$N_{\mathrm{hit}}^{(29)}(G) \leq 2m+1 < N_{\mathrm{dir}}^{BC}(G).$$

Put $s_j = m_j^{\mathrm{st}}$ and $m_j^G = s_j + \delta_j^G$. If

$$-\frac{1}{2} s_j \leq \delta_j^G \leq 0 \qquad (2\,\mathrm{rank}(G) + 2 \leq j \leq 2m+5),$$

then $D_G(2m+1) \geq 0$.

Proof. Substitution in lemma 2.3 and definition 2.10 gives

$$D_G(2m+1) = D_{\mathrm{st}}(2m+1) + \sum_j L_{m,j} \delta_j^G + \sum_{i,j} Q_{m,i,j} \delta_i^G \delta_j^G.$$

The two moment-formula diagonals use indices only through $2m+5$. Under the displayed box constraints, discarding sign-favourable terms gives the exact sign-truncated lower bound

$$\mathcal{L}_m = D_{\mathrm{st}}(2m+1) - \frac{1}{2} \sum_{L_{m,j} > 0} L_{m,j} s_j + \frac{1}{4} \sum_{Q_{m,i,j} < 0} Q_{m,i,j} s_i s_j.$$

The replay uses the still smaller bound $\widetilde{\mathcal{L}}_m$ obtained from $[x+y]_+ \leq [x]_+ + [y]_+$ on the two linear diagonals; thus $D_G(2m+1) \geq \mathcal{L}_m \geq \widetilde{\mathcal{L}}_m$.

Here is the analytic reduction used by the source. Put $S = n+2$ and

$$A_n(k) = \binom{n}{k-2} + \binom{n}{k} - 2\binom{n}{k-1} = \frac{n!}{k!(S-k)!}\big((2k-S)^2 - S\big), \qquad q_n = \sum_{k=0}^{S} A_n(k) s_k s_{S-k}.$$

If $M(x) = \sum_{j \geq 0} s_j x^j / j!$, the stable recurrence gives

$$M(x) = \frac{\exp(-x/2 + x^2/4)}{\sqrt{1-x}}, \qquad \sum_{n \geq 0} q_n \frac{x^n}{n!} = 2\big(M'' M - (M')^2\big) = e^{-x + x^2/2}\left((1-x)^{-3} + (1-x)^{-1}\right).$$

Thus all stable diagonal values are generated by a fixed finite recurrence; equivalently, for $Q(x)=\sum q_n x^n/n!$,

$$(-x^3+3x^2-4x+2)Q'=(-x^4+4x^3-6x^2+4x+2)Q,\qquad q_0=q_1=2.$$

Moreover, $A_n(k)<0$ only when $|2k-S|<\sqrt{S}$. Consequently the negative parts require only $O(\sqrt{n})$ central terms. The positive active parts are recovered from q_n by subtracting that central sum and the excluded boundary sum $k<30$. This fixed frontier enlarges every active row's correction box and therefore produces a no-larger lower bound. To reduce the remaining central arithmetic, put $t_j=s_j/j!$. The source checks exactly that

$$j s_{j+1}s_{j-1}\ge (j+1)s_j^2\qquad (9\le j\le 30898),$$

which is precisely the log-convexity of (t_j). The central-band argument uses the first half of this range; the full audit also supplies the power-loss reduction below. Each product comparison is certified from rigorous 192-bit dyadic enclosures: truncating a positive integer gives an exact lower prefix and adding one unit gives an exact upper prefix. Lower left products exceed upper right products in all 30,890 cases, so no full-width fallback multiplication is needed. If k_0 is the left endpoint of $\{k:(2k-S)^2<S\}$, log-convexity gives $t_k t_{S-k}\le t_{k_0}t_{S-k_0}$ throughout that band. Hence its negative part is bounded below by the negative of

$$\frac{\binom{S}{k_0}s_{k_0}s_{S-k_0}}{S(S-1)}\sum_{\substack{|d|<\sqrt{S}\\ d\equiv S\pmod 2}}(S-d^2).$$

The final weight sum is evaluated by the closed formulas for consecutive even or odd squares, and the rational result is rounded upward to an integer penalty. A startup audit compares this bound with the original exact central sum through $n=128$. This same endpoint majorant is used wherever the negative central sum enters the quadratic penalty. Finally, since

$$L_{m,k}s_k=2A_{2m+3}(k)s_k s_{2m+5-k}-8A_{2m+1}(k)s_k s_{2m+3-k},$$

the inequality $[x+y]_+\le[x]_++[y]_+$ supplies the asserted separable linear majorant. This replaces the former full diagonal traversal by one large-integer endpoint product per index, plus fixed-width boundary work. The GMP/OpenMP source `character_ring_iter/post_m29_bc_interval_bridge_frontier_gmp.cpp` checks $\widetilde{\mathcal{L}}_m>0$ for every $m=31,\dots,15447$. The accepted transcript `certificates/post_m29/bc_lambda_half_bridge_gmp_certificate.log` reports no failures and has SHA-256

`ac2e805c5fb7f039ed03849ad651528e`
`d0dd957abea4f9643c416d995c25cf41`.

This range contains every bridge index in the displayed rows, and proves the claim. □

Lemma 4.64 (Exact finite type-B/C adjoint moments)**.** *Write*

$$e_2^j=\sum_{\lambda\vdash 2j}c_j(\lambda)s_\lambda,\qquad c_j(\lambda)\in\mathbb{Z}_{\ge 0},$$

and let s_j be the stable classical adjoint moment. Then

$$m_j^{B_b}=s_j-\sum_{\substack{\nu\vdash j\\ \ell(\nu)>2b+1}}c_j(2\nu)$$

and

$$m_j^{C_b}=s_j-\sum_{\substack{\nu\vdash j\\ \ell(\nu)>b}}c_j((\nu_1,\nu_1,\nu_2,\nu_2,\dots)')\,.$$

An empty partition sum is zero.

Proof. For type B, the adjoint representation is $\bigwedge^2 V$, so the Schur expansion of e_2^j is the character decomposition of $(\bigwedge^2 V)^{\otimes j}$; iterated vertical-two-strip Pieri gives $c_j(\lambda)\in\mathbb{Z}_{\ge 0}$. The orthogonal Schur integral criterion in Rains [5, Section 3, paragraph preceding Lemma 3.1] says that the $\mathrm{O}(N)$ integral of s_λ is one precisely for even λ of length at most N, and is zero otherwise. For odd $N=2b+1$, the central element $-I$ identifies the $\mathrm{O}(N)$ and $\mathrm{SO}(N)$ integrals of every power of $\bigwedge^2 V$. Removing the even partitions excluded by the length wall gives the first formula.

For type C, the adjoint representation is $\mathrm{Sym}^2 V$. The standard involution ω sends h_2 to e_2 and s_λ to $s_{\lambda'}$, so the coefficient of s_λ in h_2^j is $c_j(\lambda')$. Rains' symplectic criterion in the same paragraph admits exactly the partitions of length at most $2b$ in which every row length has even multiplicity. Such a partition is uniquely

$\lambda = (\nu_1, \nu_1, \nu_2, \nu_2, \ldots)$, and its length constraint is $\ell(\nu) \le b$. Removing the excluded partitions gives the second formula. $\square$

Lemma 4.65 (Bounded-Littlewood determinant specification for the active B/C moments)**.** *Work degree by degree in the completed ring of symmetric functions over $\mathbb{Q}$, put $e_r = h_r = 0$ for $r < 0$, and define*

$$f_r^e = \sum_{k\in\mathbb{Z}} e_k e_{k+r}, \qquad f_r^h = \sum_{k\in\mathbb{Z}} h_k h_{k+r}, \qquad e = \sum_{k\ge 0} e_k, \qquad \bar{e} = \sum_{k\ge 0} (-1)^k e_k.$$

For $b \ge 1$, let

$$\mathcal{B}_b = \frac{1}{2}\left\{ e \det_{1\le u,v\le b}(f^e_{u-v} - f^e_{u+v-1}) + \bar{e} \det_{1\le u,v\le b}(f^e_{u-v} + f^e_{u+v-1}) \right\},$$
$$\mathcal{C}_b = \det_{1\le u,v\le b}(f^h_{u-v} - f^h_{u+v}).$$

Then, for every $j \ge 0$,

$$m_j^{B_b} = [m_{(2^j)}]\mathcal{B}_b, \qquad m_j^{C_b} = [m_{(2^j)}]\mathcal{C}_b.$$

Proof. Hall duality gives $\langle h_2^j, F\rangle = [m_{(2^j)}]F$. Transposing the admitted partitions in the type-B formula of lemma 4.64, and using the involution $\omega(e_2) = h_2$, gives

$$m_j^{B_b} = \left\langle h_2^j, \sum_{\substack{\lambda_1 \le 2b+1 \\ c(\lambda)=0}} s_\lambda \right\rangle,$$

where $c(\lambda)$ is the number of odd columns. Huh–Kim–Krattenthaler– Okada [10, Theorem 1.3, equations (1.9)–(1.10)], at $u = 0$, give the sum and difference of the sectors $c(\lambda) = 0$ and $c(\lambda) = 2b + 1$. Their half-sum is $\mathcal{B}_b$.

For type C, transposition gives

$$m_j^{C_b} = \left\langle e_2^j, \sum_{\substack{\lambda_1 \le 2b \\ r(\lambda)=0}} s_\lambda \right\rangle,$$

where $r(\lambda)$ is the number of odd rows. Equation (1.6) of the same source identifies the Schur sum with $\det(f^e_{u-v} - f^e_{u+v})$. Applying the Hall-isometric involution ω, which sends f_r^e to f_r^h, and then Hall duality proves the formula for $\mathcal{C}_b$. $\square$

Lemma 4.66 (Exact coefficient functional used by the determinant replay)**.** *Let F be homogeneous of symmetric-function degree $2j$. Define π_2 on power sums by*

$$\pi_2(p_1) = p, \qquad \pi_2(p_2) = q, \qquad \pi_2(p_r) = 0 \quad (r \ge 3).$$

Then

$$[m_{(2^j)}]F = j! \sum_{a=0}^{j} (2(j-a)-1)!! [p^{2(j-a)} q^a] \pi_2(F).$$

Moreover,

$$\sum_{r\ge 0} \pi_2(e_r) u^r = \exp(pu - qu^2/2), \qquad \sum_{r\ge 0} \pi_2(h_r) u^r = \exp(pu + qu^2/2),$$

so, with $e_{-1} = h_{-1} = 0$ and with e_r, h_r abbreviating their specialized images,

$$re_r = pe_{r-1} - qe_{r-2}, \qquad rh_r = ph_{r-1} + qh_{r-2}.$$

Proof. Expand F in the power-sum basis. A monomial containing p_r with $r \ge 3$ cannot contain the ordinary monomial $x_1^2 \cdots x_j^2$ and hence does not contribute to $[m_{(2^j)}]F$. In $p_1^{2(j-a)} p_2^a$, the a ordered p_2 factors choose distinct variables and the other ordered factors hit each remaining variable twice. The coefficient of $x_1^2 \cdots x_j^2$ is

$$\frac{j!}{(j-a)!} \frac{(2(j-a))!}{2^{j-a}} = j!(2(j-a)-1)!!.$$

This proves the coefficient formula. Applying π_2 to the standard power-sum generating series for elementary and complete symmetric functions gives the two exponentials; differentiation and coefficient comparison give the recurrences. $\square$

Proposition 4.67 (Active residual B/C correction-prefix contract)**.** *Put $q = b+1$, write $s_j = m_j^{\mathrm{st}}$, and write $m_j^G = s_j + \delta_j^G$. The following bounds hold:*

$$-\frac{1}{2}s_{2q+r} \le \delta_{2q+r}^{B_b} \le 0 \quad (14 \le b \le 123, 0 \le r \le 27), \tag{4.67.1}$$

and

$$-\frac{1}{2}s_{2q+r} \le \delta_{2q+r}^{C_b} \le 0 \quad (20 \le b \le 217, 0 \le r \le 27). \tag{4.67.2}$$

The common upper bound is proposition 4.62. For audit purposes the derivation archive gives an offset-by-offset proposition table; the submission proof instead uses the direct finite certificate contract printed below.

domain	*exact lower-bound supplier*
type B nonstable frontiers	*the $B_{14}, j = 37$ seed; exact hook-length bounds for 296 active H8–H27 cases; and a six-row determinant/CRT residual reconstruction for the other 41 cases through moment 67*
type C nonstable frontiers	*closed fixed-point-free and arithmetic formulas, checked in 60 and 179 cases respectively*
all caller rows	*exact stable-recurrence and power-loss arithmetic on 402 rows and 3,904,626 integer comparisons, followed by an independent domain audit of every offset $0 \le r \le 27$*

The additional offset-28 and offset-29 results in the source archive are not needed by the main-theorem propagation below.

Proof. The finite lower-bound claim is checked directly, without importing an archive proposition. The mathematical values computed by the finite programs are fixed, independently of their implementations, by lemmas 4.64 to 4.66. The source

`character_ring_iter/verify_active_bc_b_frontiers_bounded_littlewood_gmp.cpp`

reconstructs the required type-B moments from the displayed bounded-Littlewood determinant and coefficient recurrence modulo enough primes for unique integer recovery, subtracts them from the stable moments, and checks the half-stable inequality in every nonautomatic active frontier case. The determinant evaluations are flattened over the prime–interpolation-node grid, and the independent exact hook-length sums over paired-row partitions are reduced in a fixed order. It fails closed unless the ledger has exactly 337 cases and maximum moment 121. The $B_{14}, j = 37$ seed is checked separately by `post_m29_bc_b14_j37_badshape_gmp`. The type-C frontier programs evaluate with GMP the fixed-point-free and arithmetic specializations of the type-C partition sum displayed in lemma 4.64; the determinant identity gives an exact alternative specification of the same moments. The programs fail closed unless their combined scopes are 60 and 179 cases.

The stable recurrence and the remaining exact dominance comparisons are checked by `post_m29_bc_pieri_powerloss_gmp`: its accepted run covers all 402 caller rows, 3,904,626 integer comparisons, and moment indices through 30,899. Finally, `verify_parts_i_ii_contract.py` independently reconstructs the caller domains from the proposition statement and requires both complete offset sets $\{0, \dots, 27\}$, exactly 2,834 propagated odd targets, and no consumed offset beyond 27. Thus the direct determinant, formula, and recurrence suppliers exhaust (4.67).1–.2; no statement from the derivation archive is a premise. The Pieri omitted-summand formula in proposition 4.62 gives $\delta_j^G \le 0$ for both families. Combining the two sides proves the contract. □

Proposition 4.68 (Residual B/C post-$m = 29$ closure)**.** *For*

$$G = B_b \quad (14 \le b \le 123), \qquad \text{or} \qquad G = C_b \quad (20 \le b \le 217),$$

one has

$$Q_3^G(n) \ge 0$$

for every odd integer $n \ge N_{\mathrm{hit}}^{(29)}(G)$.

Proof. Write $m_j^G = s_j + \delta_j^G$, where s_j is the stable moment. The complete active correction import is proposition 4.67; in particular, its table replaces the former prose reference to a suite of unnamed contracts.

We record why this prefix is sufficient. The analytic suppliers used below start no later than the odd exponent

$$N_{\mathrm{tail}}(b) = 2b + 29.$$

To propagate to an odd target $t < N_{\mathrm{tail}}(b)$, the last Chain difference used is $D_G(t-2)$. By lemma 2.3 and definition 2.10, that difference involves moments only through m^G_{t+2}. Since $t \le 2b+27$, its largest possible correction index is

$$t+2 \le 2b+29 = 2q+27.$$

Thus the offsets $0 \le r \le 27$ in proposition 4.67 are exactly the correction prefix consumed by the main theorem. The offset-28 and offset-29 results and the longer conditional development routes retained in the source archive are not premises of this proposition.

Apply the local half-stable bridge proposition 4.63. To make its finite predicate explicit, write the exact moment-formula expansion as

$$D_G(2m+1) = D_{\mathrm{st}}(2m+1) + \sum_j L_{m,j}\delta^G_j + \sum_{i,j} Q_{m,i,j}\delta^G_i\delta^G_j,$$

where the integer coefficients are obtained by substituting $m^G_j = s_j + \delta^G_j$ into lemma 2.3 and definition 2.10. Under $-s_j/2 \le \delta^G_j \le 0$, sign truncation gives the exact lower bound $\mathcal{L}_m$ displayed in proposition 4.63. The separable linear majorant proved there then gives the certified bound

$$\widetilde{\mathcal{L}}_m \le \mathcal{L}_m \le D_G(2m+1).$$

Its generating-function recurrence, fixed rank-14 frontier, and audited log-convex central-band majorant reduce the large-integer evaluation to one endpoint product per index.
The exact GMP/OpenMP source

`character_ring_iter/post_m29_bc_interval_bridge_frontier_gmp.cpp`

checks $\widetilde{\mathcal{L}}_m > 0$ for $m = 31, \dots, 15447$, reports no failures, and has replay SHA-256

`ac2e805c5fb7f039ed03849ad651528e`
`d0dd957abea4f9643c416d995c25cf41`.

Together with the first-hit proposition proposition 4.44 and Chain propagation, this reaches the applicable analytic supplier. The layered-MGF proposition proposition 4.43 supplies $B_{14}, \dots, B_{61}$ and $C_{20}, \dots, C_{61}$; the tilted-MGF propositions propositions 4.29 and 4.33 and the direct interval proposition proposition 4.59 supply $B_{62}, \dots, B_{123}$ and $C_{62}, \dots, C_{217}$. For $B_{14}, \dots, B_{17}$ the layered-MGF onset already precedes $N^{(29)}_{\mathrm{hit}}(G)$. These ranges are disjoint and exhaust the displayed rows, proving every stated odd exponent. No inactive archival route is used. ☐

Theorem 4.69 (Classical B, C, D closure)**.** *For every compact connected simple group whose Lie algebra has type B_b, C_b, or D_b, and for every $n \ge 0$,*

$$Q^G_3(n) \ge 0.$$

Proof. Even n is lemma 2.9. For odd n, the moment formula and lemma 2.6 give the strict base value $Q^G_3(1) = 2m^G_3 > 0$ in each adjoint classical row. The stable edge lemma 4.3 and row-gated bridge proposition 4.16, followed by lemma 2.11, supply all pre-tail odd exponents. The high-rank rows are proposition 4.14, and the post-bridge finite rows listed in proposition 4.17 follow from the trace-tail criterion.

It remains only to partition the post-$m = 29$ rows. The low-rank rows are proposition 4.18; the type-D rows $D_{25}, \dots, D_{52}$ are proposition 4.51, the rows $D_{53}, \dots, D_{295}$ are proposition 4.49, while D_{297} is proposition 4.17 and the remaining higher type-D rows are proposition 4.14. The high-edge B/C rows are proposition 4.19. The residual ranges $B_{14}, \dots, B_{123}$ and $C_{20}, \dots, C_{217}$ are proposition 4.68. Finally, $B_{124}, \dots, B_{217}$ are proposition 4.21. These alternatives exhaust every non-A classical row and every odd exponent. ☐

5. Exceptional Lie Groups

The exceptional proof uses the exact Chain difference of definition 2.10. For every exceptional adjoint family, the moment formula and lemma 2.6 give $Q^G_3(1) = 2m^G_3 > 0$. Nonnegative Chain differences therefore propagate positivity by lemma 2.11.

Lemma 5.1 (Exceptional adjoint endpoint and Chain negative region)**.** *Let G be a compact exceptional simple group of rank r, and put $C_G = r$. Then*

$$\chi_{\mathrm{ad}}(g) \ge -C_G \qquad (g \in G).$$

For odd n, with the pushforward measure ψ_G of lemma 4.5,

$$D_G(n) = \int_{\mathbb{R}} u^n(u^2-4)\, d\psi_G(u).$$

Group	exact prefix	tail/bridge closure
G_2	Chain $m = 0, \dots, 97$	direct rectangular tail from odd $n \geq 21$
F_4	Chain $m = 0, \dots, 107$	direct rectangular tail from odd $n \geq 65$
E_6	Chain $m = 0, \dots, 37$	direct Chain tail from odd $n \geq 75$
E_7	Chain $m = 0, \dots, 32$	direct Chain tail from odd $n \geq 65$
E_8	Chain $m = 0, \dots, 32$	tail from odd $n \geq 133$ plus bridges to $n = 77$

TABLE 1. Exceptional exact prefixes and tail certificates.

Consequently, if $L(n)$ is any lower bound for a positive part of this integral, then

$$D_G(n) \geq L(n) - U_G(n), \qquad U_G(n) = 2\big((2C_G)^2 + 4\big)(2C_G)^n. \tag{5.1.1}$$

Proof. Garibaldi–Guralnick–Rains [13, Theorem 2.1] prove $\operatorname{Tr}\operatorname{Ad}(g) \geq -\operatorname{rank} G$ for every element of every compact Lie group. The displayed identity follows from definition 2.10 and lemma 4.5. For odd n, its integrand is negative only on

$$[-2C_G, -2] \cup [0, 2].$$

On this union its absolute value is at most $((2C_G)^2 + 4)(2C_G)^n$. Since $\psi_G(\mathbb{R}) = 2$ by lemma 4.5, integration gives (5.1).1. ☐

Lemma 5.2 (Normalized exceptional Weyl–rectangle coefficient)**.** *Normalize long roots to have squared length two. Let $Q^\vee \subset P^\vee$ be the coroot and coweight lattices, put*

$$z_R = \prod_{\alpha \in R_+} \frac{|\alpha|^2}{2}, \qquad \nu_R = \frac{|P^\vee/Q^\vee|^2}{\operatorname{covol}(Q^\vee)^2},$$

and let d_i be the invariant degrees. For the exceptional root systems,

$$\begin{array}{c|ccccc} R & G_2 & F_4 & E_6 & E_7 & E_8 \\ \hline z_R & 1/27 & 1/4096 & 1 & 1 & 1 \\ \operatorname{covol}(Q^\vee)^2 & 3 & 4 & 3 & 2 & 1 \\ |P^\vee/Q^\vee| & 1 & 1 & 3 & 2 & 1 \\ \nu_R & 1/3 & 1/4 & 3 & 2 & 1 \\ \min_{0 \neq \lambda \in P^\vee} |\lambda|^2 & 2 & 2 & 4/3 & 3/2 & 2 \end{array} \tag{5.2.1}$$

Let $a = d/2$, where $d = \dim G$, and let

$$A_R = \nu_R z_R^2 \frac{8ad^2(d/\kappa)^d \prod_i ((d_i - 1)!)^2}{(2\pi)^r}. \tag{5.2.2}$$

For a retained radial cell $[s_0, s_1] \times [t_0, t_1]$ and its transpose, suppose the local character bounds are g_R, h_R and the product of the certified Weyl-sine and optional radial exponential losses is $\eta_R(n) > 0$. Then their combined positive contribution to $D_G(n)$ is at least

$$\frac{A_R(2d)^n}{n^{d+2}} \frac{g_R(n)^2 \eta_R(n)}{a\Gamma(a)^2} \left(\int_{s_0}^{s_1} s^{a-1}\, ds\right) \left(\int_{t_0}^{t_1} t^{a-1}\, dt\right) h_R(n)^n \big((2dh_R(n))^2 - 4\big). \tag{5.2.3}$$

All exceptional caps used below lie in an injective exponential chart of the adjoint maximal torus.

Proof. For the adjoint global form the exponential kernel is $2\pi P^\vee$. Thus normalized torus Haar measure in Euclidean exponential coordinates is

$$\frac{dv}{(2\pi)^r \operatorname{covol}(P^\vee)}, \qquad \operatorname{covol}(P^\vee) = \frac{\operatorname{covol}(Q^\vee)}{|P^\vee/Q^\vee|}. \tag{5.2.4}$$

Apply the Weyl integration formula [1, Chapter IV, (1.11)] in both variables. If $\Delta_R(v) = \prod_{\alpha>0} \alpha(v)$ and Δ_{std} uses the squared-length-two normal to every reflecting hyperplane, then $\Delta_R^2 = z_R \Delta_{\mathrm{std}}^2$. The Macdonald–Mehta identity [12, §2.1 and Theorem 3.1(i)], scaled by $(d/\kappa)^{1/2}$, gives

$$\int e^{-\kappa|v|^2/(2d)} \Delta_R(v)^2\, dv = z_R (2\pi)^{r/2} (d/\kappa)^a \prod_i d_i!.$$

The square of this identity, the two factors in (5.2).4, and $|W| = \prod_i d_i$ give

$$\nu_R z_R^2 \frac{(d/\kappa)^d \prod_i ((d_i - 1)!)^2}{(2\pi)^r}.$$

In the scaled radial variable $s = n\kappa|v|^2/(2d)$, homogeneity of Δ_R^2 gives the radial density $s^{a-1}/\Gamma(a)$. The character-gap square contributes $(2d/n)^2 g_R^2$; adjoining the transposed cell contributes the factor two. Writing $8d^2$ as $(8ad^2)/a$ yields exactly (5.2).3. The retained cells have $s_0 \geq t_1$, so the cells and their transposes are disjoint up to null boundaries.

The Cartan and simple-coroot Gram determinants give the middle three rows of (5.2).1; the short-root products give the first row. For E_6, E_7, inversion of the Cartan matrix and exact enumeration of the simple-root pairings in $\{-1, 0, 1\}$ give the displayed coweight minima. This enumeration is global: if one pairing has absolute value at least two, Cauchy's inequality and $|\alpha_i|^2 = 2$ give $|\lambda|^2 \geq 2$. The same calculation gives the G_2, F_4 minima; for E_8, determinant one and the even integral Cartan form give minimum two. These determinant and minimum calculations are repeated in the exact C++ checkers.

Finally, a cap point satisfies $|v|^2 < \delta^2/2$. For G_2, F_4 one has $\delta \leq \pi$; for E_6, E_7, E_8 one has $\delta \leq 14/3$. The last row of (5.2).1 and the rational inequality $\pi > 333/106$ show that the diameter of every cap is strictly smaller than $2\pi \min_{0 \neq \lambda \in P^\vee} |\lambda|$. Hence no two cap points differ by a nonzero exponential-kernel vector. ☐

Lemma 5.3 (Rectangular Chain certificate template)**.** *Let G be exceptional, let $d = \dim G$, and write $D_G(n) = Q_3^G(n+2) - 4Q_3^G(n)$. Suppose a finite union of rational rectangles $R = [s_0, s_1] \times [t_0, t_1]$ in radial variables satisfies, for all odd $n \geq n_0$,*

$$\chi(y) - \chi(x) \geq \frac{2d}{n} g_R, \qquad \frac{\chi(x) + \chi(y)}{2d} \geq h_R(n),$$

with $g_R > 0$, $h_R(n) > 0$, and with every rectangle contained in the specified root-angle cap $|\alpha \cdot v| \leq \delta$ on which the stated scalar trigonometric minorants have been proved. Let $L(n) > 0$ be the resulting sum of integrated rectangular lower bounds, and suppose an explicit negative-region estimate $U(n) > 0$ satisfies, for every odd $n \geq n_0$,

$$D_G(n) \geq L(n) - U(n), \qquad L(n_0) \geq U(n_0), \qquad L(n+2)U(n) \geq L(n)U(n+2) \tag{5.3.1}$$

Then $D_G(n) \geq 0$ for every such n.

Proof. The two local bounds supply the squared gap and positive-power factors; the cap makes the stated scalar bounds uniform. The middle comparison gives $L(n_0)/U(n_0) \geq 1$, and the last makes this ratio nondecreasing along odd n. Hence $L(n) \geq U(n)$ and $D_G(n) \geq 0$. ☐

Proposition 5.4 (Direct rectangular tails for E_6 and E_7)**.** *The following exact rectangular certificates hold:*

G	n_0	*grid*	*retained cells*	$\log_{10}(L(n_0)/U(n_0))$	*odd-step lower bound*
E_6	75	1/10	23088	8.904	$> 10^{0.573}$
E_7	65	1/20	66925	> 0.42	$> 101/100$

Consequently $D_{E_6}(n) \geq 0$ for every odd $n \geq 75$, and $D_{E_7}(n) \geq 0$ for every odd $n \geq 65$.

Proof. Put $\delta = 14/3$. Exact reflection generation gives

$$(d, r, |R_+|, \kappa, q) = (78, 6, 36, 12, 16) \quad (E_6), \qquad (133, 7, 63, 18, 27) \quad (E_7),$$

where $Q = \kappa|u|^2$ and $\sum_{\alpha>0} (\alpha \cdot u)^4 = Q(u)^2/q$; the checker verifies both polynomial identities coefficientwise. It also proves the rational wide-cap cosine and Weyl-sine minorants by exact Bernstein coefficients; with Cauchy's radial sixth-power bounds these give exact $g_R(n), h_R(n)$ satisfying lemma 5.3 on every retained cell.

For E_6 use $U(n) = 296 \cdot 12^n$. For E_7 put

$$P(S) = \frac{T_{26}((2S - 273)/259)}{T_{26}(-301/259)}, \quad I = \int S^{14}(S^2 + 4)P(S)^2 \, d\psi_{E_7}(S), \quad U(n) = \frac{24}{25} 14^{n-14} I.$$

Exact Bernstein coefficients prove the associated pointwise majorant on $[-14, -2] \cup [0, 2]$ at $n = 65$; it propagates since $|S| \leq 14$, and I uses only $m_0, \ldots, m_{70}$. The OpenMP/GMP checker verifies the cell counts, the normalized Weyl factors 3, 2, the coweight minima, transposed-cell disjointness, all cleared monotonicity inequalities, and the stated base and odd-step bounds. Hence (5.3).1 holds for both tails. ☐

Proposition 5.5 (E_8 rectangular bridge and tail ledger)**.** *For E_8, the following exact rectangular certificates prove $D_{E_8}(n) \geq 0$ at the listed odd exponents:*

n	*cells*	$\log_{10}$ *ratio*	n	*cells*	$\log_{10}$ *ratio*
77	25493	.09	105	15875	.05
79	7518	.40	107	31163	.00
81	19706	.08	109	4762	.74
83	22609	.10	111	21133	.26
85	25688	.11	113	5667	1.50
87	28933	.27	115	24567	.39
89	32597	.30	117	71087	.02
91	30748	.04	119	3720	.39
93	34241	.00	121	18367	.01
95	3217	.38	123	1741	.88
97	8930	.13	125	517	.89
99	2595	1.26	127	621	.80
101	5459	.40	129	3153	.28
103	6185	.39	131	1074	1.39

The analytic tail from $n \geq 133$ uses 8706 *retained cells, base-ratio logarithm* 0.01*, and odd-step logarithm* 0.85*.*

Proof. The E_8 constants are $d = 248$, $|R_+| = 120$, rank 8, and $C_G = 8$. For the analytic tail the negative-region estimate is $U(n) = 2((2C_G)^2 + 4)(2C_G)^n = 520 \cdot 16^n$. The certificates use the quartic Weyl identity

$$\sum_{\alpha>0} (\alpha \cdot u)^4 = \frac{1}{50} \left(\sum_{\alpha>0} (\alpha \cdot u)^2 \right)^2 .$$

On each retained rational cell the checker proves the two local character bounds of lemma 5.3; all retained cells are inside the root-angle cap. The mesh widths are enlarged by the audited factors $5/4, 3/2, 2, 3$, or 4 according to the row. The same endpoint monotonicity inequalities apply to each enlarged rectangle, and exact retained-cell counts and transposed-cell disjointness ensure that only endpoint bounds and nonoverlapping cells enter the sum. It also verifies the factor-one normalized Weyl coefficient and transposed-cell disjointness required by lemma 5.2. For the tail certificate at $n = 133$, for instance, the cell formula is

$$g_R = s_0 - t_1 - \frac{248}{300n} s_1^2, \qquad h_R(n) = 1 - \frac{s_1 + t_1}{n} + \frac{1}{18} \frac{496}{50} \frac{s_0^2 + t_0^2}{n^2},$$

and the retained cells have

$$\min h_R(133) = \frac{62706243}{113209600}, \qquad \min(496^2 h_R(133)^2 - 4) > 0.$$

An exact source audit first rederives all 29 negative-bound rows from their moment sources and pointwise majorants and requires exact equality with the manifest-pinned table. The OpenMP replay then consumes it and checks all cell counts, the factor-one normalized E_8 Weyl coefficient of lemma 5.2, transposed-cell disjointness, and the base comparisons by 384-bit outward rounding of exact rational factors. For the tail propagation, on the fixed $n = 133$ tail cells one has $s_1 < 40$, $t_1 \leq 31$, and $s_1 + t_1 > 63$. With $A_R = s_1 + t_1$ and $B_R = (124/225)(s_0^2 + t_0^2)$, one has $133A_R > 8379 > 2824 > 2B_R$, so $h_R(n) = 1 - A_R/n + B_R/n^2$ increases thereafter. The gap, cap, sine-density, and direct factors also improve. Hence every cell has odd-step quotient at least $((248/8) \min_R h_R(133))^2 (133/135)^{250} > 1$, the logged exact value. Thus (5.3).1 holds throughout the tail, proving the listed differences and every odd $n \geq 133$. □

Proposition 5.6 (G_2 and F_4)**.** *For $G = G_2$ or $G = F_4$ and every $n \geq 0$,*

$$Q_3^G(n) \geq 0.$$

Proof. The base value $Q_3^G(1) > 0$ is lemmas 2.3 and 2.6. Put $Q(u) = \sum_{\alpha>0} (\alpha \cdot u)^2$. Exact expansion of the positive roots gives

G	d	r	$\lvert R_+\rvert$	$\kappa : Q = \kappa \lvert u\rvert^2$	$q : \sum_{\alpha>0} (\alpha \cdot u)^4 = Q(u)^2/q$	(d_i)
G_2	14	2	6	4	16/5	$(2, 6)$
F_4	52	4	24	9	54/5	$(2, 6, 8, 12)$.

The Macdonald–Mehta identity [12, Theorem 3.1(i)], whose root normals have squared length 2, gives the positive-saddle normalization

$$A_0 = \nu_G z_G^2 \frac{8ad^2(d/\kappa)^d \prod_i((d_i-1)!)^2}{(2\pi)^r}, \qquad a = d/2.$$

Here the actual short roots change the squared discriminant by $z_{G_2} = 1/27$ and $z_{F_4} = 1/4096$; these are respectively the products of $|\alpha|^2/2$ over the three and twelve positive short roots. The normalized-torus factors from lemma 5.2 are $\nu_{G_2} = 1/3$ and $\nu_{F_4} = 1/4$.

Apply lemma 5.3 with $C = 2, 4$ and with

G	n_0	$[s_0, s_1] \times [t_0, t_1]$	g	$h(n_0)$
G_2	21	$[19/5, 21/5] \times [7/5, 9/5]$	111/80	1661/2268
F_4	65	$[51/4, 53/4] \times [31/4, 33/4]$	3023/1296	44063/63180.

Writing q for the preceding quartic denominator, the exact local bounds are

$$g = s_0 - t_1 - \frac{ds_1^2}{6qn_0}, \qquad h(n) = 1 - \frac{s_1+t_1}{n} + \frac{2d(s_0^2+t_0^2)}{18qn^2}.$$

The normalized rectangular radial mass is bounded below by

$$g^2\, 3^{-\lceil s_1+t_1 \rceil} \frac{(s_1^a - s_0^a)(t_1^a - t_0^a)}{a^3((a-1)!)^2}.$$

Using $\pi < 355/113$, the Weyl-sine lower factors are 3^{-1} and 3^{-4}. The resulting exact base comparisons with $U(n) = 2((2C)^2+4)(2C)^n$ exceed 1 and 10^{13}, respectively. Moreover the odd-step quotients are bounded below by

$$\left(\frac{d}{C}h(n_0)\right)^2 \left(\frac{n_0}{n_0+2}\right)^{d+2} > 6, 10,$$

respectively. The exact checks also give $n_0(s_1+t_1) > 2d(s_0^2+t_0^2)/(9q)$, so $h(n)$ increases; the gap and cap improve with n. They also verify the coroot Gram and Cartan determinants, the coweight minima, and transposed-cell disjointness required by lemma 5.2. Thus the rectangular Chain tails begin at odd $n = 21, 65$. The exact prefixes reach odd $n = 197, 217$ and overlap both tails. Lemma 2.11 closes all odd exponents; even exponents are lemma 2.9. ☐

Proposition 5.7 (E_6 and E_7)**.** *For $G = E_6$ or $G = E_7$ and every $n \geq 0$,*

$$Q_3^G(n) \geq 0.$$

Proof. The base value $Q_3^G(1) > 0$ is lemmas 2.3 and 2.6. The exact prefixes verify the Chain differences through $m = 37$ for E_6 and $m = 32$ for E_7, reaching the odd exponents 75 and 65, respectively. The direct rectangular certificates of proposition 5.4 prove the corresponding Chain tails from those same odd exponents onward. The prefix endpoint and the tail start overlap exactly, so lemma 2.11 gives $Q_3^G(n) \geq 0$ for every odd n. Even n is lemma 2.9. ☐

Proposition 5.8 (E_8 finite prefix and bridge)**.** *For E_8, the finite prefix Chain differences*

$$D_{E_8}(n) \quad (n = 1, 3, \ldots, 65)$$

are positive. The finite bridge values

$$Q_3^{E_8}(n) \quad (n = 69, 71, 73, 75, 77)$$

are positive, and the Chain differences

$$D_{E_8}(n) \quad (n = 67, 69, 71, 73, 75, 77, 79, 81, 83, 85, 87, 89, 91, 93, 95)$$

are nonnegative.

Proof. The exact source-pairing replay applies lemmas 2.4 and 2.5 to regenerate the complete range $m_0, \ldots, m_{100}$ from the E_8 Cartan datum. It compares every regenerated value with the local ledger. As a logically separate control, it then compares the complete range termwise with the Bourjaily–Plesser–Vergu ancillary block; the maintained $m_{71}, \ldots, m_{100}$ transcription is also compared directly with the primary ancillary file. All comparisons have zero mismatches. For odd $n \le 98$, lemma 2.3 computes $Q_3^{E_8}(n)$ using only these moments. Exact integer evaluation gives

$$\begin{aligned} &\min D_{E_8}(1,3,\ldots,65) &&= 20,\\ &\log_{10}\min Q_3^{E_8}(69,71,73,75,77) &&\approx 100.94,\\ &\log_{10}\min D_{E_8}(67,69,\ldots,77) &&\approx 100.94,\\ &\log_{10}\min D_{E_8}(67,69,\ldots,95) &&\approx 100.94. \end{aligned}$$

The replay flags are

```
moment_log_0_100_arxiv_extension    OK,
prefix_chain_diff_1_65_positive     OK,
bridge_q3_69_77_positive            OK,
chain_diff_67_77_positive           OK,
extended_chain_diff_67_95_positive  OK.
```

□

Proposition 5.9 (E_8 tail)**.** *For E_8, the Chain difference is nonnegative for every odd $n \ge 77$.*

Proof. The direct rectangular certificates in proposition 5.5 prove the Chain tail for odd $n \ge 133$ and the successive bridge steps

$$n = 131, 129, 127, 125, 123, 121, 119, 117, 115, 113, 111, 109, 107, 105, 103, 101,$$

$$n = 99, 97, 95, 93, 91, 89, 87, 85, 83, 81, 79, 77.$$

Together with proposition 5.8, these bridge steps connect the exact prefix to the $n = 133$ tail. Therefore $D_{E_8}(n) \ge 0$ for every odd $n \ge 77$. □

Theorem 5.10 (Exceptional closure)**.** *For every $G \in \{G_2, F_4, E_6, E_7, E_8\}$ and every $n \ge 0$,*

$$Q_3^G(n) \ge 0.$$

Proof. Use proposition 5.6 and proposition 5.7 for G_2, F_4, E_6, E_7. For E_8, proposition 5.8 supplies the prefix Chain differences through $n = 65$, the bridge values at $69, 71, 73, 75, 77$, and the bridge Chain differences through 95; then proposition 5.9 supplies every later Chain difference. Applying lemma 2.11 from the strict base value $Q_3^{E_8}(1) > 0$ of lemmas 2.3 and 2.6 gives every odd exponent. Even exponents are covered by lemma 2.9. □

6. Assembly of the Classification

Proof of theorem 2.2. By lemma 2.8, $Q_3^G(n)$ depends only on the compact simple Lie algebra. In the irreducible-root-system classification, $B_1 = C_1 = A_1$ and $D_3 = A_3$ are already in the type-A branch, D_2 is not simple, and the remaining classical types have B_b, C_b with $b \ge 2$ and D_b with $b \ge 4$. The classification therefore leaves type A, the non-A classical types B, C, D, and the exceptional types G_2, F_4, E_6, E_7, E_8. Type A is theorem 3.16; types B, C, D are theorem 4.69; the exceptional types are theorem 5.10. These cases exhaust all compact connected simple Lie groups. □

7. Conclusion and limitations

The proof establishes one uniform sign inequality for the adjoint tensor category of every compact simple type. Its common structure is the passage from the moment identity in lemma 2.3 to a stable prefix, a finite-rank correction interval, and an analytic tail. The classification is needed because the mechanism controlling the middle interval differs between type A, the three other classical families, and the exceptional groups.

The theorem does not compare unequal characters and does not prove Ginibre's condition on the cone of all central positive-definite functions. It also does not assert that the exactly-two-minus argument extends directly to four or more minus factors. Part III addresses the latter hierarchy only on the closed multiplicative convex cone generated by χ_{ad}, using the present theorem as an input. These scope boundaries separate the mathematical conclusion from possible applications to larger character cones or to model-specific correlation inequalities.

Remark 7.1 (Final-facing dependency map)**.** The proof graph is the following; it deliberately omits superseded diagnostic routes retained in the expanded archive. All incoming nodes are proved in the manuscript or discharged by the authenticated exact replays.

closure	load-bearing incoming results
theorem 3.16	theorems 3.1, 3.5, 3.7 and 3.14, lemma 3.9, and corollary 3.15
theorem 4.69	lemma 4.3 and propositions 4.14, 4.16 to 4.19, 4.21, 4.49 and 4.67, proposition 4.63, proposition 4.68
theorem 5.10	propositions 5.6 to 5.9
theorem 2.2	lemma 2.8 and theorems 3.16, 4.69 and 5.10

Inside proposition 4.68, the only live residual route is the proved correction-prefix contract and local half-stable bridge, followed by the layered/tilted MGF or direct interval supplier. Local fixed-witness lemmas used to prove the boundary-through-twenty-seventh correction contract are represented in the accepted correction-prefix suite. The expanded archive supplies a line-by-line audit of those contracts. Global all-row fixed-witness/compression interfaces explicitly marked superseded have no incoming edge from this map.

Appendix A. Certificate and Replay Contract

Definition A.1 (Accepted certificate)**.** An accepted certificate has manifested source and output, recorded status zero, and either exact integer/rational arithmetic or an outward-rounded interval with positive lower endpoint. Diagnostic, interrupted, nonzero, and superseded artifacts are excluded. A certificate may verify a conditional arithmetic implication only when its hypotheses are stated in a numbered result and discharged at every main-theorem-reachable invocation; an undischarged implication is not accepted proof evidence. Every character-moment source passed to an active root-datum source-replay stage is regenerated during the clean-room replay and compared termwise with the consumed ledger. The finite row-gated bridge is likewise regenerated by the exact Pieri/CRT and type-D interval verifiers of propositions 4.15 and 4.48. The superseded historical bridge archive is provenance only. A content hash establishes integrity and provenance only; it never by itself decides a sign.

Proposition A.2 (Certificate coverage and replay contract)**.** *Every computer-assisted sign or threshold claim used in theorems 3.16, 4.69 and 5.10 is decided by an accepted certificate in the sense of definition A.1; no unenclosed floating-point value and no hash-only source decides a sign. Moreover,* `make -C ginibre_q3 clean-room-replay` *returns success only after the manifest, source-regeneration, status, source-structure, accepted-margin, fresh-build, main-theorem certificate-reachability coverage, and three-pass document checks stated in* `REPLAY.md` *all pass.*

Proof. The discrete checkers form lemma 2.3 and definition 2.10 with GMP integers or rationals; analytic checkers use directed MPFR and report positive lower margins. Each active character source is regenerated either by the checked root-datum Racah–Speiser/character-pairing path or by the proved bounded-Littlewood determinant over enough prime fields for unique CRT recovery before any stored moment is consumed. The isolated driver deletes copied build products, rebuilds the active checkers and both documents, extracts all certificate logs cited on the main-theorem dependency walk, and requires exact equality with the set of recomputed certificate stages. It also requires every such log to be hash-authenticated and every reachable post-$m = 29$ log to have accepted classification. It rejects any failed prerequisite listed in the statement. ☐

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